Integration Part 1 Integration Basics - Homework Answers

Revision | Warm-up

Throughout, $x > 0$ wherever there is a square root or a fractional index.

1 Year 11 Exercise 7B Q13

a $\dfrac{1}{\sqrt{x}} = \dfrac{1}{x^{\frac{1}{2}}} = x^{-\frac{1}{2}}$

b $\dfrac{12}{\sqrt{x}} = 12 \times \dfrac{1}{x^{\frac{1}{2}}} = 12x^{-\frac{1}{2}}$

c $-\dfrac{5}{\sqrt{x}} = -5 \times \dfrac{1}{x^{\frac{1}{2}}} = -5x^{-\frac{1}{2}}$

d $\dfrac{15}{\sqrt[3]{x}} = 15 \times \dfrac{1}{x^{\frac{1}{3}}} = 15x^{-\frac{1}{3}}$

e $-\dfrac{4}{\sqrt[3]{x^2}} = -4 \times \dfrac{1}{x^{\frac{2}{3}}} = -4x^{-\frac{2}{3}}$

f $x\sqrt{x} = x^1 \times x^{\frac{1}{2}} = x^{\frac{3}{2}}$
(also written $x^{1\frac{1}{2}}$)

g $\dfrac{5}{x\sqrt{x}} = \dfrac{5}{x^{\frac{3}{2}}} = 5x^{-\frac{3}{2}}$
(also written $5x^{-1\frac{1}{2}}$)

h $8x^2\sqrt{x} = 8x^2 \times x^{\frac{1}{2}} = 8x^{\frac{5}{2}}$
(also written $8x^{2\frac{1}{2}}$)

2 Year 11 Exercise 8E Q5

a $y = \dfrac{x^3}{x} + \dfrac{x^2}{x} = x^2 + x$, so $\dfrac{dy}{dx} = 2x + 1$.

b $y = \dfrac{x^3}{x^5} + \dfrac{x^2}{x^5} = x^{-2} + x^{-3}$, so $\dfrac{dy}{dx} = -2x^{-3} - 3x^{-4}$.

c $y = \dfrac{4x}{x^2} - \dfrac{5x^4}{x^2} = 4x^{-1} - 5x^2$, so $\dfrac{dy}{dx} = -4x^{-2} - 10x$.

d $y = \dfrac{6x^2}{2x^6} + \dfrac{6x^6}{2x^6} = 3x^{-4} + 3$, so $\dfrac{dy}{dx} = -12x^{-5}$.

3 Year 11 Exercise 8G Q4

a $\dfrac{dy}{dx} = 5 \times 5 \times (5x - 7)^4 = 25(5x - 7)^4$

b $\dfrac{dy}{dx} = 9 \times 5 \times 4 \times (5x + 3)^3 = 180(5x + 3)^3$

c $\dfrac{dy}{dx} = (-3) \times 7 \times (4 - 3x)^6 = -21(4 - 3x)^6$

d $\dfrac{dy}{dx} = 4 \times (-7) \times (4x - 5)^{-8} = -28(4x - 5)^{-8}$

e $\dfrac{dy}{dx} = 2 \times 3 \times (-5) \times (3x + 7)^{-6} = -30(3x + 7)^{-6}$

f $\dfrac{dy}{dx} = 3 \times (-7) \times (-4) \times (5 - 7x)^{-5} = 84(5 - 7x)^{-5}$

4 Year 11 Exercise 8G Q8

a Let $u = 5 - x^2$. Then $y = u^3$, $\frac{du}{dx} = -2x$ and $\frac{dy}{du} = 3u^2$.

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} = 3(5 - x^2)^2 \times (-2x) \\ &= -6x(5 - x^2)^2\end{aligned}$$

b Let $u = 3x^2 + 7$. Then $y = u^7$, $\frac{du}{dx} = 6x$ and $\frac{dy}{du} = 7u^6$.

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} = 7(3x^2 + 7)^6 \times 6x \\ &= 42x(3x^2 + 7)^6\end{aligned}$$

c Let $u = 3x^3 - 7$. Then $y = u^5$, $\frac{du}{dx} = 9x^2$ and $\frac{dy}{du} = 5u^4$.

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} = 5(3x^3 - 7)^4 \times 9x^2 \\ &= 45x^2(3x^3 - 7)^4\end{aligned}$$

d Let $u = x^3 - x^2$. Then $y = u^{-5}$, $\frac{du}{dx} = 3x^2 - 2x$ and $\frac{dy}{du} = -5u^{-6}$.

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} = -5(x^3 - x^2)^{-6} \times (3x^2 - 2x) \\ &= -5(3x^2 - 2x)(x^3 - x^2)^{-6}\end{aligned}$$

5 Year 12 Exercise 1E Q4

a $\displaystyle\sum_{n=1}^{6} 2n = 2 + 4 + 6 + 8 + 10 + 12 = 42$

b $\displaystyle\sum_{n=1}^{6} (3n + 2) = 5 + 8 + 11 + 14 + 17 + 20 = 75$

c $\displaystyle\sum_{n=5}^{8} n^2 = 25 + 36 + 49 + 64 = 174$

d $\displaystyle\sum_{n=0}^{5} 2^n = 1 + 2 + 4 + 8 + 16 + 32 = 63$

e $\displaystyle\sum_{n=2}^{4} 3^n = 9 + 27 + 81 = 117$

f $\displaystyle\sum_{n=5}^{105} 4 = 4 + 4 + 4 + \cdots + 4$. There are $105 - 5 + 1 = 101$ terms, so the sum is $101 \times 4 = 404$.

g $\displaystyle\sum_{n=0}^{4} (-1)^n(n + 5) = 5 - 6 + 7 - 8 + 9 = 7$

Integration Part 1 Integration Basics - Homework Answers

Lesson 4J | Primitive functions

Lesson Examples

Throughout, C and D are constants, and $x > 0$ wherever there is a square root or a fractional index of x.

C1 **Tutor demo.**

a $f(x) = (x+1)^2 = x^2 + 2x + 1$, so $f'(x) = 2x + 2$. Also $g'(x) = 2x + 2$.

b $f(x) - g(x) = (x^2 + 2x + 1) - (x^2 + 2x) = 1$.

The two functions have the same derivative for all x, and they differ by the constant 1, as Box 14B says.

Your turn.

a $f(x) = (x-3)^2 = x^2 - 6x + 9$, so $f'(x) = 2x - 6$. Also $g'(x) = 2x - 6$.

b $f(x) - g(x) = (x^2 - 6x + 9) - (x^2 - 6x + 2) = 7$.

The two functions have the same derivative for all x, and they differ by the constant 7, as Box 14B says.

C2 **Tutor demo.**

a For example, x^3, $x^3 + 1$, $x^3 - 4$ and $x^3 + \pi$:

$$\frac{d}{dx}(x^3) = 3x^2, \quad \frac{d}{dx}(x^3 + 1) = 3x^2 + 0 = 3x^2,$$
$$\frac{d}{dx}(x^3 - 4) = 3x^2 - 0 = 3x^2, \quad \frac{d}{dx}(x^3 + \pi) = 3x^2 + 0 = 3x^2.$$

b If $f'(x) = 3x^2$, then $f(x) = x^3 + C$, for some constant C.

c Substitute the point $(2, 5)$ into $f(x) = x^3 + C$: $5 = 2^3 + C$, $5 = 8 + C$, so $C = -3$. Hence $f(x) = x^3 - 3$.

Your turn.

a For example, x^4, $x^4 + 1$, $x^4 - 7$ and $x^4 + \pi$:

$$\frac{d}{dx}(x^4) = 4x^3, \quad \frac{d}{dx}(x^4 + 1) = 4x^3 + 0 = 4x^3,$$
$$\frac{d}{dx}(x^4 - 7) = 4x^3 - 0 = 4x^3, \quad \frac{d}{dx}(x^4 + \pi) = 4x^3 + 0 = 4x^3.$$

b If $f'(x) = 4x^3$, then $f(x) = x^4 + C$, for some constant C.

c Substitute the point $(1, -2)$ into $f(x) = x^4 + C$: $-2 = 1^4 + C$, $-2 = 1 + C$, so $C = -3$. Hence $f(x) = x^4 - 3$.

In part **a** any four functions of the form $x^3 + C$ (or $x^4 + C$) are correct.

C3 **Tutor demo.**

a Let $\frac{dy}{dx} = x^4 + x^3 + x^2 + 3$. Then

$y = \frac{x^5}{5} + \frac{x^4}{4} + \frac{x^3}{3} + 3x + C = \frac{1}{5}x^5 + \frac{1}{4}x^4 + \frac{1}{3}x^3 + 3x + C.$

b Let $f'(x) = 4x^5 - 2$. Then $f(x) = 4 \times \frac{x^6}{6} - 2x + C = \frac{2}{3}x^6 - 2x + C.$

c Expand first: $\frac{dy}{dx} = (2x+1)(x-3) = 2x^2 - 6x + x - 3 = 2x^2 - 5x - 3.$

Then $y = 2 \times \frac{x^3}{3} - 5 \times \frac{x^2}{2} - 3x + C = \frac{2}{3}x^3 - \frac{5}{2}x^2 - 3x + C.$

Your turn.

a Let $\frac{dy}{dx} = x^5 + x^2 + x + 4$. Then

$y = \frac{x^6}{6} + \frac{x^3}{3} + \frac{x^2}{2} + 4x + C = \frac{1}{6}x^6 + \frac{1}{3}x^3 + \frac{1}{2}x^2 + 4x + C.$

b Let $f'(x) = 3x^4 + 5$. Then $f(x) = 3 \times \frac{x^5}{5} + 5x + C = \frac{3}{5}x^5 + 5x + C.$

c Expand first: $\frac{dy}{dx} = (x+3)(x-1) = x^2 - x + 3x - 3 = x^2 + 2x - 3.$

Then $y = \frac{x^3}{3} + 2 \times \frac{x^2}{2} - 3x + C = \frac{1}{3}x^3 + x^2 - 3x + C.$

C4 **Tutor demo.**

a Let $f'(x) = \frac{4}{x^3} = 4x^{-3}$. Then

$f(x) = 4 \times \frac{x^{-2}}{-2} + C = -2x^{-2} + C = -\frac{2}{x^2} + C.$

b Let $\frac{dy}{dx} = \sqrt[3]{x^2} = x^{\frac{2}{3}}$. Then $y = \frac{x^{\frac{5}{3}}}{\frac{5}{3}} + C = \frac{3}{5}x^{\frac{5}{3}} + C.$

Your turn.

a Let $f'(x) = \frac{6}{x^4} = 6x^{-4}$. Then

$f(x) = 6 \times \frac{x^{-3}}{-3} + C = -2x^{-3} + C = -\frac{2}{x^3} + C.$

b Let $\frac{dy}{dx} = x\sqrt{x} = x^1 \times x^{\frac{1}{2}} = x^{\frac{3}{2}}$. Then $y = \frac{x^{\frac{5}{2}}}{\frac{5}{2}} + C = \frac{2}{5}x^{\frac{5}{2}} + C.$

C5 **Tutor demo.**

a Here $a = 2$ and $n = 3$: $y = \dfrac{(2x+5)^4}{4 \times 2} + C = \dfrac{(2x+5)^4}{8} + C.$

b Here $a = -2$ and $n = 5$: $y = \dfrac{(3-2x)^6}{6 \times (-2)} + C = -\dfrac{(3-2x)^6}{12} + C.$

c $\dfrac{dy}{dx} = \dfrac{1}{(x+3)^3} = (x+3)^{-3}$, with $a = 1$ and $n = -3$:

$y = \dfrac{(x+3)^{-2}}{-2} + C = -\dfrac{1}{2(x+3)^2} + C.$

d $\dfrac{dy}{dx} = \sqrt{4x+1} = (4x+1)^{\frac{1}{2}}$, with $a = 4$ and $n = \frac{1}{2}$:

$y = \dfrac{(4x+1)^{\frac{3}{2}}}{\frac{3}{2} \times 4} + C = \dfrac{(4x+1)^{\frac{3}{2}}}{6} + C.$

Your turn.

a Here $a = 4$ and $n = 5$: $y = \dfrac{(4x+3)^6}{6 \times 4} + C = \dfrac{(4x+3)^6}{24} + C.$

b Here $a = -5$ and $n = 4$: $y = \dfrac{(2-5x)^5}{5 \times (-5)} + C = -\dfrac{(2-5x)^5}{25} + C.$

c $\dfrac{dy}{dx} = \dfrac{1}{(x-4)^3} = (x-4)^{-3}$, with $a = 1$ and $n = -3$:

$y = \dfrac{(x-4)^{-2}}{-2} + C = -\dfrac{1}{2(x-4)^2} + C.$

d $\dfrac{dy}{dx} = \sqrt{x+3} = (x+3)^{\frac{1}{2}}$, with $a = 1$ and $n = \frac{1}{2}$:

$y = \dfrac{(x+3)^{\frac{3}{2}}}{\frac{3}{2}} + C = \frac{2}{3}(x+3)^{\frac{3}{2}} + C.$

C6 **Tutor demo.**

a $y = 12 \times \dfrac{x^3}{3} - 2x + C = 4x^3 - 2x + C$. Substituting $x = 1$ and $y = 5$:
$5 = 4 \times 1^3 - 2 \times 1 + C$, $5 = 4 - 2 + C$, so $C = 3$. Hence $y = 4x^3 - 2x + 3$.

b $y = \dfrac{(2x-1)^4}{4 \times 2} + C = \dfrac{(2x-1)^4}{8} + C$. Substituting $x = 1$ and $y = 2$:
$2 = \dfrac{(2-1)^4}{8} + C$, $2 = \frac{1}{8} + C$, so $C = \frac{15}{8}$. Hence $y = \dfrac{(2x-1)^4}{8} + \dfrac{15}{8}$.

Your turn.

a $y = 6 \times \dfrac{x^3}{3} + 2 \times \dfrac{x^2}{2} + C = 2x^3 + x^2 + C$. Substituting $x = 1$ and $y = 7$:
$7 = 2 + 1 + C$, so $C = 4$. Hence $y = 2x^3 + x^2 + 4$.

b $y = \dfrac{(3x+1)^3}{3 \times 3} + C = \dfrac{(3x+1)^3}{9} + C$. Substituting $x = 0$ and $y = 1$:
$1 = \dfrac{1^3}{9} + C$, $1 = \frac{1}{9} + C$, so $C = \frac{8}{9}$. Hence $y = \dfrac{(3x+1)^3}{9} + \dfrac{8}{9}$.

C7 **Tutor demo.**

a $f'(x) = 12 \times \dfrac{(x-2)^3}{3 \times 1} + C = 4(x-2)^3 + C$, then
$f(x) = 4 \times \dfrac{(x-2)^4}{4 \times 1} + Cx + D = (x-2)^4 + Cx + D$.
Because $f(0) = 0$: $0 = (0-2)^4 + C \times 0 + D$, $0 = 16 + D$, so $D = -16$.
Because $f(2) = 0$: $0 = (2-2)^4 + 2C + D$, $0 = 2C - 16$, so $C = 8$.
Hence $f(x) = (x-2)^4 + 8x - 16$, and
$f(3) = (3-2)^4 + 8 \times 3 - 16 = 1 + 24 - 16 = 9$.

b $y' = 6 \times \dfrac{x^2}{2} - 2x + C = 3x^2 - 2x + C$. When $x = 1$, $y' = 4$:
$4 = 3 - 2 + C$, so $C = 3$ and $y' = 3x^2 - 2x + 3$.
Then $y = 3 \times \dfrac{x^3}{3} - 2 \times \dfrac{x^2}{2} + 3x + D = x^3 - x^2 + 3x + D$. When $x = 1$, $y = 3$: $3 = 1 - 1 + 3 + D$, so $D = 0$.
Hence $y = x^3 - x^2 + 3x$.

Your turn.

a $f'(x) = 6 \times \dfrac{(x+1)^2}{2 \times 1} + C = 3(x+1)^2 + C$, then
$f(x) = 3 \times \dfrac{(x+1)^3}{3 \times 1} + Cx + D = (x+1)^3 + Cx + D$.
Because $f(0) = 0$: $0 = (0+1)^3 + C \times 0 + D$, $0 = 1 + D$, so $D = -1$.
Because $f(1) = 6$: $6 = (1+1)^3 + C \times 1 + D$, $6 = 8 + C - 1$, so $C = -1$.

Hence $f(x) = (x+1)^3 - x - 1$, and
$f(2) = (2+1)^3 - 2 - 1 = 27 - 2 - 1 = 24$.

b $y' = 12 \times \dfrac{x^2}{2} + 2x + C = 6x^2 + 2x + C$. When $x = 0$, $y' = 1$:
$1 = 0 + 0 + C$, so $C = 1$ and $y' = 6x^2 + 2x + 1$.

Then $y = 6 \times \dfrac{x^3}{3} + 2 \times \dfrac{x^2}{2} + x + D = 2x^3 + x^2 + x + D$. When $x = 0$,
$y = -2$: $-2 = 0 + 0 + 0 + D$, so $D = -2$.
Hence $y = 2x^3 + x^2 + x - 2$.

Textbook Homework

Each answer shows the rule applied, then the simplified primitive. C and D are constants.

1

a $\dfrac{x^{6+1}}{6+1} + C = \frac{1}{7}x^7 + C$

b $\dfrac{x^{3+1}}{3+1} + C = \frac{1}{4}x^4 + C$

c $\dfrac{x^{10+1}}{10+1} + C = \frac{1}{11}x^{11} + C$

d $3 \times \dfrac{x^2}{2} + C = \frac{3}{2}x^2 + C$

e $5 \times \dfrac{x^1}{1} + C = 5x + C$

f $5 \times \dfrac{x^{10}}{10} + C = \frac{1}{2}x^{10} + C$

g $21 \times \dfrac{x^7}{7} + C = 3x^7 + C$

h The derivative of a constant is zero, so the primitive is $0x + C = C$.

2

a $\dfrac{x^3}{3} + \dfrac{x^5}{5} + C = \frac{1}{3}x^3 + \frac{1}{5}x^5 + C$

b $4 \times \dfrac{x^4}{4} - 5 \times \dfrac{x^5}{5} + C = x^4 - x^5 + C$

c $2 \times \dfrac{x^3}{3} + 5 \times \dfrac{x^8}{8} + C = \frac{2}{3}x^3 + \frac{5}{8}x^8 + C$

d $\dfrac{x^3}{3} - \dfrac{x^2}{2} + x + C = \frac{1}{3}x^3 - \frac{1}{2}x^2 + x + C$

e $3x - 4 \times \dfrac{x^2}{2} + 16 \times \dfrac{x^8}{8} + C = 3x - 2x^2 + 2x^8 + C$

f $3 \times \dfrac{x^3}{3} - 4 \times \dfrac{x^4}{4} - 5 \times \dfrac{x^5}{5} + C = x^3 - x^4 - x^5 + C$

3

a $x(x-3) = x^2 - 3x$. Primitive: $\dfrac{x^3}{3} - 3 \times \dfrac{x^2}{2} + C = \frac{1}{3}x^3 - \frac{3}{2}x^2 + C$

b $(x+1)(x-2) = x^2 - 2x + x - 2 = x^2 - x - 2$. Primitive:

$\dfrac{x^3}{3} - \dfrac{x^2}{2} - 2x + C = \frac{1}{3}x^3 - \frac{1}{2}x^2 - 2x + C$

c $(3x-1)(x+4) = 3x^2 + 12x - x - 4 = 3x^2 + 11x - 4$. Primitive:

$3 \times \dfrac{x^3}{3} + 11 \times \dfrac{x^2}{2} - 4x + C = x^3 + \frac{11}{2}x^2 - 4x + C$

d $x^2(5x^3 - 4x) = 5x^5 - 4x^3$. Primitive:

$5 \times \dfrac{x^6}{6} - 4 \times \dfrac{x^4}{4} + C = \frac{5}{6}x^6 - x^4 + C$

e $2x^3(4x^4 + 1) = 8x^7 + 2x^3$. Primitive:

$8 \times \dfrac{x^8}{8} + 2 \times \dfrac{x^4}{4} + C = x^8 + \frac{1}{2}x^4 + C$

f $(x-3)(1+x^2) = x + x^3 - 3 - 3x^2$. Primitive:

$\dfrac{x^2}{2} + \dfrac{x^4}{4} - 3x - 3 \times \dfrac{x^3}{3} + C = \frac{1}{2}x^2 + \frac{1}{4}x^4 - 3x - x^3 + C$

4

a $y = x^2 + 3x + C$.

i $3 = 0 + 0 + C$, so $C = 3$ and $y = x^2 + 3x + 3$. **ii** $8 = 1 + 3 + C$, so $C = 4$ and $y = x^2 + 3x + 4$.

b $y = 9 \times \dfrac{x^3}{3} + 4x + C = 3x^3 + 4x + C$.

i $1 = 0 + 0 + C$, so $C = 1$ and $y = 3x^3 + 4x + 1$. **ii** $5 = 3 + 4 + C$, so $C = -2$ and $y = 3x^3 + 4x - 2$.

c $y = 3 \times \dfrac{x^3}{3} - 4 \times \dfrac{x^2}{2} + 7x + C = x^3 - 2x^2 + 7x + C$.

i $0 = 0 - 0 + 0 + C$, so $C = 0$ and $y = x^3 - 2x^2 + 7x$. **ii** $-1 = 1 - 2 + 7 + C$, so $C = -7$ and $y = x^3 - 2x^2 + 7x - 7$.

5

a $\dfrac{1}{x^2} = x^{-2}$. Primitive: $\dfrac{x^{-1}}{-1} + C = -\dfrac{1}{x} + C$

b $\dfrac{1}{x^3} = x^{-3}$. Primitive: $\dfrac{x^{-2}}{-2} + C = -\dfrac{1}{2x^2} + C$

c $-\dfrac{2}{x^3} = -2x^{-3}$. Primitive: $-2 \times \dfrac{x^{-2}}{-2} + C = x^{-2} + C = \dfrac{1}{x^2} + C$

d $-\dfrac{3}{x^4} = -3x^{-4}$. Primitive: $-3 \times \dfrac{x^{-3}}{-3} + C = x^{-3} + C = \dfrac{1}{x^3} + C$

e $\dfrac{1}{x^2} - \dfrac{1}{x^3} = x^{-2} - x^{-3}$. Primitive: $\dfrac{x^{-1}}{-1} - \dfrac{x^{-2}}{-2} + C = -\dfrac{1}{x} + \dfrac{1}{2x^2} + C$

6

a $\sqrt{x} = x^{\frac{1}{2}}$. Primitive: $\dfrac{x^{\frac{3}{2}}}{\frac{3}{2}} + C = \frac{2}{3}x^{\frac{3}{2}} + C$

b $\dfrac{1}{\sqrt{x}} = x^{-\frac{1}{2}}$. Primitive: $\dfrac{x^{\frac{1}{2}}}{\frac{1}{2}} + C = 2x^{\frac{1}{2}} + C = 2\sqrt{x} + C$

c $\sqrt[3]{x} = x^{\frac{1}{3}}$. Primitive: $\dfrac{x^{\frac{4}{3}}}{\frac{4}{3}} + C = \frac{3}{4}x^{\frac{4}{3}} + C$

d $\dfrac{2}{\sqrt{x}} = 2x^{-\frac{1}{2}}$. Primitive: $2 \times \dfrac{x^{\frac{1}{2}}}{\frac{1}{2}} + C = 4x^{\frac{1}{2}} + C = 4\sqrt{x} + C$

e $\sqrt[5]{x^3} = x^{\frac{3}{5}}$. Primitive: $\dfrac{x^{\frac{8}{5}}}{\frac{8}{5}} + C = \frac{5}{8}x^{\frac{8}{5}} + C$

7

a and **b**: $\dfrac{dy}{dx} = x^{\frac{1}{2}}$, so $y = \dfrac{x^{\frac{3}{2}}}{\frac{3}{2}} + C = \frac{2}{3}x^{\frac{3}{2}} + C$.

a $1 = 0 + C$, so $C = 1$ and $y = \frac{2}{3}x^{\frac{3}{2}} + 1$.

b $2 = \frac{2}{3} \times 9^{\frac{3}{2}} + C = \frac{2}{3} \times 27 + C = 18 + C$, so $C = -16$ and $y = \frac{2}{3}x^{\frac{3}{2}} - 16$.

8

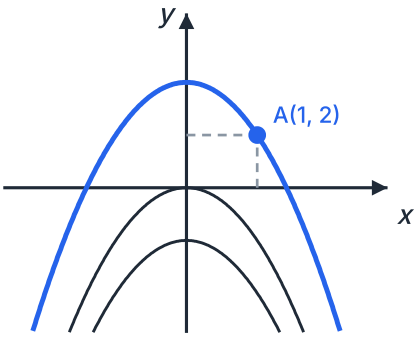
a $y = -4 \times \dfrac{x^2}{2} + C = -2x^2 + C$, the parabola $y = -2x^2$ translated up or down. At $A(1, 2)$: $2 = -2 + C$, so $C = 4$ and $y = 4 - 2x^2$.

b $y = 3x + C$, the parallel lines with gradient 3. At $A(1, 2)$: $2 = 3 + C$, so $C = -1$ and $y = 3x - 1$.

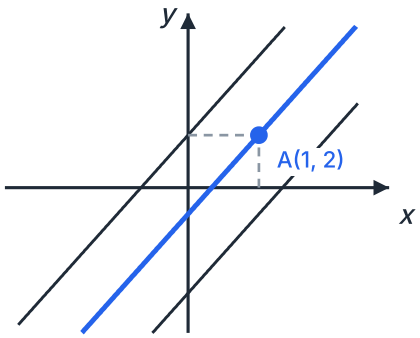
c $y = 3 \times \dfrac{x^3}{3} + C = x^3 + C$, the cubic $y = x^3$ translated up or down. At $A(1, 2)$: $2 = 1 + C$, so $C = 1$ and $y = x^3 + 1$.

d $\dfrac{dy}{dx} = -x^{-2}$, so $y = -\dfrac{x^{-1}}{-1} + C = \dfrac{1}{x} + C$, the hyperbola $y = \dfrac{1}{x}$ translated up or down. At $A(1, 2)$: $2 = 1 + C$, so $C = 1$ and $y = \dfrac{1}{x} + 1$.

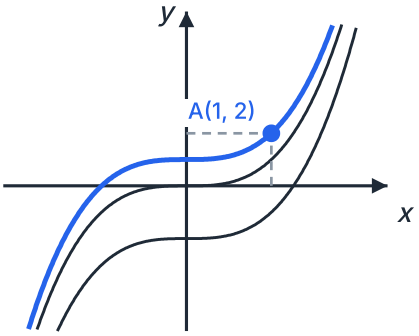
In each sketch the member through $A(1, 2)$ is blue.



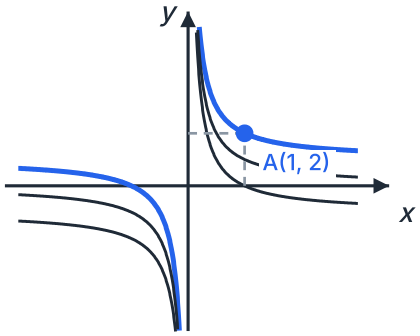
(a) $y = -2x^2 + C$, blue: $C = 4$



(b) $y = 3x + C$, blue: $C = -1$



(c) $y = x^3 + C$, blue: $C = 1$



(d) $y = x^{-1} + C$, blue: $C = 1$

9

a $\dfrac{(x+1)^4}{4\times 1} + C = \frac{1}{4}(x+1)^4 + C$

b $\dfrac{(x-2)^6}{6\times 1} + C = \frac{1}{6}(x-2)^6 + C$

c $\dfrac{(x+5)^3}{3\times 1} + C = \frac{1}{3}(x+5)^3 + C$

d $\dfrac{(2x+3)^5}{5\times 2} + C = \frac{1}{10}(2x+3)^5 + C$

e $\dfrac{(3x-4)^7}{7\times 3} + C = \frac{1}{21}(3x-4)^7 + C$

f $\dfrac{(5x-1)^4}{4\times 5} + C = \frac{1}{20}(5x-1)^4 + C$

g $\dfrac{(1-x)^4}{4\times(-1)} + C = -\frac{1}{4}(1-x)^4 + C$

h $\dfrac{(1-7x)^4}{4\times(-7)} + C = -\frac{1}{28}(1-7x)^4 + C$

i $(x-2)^{-4}$ has primitive $\dfrac{(x-2)^{-3}}{(-3)\times 1} + C = \dfrac{-1}{3(x-2)^3} + C$

j $(1-x)^{-10}$ has primitive $\dfrac{(1-x)^{-9}}{(-9)\times(-1)} + C = \dfrac{1}{9(1-x)^9} + C$

10

a $\dfrac{(x+1)^{\frac{3}{2}}}{\frac{3}{2}\times 1}+C=\frac{2}{3}(x+1)^{\frac{3}{2}}+C$

b $\dfrac{(x-5)^{\frac{3}{2}}}{\frac{3}{2}\times 1}+C=\frac{2}{3}(x-5)^{\frac{3}{2}}+C$

c $\dfrac{(1-x)^{\frac{3}{2}}}{\frac{3}{2}\times(-1)}+C=-\frac{2}{3}(1-x)^{\frac{3}{2}}+C$

d $\dfrac{(2x-7)^{\frac{3}{2}}}{\frac{3}{2}\times 2}+C=\frac{1}{3}(2x-7)^{\frac{3}{2}}+C$

e $\dfrac{(3x-4)^{\frac{3}{2}}}{\frac{3}{2}\times 3}+C=\frac{2}{9}(3x-4)^{\frac{3}{2}}+C$

11

a $y=\dfrac{(x-1)^5}{5\times 1}+C=\frac{1}{5}(x-1)^5+C$. Then $0=\frac{1}{5}\times 0^5+C$, so $C=0$ and $y=\frac{1}{5}(x-1)^5$.

b $y=\dfrac{(2x+1)^4}{4\times 2}+C=\frac{1}{8}(2x+1)^4+C$. Then $-1=\frac{1}{8}\times 1^4+C$, so $C=-\frac{9}{8}$ and $y=\frac{1}{8}(2x+1)^4-\frac{9}{8}$.

c $y=\dfrac{(2x+1)^{\frac{3}{2}}}{\frac{3}{2}\times 2}+C=\frac{1}{3}(2x+1)^{\frac{3}{2}}+C$. Then $\frac{1}{3}=\frac{1}{3}\times 1^{\frac{3}{2}}+C$, so $C=0$ and $y=\frac{1}{3}(2x+1)^{\frac{3}{2}}$.

12

a $y=3\times\dfrac{x^5}{5}-\dfrac{x^4}{4}+x+C=\frac{3}{5}x^5-\frac{1}{4}x^4+x+C$. Through the origin: $0=0-0+0+C$, so $C=0$ and $y=\frac{3}{5}x^5-\frac{1}{4}x^4+x$.

b $y=2x+3\times\dfrac{x^3}{3}-\dfrac{x^4}{4}+C=2x+x^3-\frac{1}{4}x^4+C$. Through $(2,6)$: $6=4+8-4+C$, so $C=-2$ and $y=-\frac{1}{4}x^4+x^3+2x-2$.

c $y=\dfrac{(2-5x)^4}{4\times(-5)}+C=-\frac{1}{20}(2-5x)^4+C$. Through $\left(\frac{1}{5},1\right)$: $1=-\frac{1}{20}\left(2-5\times\frac{1}{5}\right)^4+C=-\frac{1}{20}+C$, so $C=\frac{21}{20}$ and $y=-\frac{1}{20}(2-5x)^4+\frac{21}{20}$.

13

$y=8\times\dfrac{t^4}{4}-6\times\dfrac{t^3}{3}+5t+C=2t^4-2t^3+5t+C$. Then $4=0-0+0+C$, so $C=4$ and $y=2t^4-2t^3+5t+4$.

When $t=2$, $y=2\times 16-2\times 8+10+4=30$.

14 The rule would give the primitive of x^{-1} as $\frac{x^0}{0}$, which is undefined, because we cannot divide by zero. This problem is addressed in Chapter 6 (Lesson 6I in Part 2 of this course).

15 $y' = 6 \times \frac{x^2}{2} + 4x + C = 3x^2 + 4x + C$. When $x = 1$, $y' = 2$: $2 = 3 + 4 + C$, so $C = -5$ and $y' = 3x^2 + 4x - 5$.
Then $y = 3 \times \frac{x^3}{3} + 4 \times \frac{x^2}{2} - 5x + D = x^3 + 2x^2 - 5x + D$. When $x = 1$, $y = 4$: $4 = 1 + 2 - 5 + D$, so $D = 6$.
Hence $y = x^3 + 2x^2 - 5x + 6$.

16 **a** $f'(x) = 2 \times \frac{x^2}{2} - 10x + C = x^2 - 10x + C$. The tangent at $x = 3$ has gradient 20, so $f'(3) = 20$: $20 = 9 - 30 + C$, so $C = 41$. Hence $f'(x) = x^2 - 10x + 41$.
b $f(x) = \frac{x^3}{3} - 10 \times \frac{x^2}{2} + 41x + D = \frac{1}{3}x^3 - 5x^2 + 41x + D$. The graph passes through $(3, -34)$: $-34 = 9 - 45 + 123 + D$, so $D = -121$.
Hence $f(x) = \frac{1}{3}x^3 - 5x^2 + 41x - 121$. Then $f(0) = -121$, so the graph cuts the y-axis at $(0, -121)$.

17 $y' = 8x - 6 \times \frac{x^2}{2} + C = 8x - 3x^2 + C$, and
$y = 8 \times \frac{x^2}{2} - 3 \times \frac{x^3}{3} + Cx + D = 4x^2 - x^3 + Cx + D$.
Through $(1, 6)$: $6 = 4 - 1 + C + D$, so $C + D = 3$. Through $(-1, 8)$: $8 = 4 + 1 - C + D$, so $-C + D = 3$.
Adding, $2D = 6$, so $D = 3$ and $C = 0$. Hence $y = -x^3 + 4x^2 + 3$.

18 $f'(x) = -x^{-2}$, so on any interval not containing 0, $f(x) = -\dfrac{x^{-1}}{-1} + C = \dfrac{1}{x} + C$.

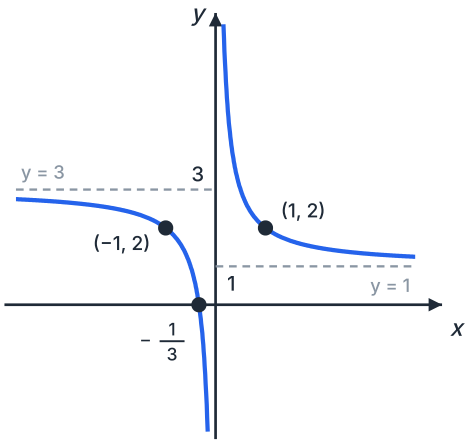
The function $\dfrac{1}{x}$ is undefined at $x = 0$, so the domain is two separate intervals, $x > 0$ and $x < 0$. Box 14 applies to each interval separately, so the constants on the two sides can be different:

for $x > 0$, $f(x) = \dfrac{1}{x} + C$, and $f(1) = 2$ gives $2 = 1 + C$, so $C = 1$;

for $x < 0$, $f(x) = \dfrac{1}{x} + D$, and $f(-1) = 2$ gives $2 = -1 + D$, so $D = 3$.

Hence $f(x) = \dfrac{1}{x} + 1$ for $x > 0$, and $f(x) = \dfrac{1}{x} + 3$ for $x < 0$.

The asymptotes are $x = 0$, with $y = 1$ on the right and $y = 3$ on the left. The left branch cuts the x-axis at $x = -\frac{1}{3}$, and the right branch does not cut it.



19 Write $p^{(k)}(x)$ for the kth derivative of $p(x)$.

a Let $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$, where $a_n \neq 0$. Differentiating lowers the index of each power of x by 1, and the derivative of a constant is zero. So after k derivatives, a term $a_j x^j$ with $j \geq k$ becomes $j(j-1)\cdots(j-k+1)\,a_j x^{j-k}$, and a term with $j < k$ has become zero.

After n derivatives every term with $j < n$ has vanished, and $a_n x^n$ has become the constant $n(n-1)\cdots 2 \times 1 \times a_n = n!\,a_n$. So $p^{(n)}(x) = n!\,a_n \neq 0$, because $a_n \neq 0$: the nth derivative does not vanish.

The $(n+1)$th derivative is the derivative of this constant, which is zero, and every higher derivative is the derivative of zero, which is zero. So the $(n+1)$th and higher derivatives vanish.

b The $(n+1)$th derivative is zero, so by Box 14A the nth derivative is a constant: $p^{(n)}(x) = c$, and $c \neq 0$ because the nth derivative does not vanish.

Now take the primitive n times. Each primitive raises every index by 1 and adds a new constant term:

$$p^{(n-1)}(x) = cx + C_1, \quad p^{(n-2)}(x) = \frac{c}{2}x^2 + C_1 x + C_2,$$

$$p^{(n-3)}(x) = \frac{c}{2 \times 3}x^3 + \frac{C_1}{2}x^2 + C_2 x + C_3, \quad \text{and so on, until}$$

$$p(x) = \frac{c}{n!}x^n + (\text{terms in lower powers of } x).$$

The coefficient $\dfrac{c}{n!}$ of x^n is not zero, so $p(x)$ has degree n.

Integration Part 1 Integration Basics - Homework Answers

Lesson 5A | Areas and the definite integral

Lesson Examples

C1 **Tutor demo.**

a $\int_1^4 x^3\,dx$. The integrand is x^3, the lower limit is $x = 1$ and the upper limit is $x = 4$. (For $1 \le x \le 4$, x^3 is positive, so Box 1 applies.)

b It is the region between the curve $y = 9 - x^2$ and the x-axis, from $x = 0$ to $x = 3$. For $0 \le x \le 3$, $x^2 \le 9$, so $9 - x^2 \ge 0$: the integrand is never negative in the interval.

c For $-2 \le x < -1$, $x + 1 < 0$, so part of the line $y = x + 1$ is below the x-axis. Box 1 only applies, for the moment, when the integrand is never negative in the interval.

Your turn.

a $\int_4^9 \sqrt{x}\,dx$. The integrand is $\sqrt{x}$, the lower limit is $x = 4$ and the upper limit is $x = 9$. ($\sqrt{x}$ is never negative.)

b It is the region between the curve $y = 4 - x^2$ and the x-axis, from $x = -2$ to $x = 2$. For $-2 \le x \le 2$, $x^2 \le 4$, so $4 - x^2 \ge 0$: the integrand is never negative in the interval.

c For $0 \le x < 1$, $x - 1 < 0$, so part of the line $y = x - 1$ is below the x-axis. Box 1 only applies, for the moment, when the integrand is never negative in the interval.

C2 **Tutor demo.**

a A triangle with base $5 - 2 = 3$ and height $5 - 2 = 3$:

$\int_2^5 (x - 2)\,dx = \frac{1}{2} \times 3 \times 3 = 4\frac{1}{2}$.

b A trapezium with width $5 - 3 = 2$ and parallel sides $3 - 2 = 1$ and $5 - 2 = 3$:

$\int_3^5 (x - 2)\,dx = \frac{1 + 3}{2} \times 2 = 2 \times 2 = 4$.

c Two triangles: base 4 and height 4 on the left, base 2 and height 2 on the right:

$\int_{-4}^2 |x|\,dx = \frac{1}{2} \times 4 \times 4 + \frac{1}{2} \times 2 \times 2 = 8 + 2 = 10$.

d $y = \sqrt{36 - x^2}$ is the upper semicircle with radius 6:

$\int_{-6}^6 \sqrt{36 - x^2}\,dx = \frac{1}{2} \times \pi \times 6^2 = 18\pi$.

Your turn.

a A triangle with base $6 - 2 = 4$ and height $6 - 2 = 4$:

$\int_2^6 (x - 2)\,dx = \frac{1}{2} \times 4 \times 4 = 8$.

b A trapezium with width $6 - 4 = 2$ and parallel sides $4 - 2 = 2$ and $6 - 2 = 4$:

$$\int_4^6 (x-2)\,dx = \frac{2+4}{2} \times 2 = 3 \times 2 = 6.$$

c Two triangles: base 2 and height 2 on the left, base 3 and height 3 on the right:

$$\int_{-2}^3 |x|\,dx = \tfrac{1}{2} \times 2 \times 2 + \tfrac{1}{2} \times 3 \times 3 = 2 + 4\tfrac{1}{2} = 6\tfrac{1}{2}.$$

d $y = \sqrt{4-x^2}$ is the upper semicircle with radius 2:

$$\int_{-2}^2 \sqrt{4-x^2}\,dx = \tfrac{1}{2} \times \pi \times 2^2 = 2\pi.$$

C3 **Tutor demo.** $y = x^3$ is increasing for $x \geq 0$, so lower rectangles use the left ends and upper rectangles use the right ends.

a For $0 \leq x \leq 1$, $0 \leq x^3 \leq 1$, so the region lies inside the unit square, which has area 1. Hence $0 < A < 1$.

b Two strips of width $\frac{1}{2}$. Lower: $\frac{1}{2} \times 0^3 + \frac{1}{2} \times \left(\frac{1}{2}\right)^3 = 0 + \frac{1}{2} \times \frac{1}{8} = \frac{1}{16}$. Upper: $\frac{1}{2} \times \left(\frac{1}{2}\right)^3 + \frac{1}{2} \times 1^3 = \frac{1}{16} + \frac{8}{16} = \frac{9}{16}$. So $\frac{1}{16} < A < \frac{9}{16}$.

c Four strips of width $\frac{1}{4}$, with heights $\left(\frac{k}{4}\right)^3 = \frac{k^3}{64}$. Lower: $\frac{1}{4} \times \frac{0+1+8+27}{64} = \frac{36}{256} = \frac{9}{64}$. Upper: $\frac{1}{4} \times \frac{1+8+27+64}{64} = \frac{100}{256} = \frac{25}{64}$. So $\frac{9}{64} < A < \frac{25}{64}$. (Exercise 5A Question 15 shows that $A = \frac{1}{4} = \frac{16}{64}$.)

Your turn.

a For $0 \leq x \leq 2$, $0 \leq x^2 \leq 4$, so the region lies inside the rectangle with width 2 and height 4, which has area 8. Hence $0 < A < 8$.

b Two strips of width 1. Lower: $1 \times 0^2 + 1 \times 1^2 = 1$. Upper: $1 \times 1^2 + 1 \times 2^2 = 1 + 4 = 5$. So $1 < A < 5$.

c Four strips of width $\frac{1}{2}$, with heights $0,\ \frac{1}{4},\ 1,\ \frac{9}{4},\ 4$ at $x = 0,\ \frac{1}{2},\ 1,\ \frac{3}{2},\ 2$. Lower: $\frac{1}{2}\left(0 + \frac{1}{4} + 1 + \frac{9}{4}\right) = \frac{1}{2} \times \frac{7}{2} = \frac{7}{4}$. Upper: $\frac{1}{2}\left(\frac{1}{4} + 1 + \frac{9}{4} + 4\right) = \frac{1}{2} \times \frac{15}{2} = \frac{15}{4}$. So $\frac{7}{4} < A < \frac{15}{4}$.

C4 **Tutor demo.** **A** The n subintervals have width $\frac{2}{n}$ and end at $x = \frac{2}{n},\ \frac{4}{n},\ \ldots,\ \frac{2n}{n} = 2$. As in the text, (sum of lower rectangles) $\leq A \leq$ (sum of upper rectangles).

B The upper heights are $\left(\frac{2}{n}\right)^2 = \frac{4 \times 1^2}{n^2},\ \left(\frac{4}{n}\right)^2 = \frac{4 \times 2^2}{n^2},\ \ldots,\ \left(\frac{2n}{n}\right)^2 = \frac{4 \times n^2}{n^2}$, so

$$\begin{aligned}\text{sum of upper rectangles} &= \frac{2}{n}\left(\frac{4\times 1^2}{n^2}+\frac{4\times 2^2}{n^2}+\cdots+\frac{4\times n^2}{n^2}\right)\\ &= \frac{2}{n}\times\frac{4}{n^2}\times\left(1^2+2^2+\cdots+n^2\right)\\ &= \frac{8}{n^3}\times\frac{n(n+1)(2n+1)}{6}\\ &= \frac{8\,n(n+1)(2n+1)}{6\times n\times n\times n}\\ &= \frac{4}{3}\times\frac{n}{n}\times\frac{n+1}{n}\times\frac{2n+1}{n}\\ &= \frac{4}{3}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right),\end{aligned}$$

which has limit $\frac{4}{3}\times 1\times 2=\frac{8}{3}$ as $n\to\infty$.

C The lower heights are $0,\ \frac{4\times 1^2}{n^2},\ \ldots,\ \frac{4(n-1)^2}{n^2}$, so, with $n-1$ in place of n in the formula,

$$\begin{aligned}\text{sum of lower rectangles} &= \frac{2}{n}\left(0+\frac{4\times 1^2}{n^2}+\cdots+\frac{4(n-1)^2}{n^2}\right)\\ &= \frac{8}{n^3}\left(1^2+2^2+\cdots+(n-1)^2\right)\\ &= \frac{8}{n^3}\times\frac{(n-1)n(2n-1)}{6}\\ &= \frac{4}{3}\times\frac{n}{n}\times\frac{n-1}{n}\times\frac{2n-1}{n}\\ &= \frac{4}{3}\left(1-\frac{1}{n}\right)\left(2-\frac{1}{n}\right),\end{aligned}$$

which also has limit $\frac{4}{3}\times 1\times 2=\frac{8}{3}$.

D A is trapped between two sums with the same limit $\frac{8}{3}$, so $A=\frac{8}{3}$.

Your turn. A The n subintervals have width $\frac{1}{n}$, and (sum of lower rectangles) $\le A\le$ (sum of upper rectangles).

B The upper heights are $\frac{1}{n},\ \frac{2}{n},\ \ldots,\ \frac{n}{n}$, so

$$\begin{aligned}\text{sum of upper rectangles} &= \frac{1}{n}\left(\frac{1}{n}+\frac{2}{n}+\cdots+\frac{n}{n}\right)\\ &= \frac{1}{n^2}\left(1+2+\cdots+n\right)\\ &= \frac{1}{n^2}\times\frac{n(n+1)}{2}\\ &= \frac{1}{2}\times\frac{n}{n}\times\frac{n+1}{n}\\ &= \frac{1}{2}\left(1+\frac{1}{n}\right),\end{aligned}$$

which has limit $\frac{1}{2}$ as $n \to \infty$.

C The lower heights are $0,\ \frac{1}{n},\ \ldots,\ \frac{n-1}{n}$, so

$$\begin{aligned}\textbf{sum of lower rectangles} &= \frac{1}{n}\left(0 + \frac{1}{n} + \cdots + \frac{n-1}{n}\right) \\ &= \frac{1}{n^2}\left(1 + 2 + \cdots + (n-1)\right) \\ &= \frac{1}{n^2} \times \frac{(n-1)n}{2} \\ &= \frac{1}{2} \times \frac{n}{n} \times \frac{n-1}{n} \\ &= \frac{1}{2}\left(1 - \frac{1}{n}\right),\end{aligned}$$

which also has limit $\frac{1}{2}$.

D Hence $A = \frac{1}{2}$. Check: the region is the triangle with base 1 and height 1, with area $\frac{1}{2} \times 1 \times 1 = \frac{1}{2}$.

Textbook Homework

1 **a** $\frac{1}{2} \times 1 \times 1 = \frac{1}{2}\ \text{u}^2$

b The area under the curve is less than the area of the triangle.

2 **a** $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{4} = \frac{1}{16}\ \text{u}^2$ **b** $\frac{1}{2}\left(\frac{1}{4} + 1\right) \times \frac{1}{2} = \frac{5}{16}\ \text{u}^2$

c The area under the curve is less than the combined area of the triangle and trapezium, $\frac{1}{16} + \frac{5}{16} = \frac{3}{8}$.

3 **b** The gaps between the upper line segments and the curve are getting smaller.

4 **a** 6 **b** 12 **c** 8 **d** 9 **e** 2 **f** $\frac{25}{2}$ **g** 6 **h** 20

5 **a** 8 **b** 25 **c** 9 **d** 24 **e** 36 **f** 24 **g** 9 **h** 8

6 **a** 15 **b** 15 **c** 25 **d** 40 **e** $\frac{25}{2}$ **f** 12 **g** 16 **h** 24
i 8 **j** 18 **k** 4 **l** 16 **m** 4 **n** 16 **o** $\frac{25}{2}$ **p** $\frac{25}{2}$

7 **a** 8π (a semicircle of radius 4) **b** $\frac{25}{4}\pi$ (a quadrant of radius 5)

8 **a** $\frac{1}{4}\left(0 + \frac{1}{16} + \frac{4}{16} + \frac{9}{16}\right) = \frac{7}{32}\ \text{u}^2$ **b** $\frac{1}{4}\left(\frac{1}{16} + \frac{4}{16} + \frac{9}{16} + 1\right) = \frac{15}{32}\ \text{u}^2$

c The sum of the areas of the lower rectangles is less than the exact area under the curve, which is less than the sum of the areas of the upper rectangles. Note that $\int_0^1 x^2\,dx = \frac{1}{3}$.

9 **a** $1 \times 2^0 = 1$ and $1 \times 2^1 = 2$ **b** $\frac{1}{2}\left(1 + \sqrt{2}\right) \doteq 1.21$ and $\frac{1}{2}\left(\sqrt{2} + 2\right) \doteq 1.71$

c about 1.32 and 1.57

d As the number of rectangles increases, the interval within which the exact area lies becomes smaller. Note that $\displaystyle\int_0^1 2^x \, dx = \frac{1}{\ln 2} \doteq 1.44$.

10 **a** $\ln 2 + \ln 3 \doteq 1.79$ and $\ln 3 + \ln 4 \doteq 2.48$

b about 1.98 and 2.33 **c** about 2.07 and 2.24

d As the number of rectangles increases, the interval within which the exact area lies becomes smaller. Note that $\displaystyle\int_2^4 \ln x \, dx = 6 \ln 2 - 2 \doteq 2.16$.

11 **e** The interval is getting smaller.

f Yes, they appear to be getting closer and closer to the exact value.

13 **a** You should count approximately 133 squares. $\frac{133}{400} \doteq 0.33$. We shall see later that $\displaystyle\int_0^1 x^2 \, dx = \frac{1}{3}$.

b The exact values are: (i) $\frac{1}{24}$ (ii) $\frac{7}{24}$

14 **a** about 314 squares **b** 0.79 **c** 3.16

17 **a** $\dfrac{1}{3} + \dfrac{1}{6n^2}$

b The lines $P_0P_1,\ P_1P_2,\ \ldots$ lie above the curve. Therefore the combined area of the trapezia is greater than the area under the curve.

Integration Part 1 Integration Basics - Homework Answers

Lesson 5A | Areas and the definite integral

Past-Paper Homework - Worked Solutions

PP 1 **James Ruse Agricultural High School, 2023, Year 12 Trial HSC Examination, Mathematics Extension 1, Q4**

From $x = 1$ to $x = 3$, the graph is the horizontal line $y = 2$. From $x = 3$ to $x = 5$, it is the line $y = x - 1$, which rises from $3 - 1 = 2$ to $5 - 1 = 4$. The two pieces meet at $(3, 2)$. So the region under the graph from $x = 1$ to $x = 5$ is a rectangle followed by a trapezium.

$$\begin{aligned}\int_1^5 f(x)\,dx &= \text{rectangle} + \text{trapezium} \\ &= (3-1) \times 2 + \frac{2+4}{2} \times (5-3) \\ &= 4 + 6 \\ &= 10.\end{aligned}$$

Answer D. Option B is the area of the trapezium alone, and option C treats the whole region as a rectangle of height 2.

Integration Part 1 Integration Basics - Homework Answers

Lesson 5B | The fundamental theorem of calculus

Lesson Examples

C1 **Tutor demo.**

a x^2 is a primitive of $2x$.

$$\begin{aligned}\int_0^3 2x\,dx &= \Big[x^2\Big]_0^3 \\ &= 3^2 - 0^2 \\ &= 9.\end{aligned}$$

Areas: a triangle with base $3 - 0 = 3$ and height $2 \times 3 = 6$, so its area is $\frac{1}{2} \times 3 \times 6 = 9$.

b

$$\begin{aligned}\int_2^5 (2x - 3)\,dx &= \Big[x^2 - 3x\Big]_2^5 \\ &= (25 - 15) - (4 - 6) \\ &= 10 - (-2) \\ &= 12.\end{aligned}$$

Areas: a trapezium with width $5 - 2 = 3$ and parallel sides $2 \times 2 - 3 = 1$ and $2 \times 5 - 3 = 7$, so its area is $\dfrac{1+7}{2} \times 3 = 4 \times 3 = 12$.

Your turn.

a $2x^2$ is a primitive of $4x$.

$$\begin{aligned}\int_0^1 4x\,dx &= \Big[2x^2\Big]_0^1 \\ &= 2 \times 1^2 - 2 \times 0^2 \\ &= 2.\end{aligned}$$

Areas: a triangle with base 1 and height $4 \times 1 = 4$, so its area is $\frac{1}{2} \times 1 \times 4 = 2$.

b

$$\begin{aligned}\int_3^5 (2x - 5)\,dx &= \Big[x^2 - 5x\Big]_3^5 \\ &= (25 - 25) - (9 - 15) \\ &= 0 - (-6) \\ &= 6.\end{aligned}$$

Areas: a trapezium with width $5 - 3 = 2$ and parallel sides $2 \times 3 - 5 = 1$ and $2 \times 5 - 5 = 5$, so its area is $\dfrac{1+5}{2} \times 2 = 3 \times 2 = 6$.

C2 Tutor demo.

a $\int_0^1 7x^2\,dx = \left[\frac{7}{3}x^3\right]_0^1 = \frac{7}{3} - 0 = 2\frac{1}{3}$

b

$$\begin{aligned}\int_0^3 (16 - x^2)\,dx &= \left[16x - \tfrac{1}{3}x^3\right]_0^3 \\ &= \left(48 - \tfrac{1}{3} \times 27\right) - (0 - 0) \\ &= 48 - 9 \\ &= 39\end{aligned}$$

c

$$\begin{aligned}\int_1^2 (3x^2 - 2x + 4)\,dx &= \left[x^3 - x^2 + 4x\right]_1^2 \\ &= (8 - 4 + 8) - (1 - 1 + 4) \\ &= 12 - 4 \\ &= 8\end{aligned}$$

d

$$\begin{aligned}\int_{-2}^1 (3x^2 + 1)\,dx &= \left[x^3 + x\right]_{-2}^1 \\ &= (1 + 1) - \left((-2)^3 + (-2)\right) \\ &= 2 - (-8 - 2) \\ &= 2 + 10 \\ &= 12\end{aligned}$$

Your turn.

a $\int_0^1 4x^2\,dx = \left[\frac{4}{3}x^3\right]_0^1 = \frac{4}{3} - 0 = 1\frac{1}{3}$

b

$$\begin{aligned}\int_0^2 (9 - x^2)\,dx &= \left[9x - \tfrac{1}{3}x^3\right]_0^2 \\ &= \left(18 - \tfrac{8}{3}\right) - (0 - 0) \\ &= \tfrac{54}{3} - \tfrac{8}{3} \\ &= 15\tfrac{1}{3}\end{aligned}$$

c

$$\begin{aligned}\int_1^3 (3x^2 + 2x - 1)\,dx &= \left[x^3 + x^2 - x\right]_1^3 \\ &= (27 + 9 - 3) - (1 + 1 - 1) \\ &= 33 - 1 \\ &= 32\end{aligned}$$

d

$$\begin{aligned}\int_{-1}^{2}(3x^2+2)\,dx &= \Big[x^3+2x\Big]_{-1}^{2}\\ &= (8+4)-\big((-1)^3+2\times(-1)\big)\\ &= 12-(-1-2)\\ &= 12+3\\ &= 15\end{aligned}$$

C3 **Tutor demo.**

a $x(x+2)=x^2+2x.$

$$\begin{aligned}\int_{1}^{3}x(x+2)\,dx &= \int_{1}^{3}(x^2+2x)\,dx\\ &= \Big[\frac{x^3}{3}+x^2\Big]_{1}^{3}\\ &= (9+9)-\Big(\frac{1}{3}+1\Big)\\ &= 18-\frac{4}{3}\\ &= 16\tfrac{2}{3}\end{aligned}$$

b $(x-5)(x-6)=x^2-6x-5x+30=x^2-11x+30.$

$$\begin{aligned}\int_{0}^{3}(x-5)(x-6)\,dx &= \Big[\frac{x^3}{3}-\frac{11x^2}{2}+30x\Big]_{0}^{3}\\ &= \Big(9-\frac{99}{2}+90\Big)-(0-0+0)\\ &= 99-49\tfrac{1}{2}\\ &= 49\tfrac{1}{2}\end{aligned}$$

Your turn.

a $x(x+3)=x^2+3x.$

$$\begin{aligned}\int_{1}^{2}x(x+3)\,dx &= \Big[\frac{x^3}{3}+\frac{3x^2}{2}\Big]_{1}^{2}\\ &= \Big(\frac{8}{3}+6\Big)-\Big(\frac{1}{3}+\frac{3}{2}\Big)\\ &= \frac{7}{3}+\frac{9}{2}\\ &= \frac{14}{6}+\frac{27}{6}\\ &= 6\tfrac{5}{6}\end{aligned}$$

b $(x-3)(x-4)=x^2-4x-3x+12=x^2-7x+12.$

$$\begin{aligned}\int_0^2 (x-3)(x-4)\,dx &= \left[\frac{x^3}{3} - \frac{7x^2}{2} + 12x\right]_0^2 \\ &= \left(\frac{8}{3} - 14 + 24\right) - (0 - 0 + 0) \\ &= \frac{8}{3} + 10 \\ &= 12\tfrac{2}{3}\end{aligned}$$

C4 **Tutor demo.**

a $\dfrac{4x^4 - 3x^2}{x^2} = \dfrac{4x^4}{x^2} - \dfrac{3x^2}{x^2} = 4x^2 - 3.$

$$\begin{aligned}\int_1^3 \frac{4x^4 - 3x^2}{x^2}\,dx &= \left[\frac{4x^3}{3} - 3x\right]_1^3 \\ &= (36 - 9) - \left(\frac{4}{3} - 3\right) \\ &= 27 - \left(-\frac{5}{3}\right) \\ &= 28\tfrac{2}{3}\end{aligned}$$

b $\dfrac{x^3 - 3x^4}{x^3} = \dfrac{x^3}{x^3} - \dfrac{3x^4}{x^3} = 1 - 3x.$

$$\begin{aligned}\int_{-3}^{-1} \frac{x^3 - 3x^4}{x^3}\,dx &= \left[x - \frac{3x^2}{2}\right]_{-3}^{-1} \\ &= \left(-1 - \frac{3}{2}\right) - \left(-3 - \frac{27}{2}\right) \\ &= -\frac{5}{2} + \frac{33}{2} \\ &= 14\end{aligned}$$

Your turn.

a $\dfrac{6x^4 + 2x^2}{x^2} = 6x^2 + 2.$

$$\begin{aligned}\int_1^2 \frac{6x^4 + 2x^2}{x^2}\,dx &= \left[2x^3 + 2x\right]_1^2 \\ &= (16 + 4) - (2 + 2) \\ &= 20 - 4 \\ &= 16\end{aligned}$$

b $\dfrac{2x^3 - x^4}{x^3} = 2 - x.$

$$\begin{aligned}\int_{-2}^{-1} \frac{2x^3 - x^4}{x^3}\,dx &= \left[2x - \frac{x^2}{2}\right]_{-2}^{-1} \\ &= \left(-2 - \frac{1}{2}\right) - (-4 - 2) \\ &= -\frac{5}{2} + 6 \\ &= 3\tfrac{1}{2}\end{aligned}$$

C5 **Tutor demo.**

a

$$\begin{aligned}\int_1^4 x^{-2}\,dx &= \left[-\frac{1}{x}\right]_1^4 \\ &= -\frac{1}{4} - (-1) \\ &= \frac{3}{4}\end{aligned}$$

b

$$\begin{aligned}\int_1^2 \frac{1}{x^3}\,dx &= \left[\frac{x^{-2}}{-2}\right]_1^2 \\ &= \left[-\frac{1}{2x^2}\right]_1^2 \\ &= -\frac{1}{8} - \left(-\frac{1}{2}\right) \\ &= -\frac{1}{8} + \frac{4}{8} \\ &= \frac{3}{8}\end{aligned}$$

c $y = \dfrac{1}{x^2}$ has an asymptote at $x = 0$, which lies between the limits -2 and 1. The function is not defined at $x = 0$, so it is not continuous in $[-2, 1]$, and the fundamental theorem cannot be used. The integral is meaningless.

Your turn.

a

$$\begin{aligned}\int_1^3 x^{-2}\,dx &= \left[-\frac{1}{x}\right]_1^3 \\ &= -\frac{1}{3} - (-1) \\ &= \frac{2}{3}\end{aligned}$$

b

$$\begin{aligned}\int_1^2 \frac{1}{x^5}\,dx &= \left[\frac{x^{-4}}{-4}\right]_1^2 \\ &= \left[-\frac{1}{4x^4}\right]_1^2 \\ &= -\frac{1}{64} - \left(-\frac{1}{4}\right) \\ &= -\frac{1}{64} + \frac{16}{64} \\ &= \frac{15}{64}\end{aligned}$$

c $y = x^{-3} = \dfrac{1}{x^3}$ has an asymptote at $x = 0$, which lies between the limits -1 and 2. The function is not continuous in $[-1, 2]$, so the integral is meaningless.

C6 **Tutor demo.**

a The circumference is $2\pi r = 2\pi \times 4 = 8\pi$. It is divided equally between the top and bottom sides, so the length is $\frac{1}{2} \times 8\pi = 4\pi$.

b The height is the radius, 4. Area $= 4\pi \times 4 = 16\pi$, and $\pi r^2 = \pi \times 4^2 = 16\pi$, which agrees.

Your turn.

a The circumference is $2\pi \times 7 = 14\pi$, so the length is $\frac{1}{2} \times 14\pi = 7\pi$.

b The height is 7. Area $= 7\pi \times 7 = 49\pi$, and $\pi r^2 = \pi \times 7^2 = 49\pi$, which agrees.

Textbook Homework

1 **a** 1 **b** 15 **c** 16 **d** 84 **e** 19 **f** 243 **g** 62 **h** 2 **i** 1

2 **a** (i) 4 (ii) 25 (iii) 1 (Note that $\int_4^5 dx$ means $\int_4^5 1\,dx$.)

b Each function is a horizontal line, so each integral is a rectangle.

3 **a** 30 **b** 6 **c** 33 **d** 18 **e** 132 **f** 2 **g** 23 **h** 44 **i** 60

4 **a** 2 **b** 2 **c** 9 **d** 30 **e** 96 **f** 10

5 **a** $13\frac{1}{2}$ **b** $4\frac{2}{3}$ **c** $29\frac{1}{4}$ **d** 2 **e** $20\frac{5}{6}$ **f** 98

6 **a** 24 **b** 18 **c** $2\frac{2}{3}$ **d** 21 **e** $\frac{1}{4}$ **f** $\frac{8}{15}$

7 **a** 42 **b** 14 **c** 62 **d** $8\frac{1}{3}$ **e** $6\frac{2}{3}$ **f** 6

8 **a** $\frac{1}{24}$ **b** $\frac{20}{27}$ **c** $\frac{7}{8}$

9 **a** (i) $\frac{1}{10}$ (ii) $\frac{5}{36}$ (iii) 15 **b** (i) $\frac{1}{2}$ (ii) $\frac{15}{32}$ (iii) 7

10 **a** (ii) $k = 8$ **b** (ii) $k = 6$

11 **a** $k = 1$ **b** $k = 4$ **c** $k = 8$ **d** $k = 3$ **e** $k = 3$ **f** $k = 2$

12 **a** $1 + \frac{\pi}{2}$ **b** $2\frac{1}{2}$

13 **a** $\frac{3}{2}$ **b** $\frac{5}{8}$ **c** $42\frac{1}{3}$

14 **a** $13\frac{1}{3}$ **b** $8\frac{59}{120}$ **c** $\frac{1}{24}$

15 **a** x^2 is never negative.

b The function has an asymptote $x = 0$, which lies in the given interval. Hence the integral is meaningless and the use of the fundamental theorem is invalid.

c Part (ii) is meaningless because it crosses the asymptote at $x = 3$.

16 **a** (i) x^2 (ii) $x^3 + 3x$ (iii) $\dfrac{1}{x}$ (iv) $(x^3 - 3)^4$

Integration Part 1 Integration Basics - Homework Answers

Lesson 5B | The fundamental theorem of calculus

Past-Paper Homework - Worked Solutions

PP 1 **North Sydney Boys High School, 2022, HSC Course Assessment Task 3-Trial Examination, Mathematics Extension 1, Q10**

Read f from the graph. From $x = -1$ to $x = 0$, the graph is the straight line through $(-1, 0)$ and $(0, 1)$, which has gradient 1 and y-intercept 1, so $f(x) = x + 1$ there.

Now look at the input $x^2 - 1$. For $-1 \leq x \leq 1$, $0 \leq x^2 \leq 1$, so $-1 \leq x^2 - 1 \leq 0$. So the input always lies on that left-hand line, and

$$f(x^2 - 1) = (x^2 - 1) + 1 = x^2 \quad \text{for } -1 \leq x \leq 1.$$

Hence, by the fundamental theorem,

$$\begin{aligned} \int_{-1}^{1} f(x^2 - 1)\, dx &= \int_{-1}^{1} x^2\, dx \\ &= \left[\frac{x^3}{3}\right]_{-1}^{1} \\ &= \frac{1}{3} - \left(-\frac{1}{3}\right) \\ &= \frac{2}{3}. \end{aligned}$$

Answer D. Option B is $\int_{0}^{1} x^2\, dx$, only half of the interval.

Integration Part 1 Integration Basics - Homework Answers

Lesson 5C | The definite integral and its properties

Lesson Examples

C1 **Tutor demo.** The line $y = x - 3$ crosses the x-axis at $x = 3$. It is below the axis for $0 \leq x < 3$ and above it for $3 < x \leq 5$.

a
$$\begin{aligned}\int_0^3 (x-3)\,dx &= \left[\tfrac{1}{2}x^2 - 3x\right]_0^3 \\ &= \left(\tfrac{1}{2} \times 3^2 - 3 \times 3\right) - \left(\tfrac{1}{2} \times 0^2 - 3 \times 0\right) \\ &= \left(4\tfrac{1}{2} - 9\right) - (0 - 0) \\ &= -4\tfrac{1}{2}\end{aligned}$$

b
$$\begin{aligned}\int_3^5 (x-3)\,dx &= \left[\tfrac{1}{2}x^2 - 3x\right]_3^5 \\ &= \left(\tfrac{1}{2} \times 5^2 - 3 \times 5\right) - \left(\tfrac{1}{2} \times 3^2 - 3 \times 3\right) \\ &= \left(12\tfrac{1}{2} - 15\right) - \left(4\tfrac{1}{2} - 9\right) \\ &= -2\tfrac{1}{2} - \left(-4\tfrac{1}{2}\right) \\ &= 2\end{aligned}$$

c
$$\begin{aligned}\int_0^5 (x-3)\,dx &= \left[\tfrac{1}{2}x^2 - 3x\right]_0^5 \\ &= \left(\tfrac{1}{2} \times 5^2 - 3 \times 5\right) - \left(\tfrac{1}{2} \times 0^2 - 3 \times 0\right) \\ &= \left(12\tfrac{1}{2} - 15\right) - (0 - 0) \\ &= -2\tfrac{1}{2}\end{aligned}$$

The triangle from $x = 0$ to $x = 3$ is below the x-axis and has area $\frac{1}{2} \times 3 \times 3 = 4\frac{1}{2}$, so the answer $-4\frac{1}{2}$ in part **a** is expected. The triangle from $x = 3$ to $x = 5$ is above the axis and has area $\frac{1}{2} \times 2 \times 2 = 2$, as in part **b**. Part **c** is the upper area minus the lower area, $2 - 4\frac{1}{2} = -2\frac{1}{2}$.

Your turn. The line $y = x - 2$ is below the x-axis for $0 \leq x < 2$ and above it for $2 < x \leq 5$.

a
$$\begin{aligned}\int_0^2 (x-2)\,dx &= \left[\tfrac{1}{2}x^2 - 2x\right]_0^2 \\ &= \left(\tfrac{1}{2} \times 2^2 - 2 \times 2\right) - \left(\tfrac{1}{2} \times 0^2 - 2 \times 0\right) \\ &= (2 - 4) - (0 - 0) \\ &= -2\end{aligned}$$

b $$\begin{aligned}\int_2^5 (x-2)\,dx &= \left[\tfrac{1}{2}x^2 - 2x\right]_2^5 \\ &= \left(\tfrac{1}{2}\times 5^2 - 2\times 5\right) - \left(\tfrac{1}{2}\times 2^2 - 2\times 2\right) \\ &= \left(12\tfrac{1}{2} - 10\right) - (2-4) \\ &= 2\tfrac{1}{2} - (-2) \\ &= 4\tfrac{1}{2}\end{aligned}$$

c $$\begin{aligned}\int_0^5 (x-2)\,dx &= \left[\tfrac{1}{2}x^2 - 2x\right]_0^5 \\ &= \left(\tfrac{1}{2}\times 5^2 - 2\times 5\right) - \left(\tfrac{1}{2}\times 0^2 - 2\times 0\right) \\ &= \left(12\tfrac{1}{2} - 10\right) - (0-0) \\ &= 2\tfrac{1}{2}\end{aligned}$$

The lower triangle has area $\frac{1}{2}\times 2\times 2 = 2$ (part **a** is -2), the upper triangle has area $\frac{1}{2}\times 3\times 3 = 4\frac{1}{2}$ (part **b**), and part **c** is $4\frac{1}{2} - 2 = 2\frac{1}{2}$.

C2 **Tutor demo. a** $y = x^3 - 9x = x(x-3)(x+3)$ has zeroes -3, 0 and 3, with a hump above the axis from -3 to 0 and an equal hump below it from 0 to 3. Since $(-x)^3 - 9(-x) = -x^3 + 9x = -(x^3 - 9x)$, the integrand is odd, so $\int_{-3}^{3}(x^3 - 9x)\,dx = 0$. Without symmetry:

$$\begin{aligned}\int_{-3}^{3}(x^3 - 9x)\,dx &= \left[\tfrac{1}{4}x^4 - \tfrac{9}{2}x^2\right]_{-3}^{3} \\ &= \left(\tfrac{1}{4}\times 3^4 - \tfrac{9}{2}\times 3^2\right) - \left(\tfrac{1}{4}\times (-3)^4 - \tfrac{9}{2}\times (-3)^2\right) \\ &= \left(\tfrac{81}{4} - \tfrac{81}{2}\right) - \left(\tfrac{81}{4} - \tfrac{81}{2}\right) \\ &= 0\end{aligned}$$

b $y = x^2 + 1$ is a parabola with vertex $(0, 1)$. Since $(-x)^2 + 1 = x^2 + 1$, the integrand is even, so

$$\begin{aligned}\int_{-3}^{3}(x^2+1)\,dx &= 2\int_0^3 (x^2+1)\,dx \\ &= 2\left[\tfrac{1}{3}x^3 + x\right]_0^3 \\ &= 2\left(\left(\tfrac{1}{3}\times 3^3 + 3\right) - \left(\tfrac{1}{3}\times 0^3 + 0\right)\right) \\ &= 2\left((9+3) - (0+0)\right) \\ &= 2\times 12 \\ &= 24\end{aligned}$$

Your turn. a $y = x^3 - x = x(x-1)(x+1)$ has zeroes -1, 0 and 1. Since $(-x)^3 - (-x) = -x^3 + x = -(x^3 - x)$, the integrand is odd, so

$\int_{-1}^{1}(x^3-x)\,dx = 0$. Without symmetry:

$$\begin{aligned}\int_{-1}^{1}(x^3-x)\,dx &= \left[\tfrac{1}{4}x^4-\tfrac{1}{2}x^2\right]_{-1}^{1}\\ &= \left(\tfrac{1}{4}\times 1^4-\tfrac{1}{2}\times 1^2\right)-\left(\tfrac{1}{4}\times(-1)^4-\tfrac{1}{2}\times(-1)^2\right)\\ &= \left(\tfrac{1}{4}-\tfrac{1}{2}\right)-\left(\tfrac{1}{4}-\tfrac{1}{2}\right)\\ &= 0\end{aligned}$$

b $y=3x^2+1$ is a parabola with vertex $(0,1)$. Since $3(-x)^2+1=3x^2+1$, the integrand is even, so

$$\begin{aligned}\int_{-1}^{1}(3x^2+1)\,dx &= 2\int_{0}^{1}(3x^2+1)\,dx\\ &= 2\Big[x^3+x\Big]_{0}^{1}\\ &= 2\big((1^3+1)-(0^3+0)\big)\\ &= 2\times 2\\ &= 4\end{aligned}$$

C3 **Tutor demo. a** Left-hand side:

$\int_{0}^{3}x^2\,dx = \left[\tfrac{1}{3}x^3\right]_{0}^{3} = \tfrac{1}{3}\times 3^3-\tfrac{1}{3}\times 0^3 = 9-0 = 9.$

Right-hand side:

$\int_{0}^{1}x^2\,dx+\int_{1}^{3}x^2\,dx = \left[\tfrac{1}{3}x^3\right]_{0}^{1}+\left[\tfrac{1}{3}x^3\right]_{1}^{3} = \left(\tfrac{1}{3}-0\right)+\left(9-\tfrac{1}{3}\right) = 9$, the same (Box 7 with $c=1$).

b $\int_{2}^{2}(x^3+5)\,dx = 0$, because the interval has zero width (Box 8).

c

$$\begin{aligned}\int_{1}^{3}(x+1)\,dx &= \left[\tfrac{1}{2}x^2+x\right]_{1}^{3}\\ &= \left(\tfrac{1}{2}\times 3^2+3\right)-\left(\tfrac{1}{2}\times 1^2+1\right)\\ &= \left(4\tfrac{1}{2}+3\right)-\left(\tfrac{1}{2}+1\right)\\ &= 7\tfrac{1}{2}-1\tfrac{1}{2}\\ &= 6\end{aligned}$$

$$\begin{aligned}\int_3^1 (x+1)\,dx &= \left[\tfrac{1}{2}x^2 + x\right]_3^1 \\ &= \left(\tfrac{1}{2}\times 1^2 + 1\right) - \left(\tfrac{1}{2}\times 3^2 + 3\right) \\ &= \left(\tfrac{1}{2} + 1\right) - \left(4\tfrac{1}{2} + 3\right) \\ &= 1\tfrac{1}{2} - 7\tfrac{1}{2} \\ &= -6\end{aligned}$$

The two integrals are opposites, because the second runs backwards from $x = 3$ to $x = 1$ (Box 9).

Your turn. a Left-hand side: $\int_0^4 x\,dx = \left[\tfrac{1}{2}x^2\right]_0^4 = \tfrac{1}{2}\times 4^2 - \tfrac{1}{2}\times 0^2 = 8 - 0 = 8.$

Right-hand side:

$\int_0^1 x\,dx + \int_1^4 x\,dx = \left[\tfrac{1}{2}x^2\right]_0^1 + \left[\tfrac{1}{2}x^2\right]_1^4 = \left(\tfrac{1}{2} - 0\right) + \left(8 - \tfrac{1}{2}\right) = 8$, the same (Box 7 with $c = 1$).

b $\int_{-3}^{-3} (x^2 - 4x)\,dx = 0$, because the interval has zero width (Box 8).

c

$$\begin{aligned}\int_2^5 (x-2)\,dx &= \left[\tfrac{1}{2}x^2 - 2x\right]_2^5 \\ &= \left(\tfrac{1}{2}\times 5^2 - 2\times 5\right) - \left(\tfrac{1}{2}\times 2^2 - 2\times 2\right) \\ &= \left(12\tfrac{1}{2} - 10\right) - (2 - 4) \\ &= 2\tfrac{1}{2} + 2 \\ &= 4\tfrac{1}{2}\end{aligned}$$

$$\begin{aligned}\int_5^2 (x-2)\,dx &= \left[\tfrac{1}{2}x^2 - 2x\right]_5^2 \\ &= \left(\tfrac{1}{2}\times 2^2 - 2\times 2\right) - \left(\tfrac{1}{2}\times 5^2 - 2\times 5\right) \\ &= (2 - 4) - \left(12\tfrac{1}{2} - 10\right) \\ &= -2 - 2\tfrac{1}{2} \\ &= -4\tfrac{1}{2}\end{aligned}$$

The two integrals are opposites, because the second runs backwards from $x = 5$ to $x = 2$ (Box 9).

C4 Tutor demo.

a $$\int_0^2 (x^2+x+1)\,dx = \left[\frac{x^3}{3}+\frac{x^2}{2}+x\right]_0^2$$
$$= \left(\frac{2^3}{3}+\frac{2^2}{2}+2\right)-(0+0+0)$$
$$= 2\tfrac{2}{3}+2+2$$
$$= 6\tfrac{2}{3}$$

b $$\int_0^2 x^2\,dx+\int_0^2 x\,dx+\int_0^2 1\,dx = \left[\frac{x^3}{3}\right]_0^2+\left[\frac{x^2}{2}\right]_0^2+\left[x\right]_0^2$$
$$= \left(2\tfrac{2}{3}-0\right)+(2-0)+(2-0)$$
$$= 6\tfrac{2}{3},\ \text{the same as in part a}$$

c $$\int_1^2 6x^3\,dx = \left[\frac{6x^4}{4}\right]_1^2$$
$$= \frac{6\times 2^4}{4}-\frac{6\times 1^4}{4}$$
$$= \frac{96}{4}-\frac{6}{4}$$
$$= \frac{90}{4}$$
$$= 22\tfrac{1}{2}$$

d $$6\int_1^2 x^3\,dx = 6\times\left[\frac{x^4}{4}\right]_1^2$$
$$= 6\times\left(\frac{16}{4}-\frac{1}{4}\right)$$
$$= 6\times\frac{15}{4}$$
$$= \frac{90}{4}$$
$$= 22\tfrac{1}{2}$$

Your turn.

a $$\int_0^3 (x^2+1)\,dx = \left[\frac{x^3}{3}+x\right]_0^3$$
$$= \left(\frac{3^3}{3}+3\right)-(0+0)$$
$$= 9+3$$
$$= 12$$

b $$\int_0^3 x^2\,dx+\int_0^3 1\,dx = \left[\frac{x^3}{3}\right]_0^3+\left[x\right]_0^3$$
$$= (9-0)+(3-0)$$
$$= 12,\ \text{the same}$$

c $$\begin{aligned}\int_1^2 4x^3\,dx &= \Big[\frac{4x^4}{4}\Big]_1^2 \\ &= \big[x^4\big]_1^2 \\ &= 2^4 - 1^4 \\ &= 16 - 1 \\ &= 15\end{aligned}$$

d $$\begin{aligned}4\int_1^2 x^3\,dx &= 4 \times \Big[\frac{x^4}{4}\Big]_1^2 \\ &= 4 \times \Big(\frac{16}{4} - \frac{1}{4}\Big) \\ &= 4 \times \frac{15}{4} \\ &= 15, \text{ the same}\end{aligned}$$

C5 **Tutor demo. a** $y = 9 - x^2$ is a parabola with vertex $(0, 9)$ and x-intercepts -3 and 3, concave down.

b Since $x^2 \geq 0$, $9 - x^2 \leq 9$. For $-3 \leq x \leq 3$, $x^2 \leq 9$, so $9 - x^2 \geq 0$. Hence $0 \leq 9 - x^2 \leq 9$, and the region lies inside the rectangle with width $3 - (-3) = 6$ and height 9, which has area $6 \times 9 = 54$. By Box 12, $\int_{-3}^{3} 0\,dx \leq \int_{-3}^{3} (9 - x^2)\,dx \leq \int_{-3}^{3} 9\,dx$, that is, $0 \leq \int_{-3}^{3} (9 - x^2)\,dx \leq 54$.

c The integrand is even, so

$$\begin{aligned}\int_{-3}^{3} (9 - x^2)\,dx &= 2\int_0^3 (9 - x^2)\,dx \\ &= 2\Big[9x - \tfrac{1}{3}x^3\Big]_0^3 \\ &= 2\Big(\big(9 \times 3 - \tfrac{1}{3} \times 3^3\big) - \big(9 \times 0 - \tfrac{1}{3} \times 0^3\big)\Big) \\ &= 2\big((27 - 9) - 0\big) \\ &= 2 \times 18 \\ &= 36,\end{aligned}$$

and $0 \leq 36 \leq 54$.

Your turn. a $y = 1 - x^2$ is a parabola with vertex $(0, 1)$ and x-intercepts -1 and 1, concave down.

b Since $x^2 \geq 0$, $1 - x^2 \leq 1$. For $-1 \leq x \leq 1$, $x^2 \leq 1$, so $1 - x^2 \geq 0$. Hence the region lies inside the rectangle with width $1 - (-1) = 2$ and height 1, which has area $2 \times 1 = 2$. By Box 12, $0 \leq \int_{-1}^{1} (1 - x^2)\,dx \leq 2$.

c The integrand is even, so

$$\begin{aligned}\int_{-1}^{1}(1-x^2)\,dx &= 2\int_{0}^{1}(1-x^2)\,dx \\ &= 2\left[x-\tfrac{1}{3}x^3\right]_0^1 \\ &= 2\Big(\big(1-\tfrac{1}{3}\big)-(0-0)\Big) \\ &= 2\times\tfrac{2}{3} \\ &= \tfrac{4}{3},\end{aligned}$$

and $0 \leq \frac{4}{3} \leq 2$.

Textbook Homework

1 $\int_4^5 (2x-3)\,dx = \left[x^2-3x\right]_4^5 = (25-15)-(16-12) = 6$ and $\int_5^4 (2x-3)\,dx = \left[x^2-3x\right]_5^4 = (16-12)-(25-15) = -6$.
The values are 6 and -6, which differ by a factor of -1.

2 **a** $LHS = RHS = 2$ **b** $LHS = RHS = 6\frac{3}{4}$ **c** $LHS = RHS = 0$

3 **a** The interval has width zero. **b** $y = x$ is an odd function.

4 **a** The area is below the x-axis. **b** The area is above the x-axis.
c The areas above and below the x-axis are equal.
d The area below the x-axis is greater than the area above.

5 **a** The area is above the x-axis. **b** The area is below the x-axis.
c $y = 1-x^2$ is an even function and so is symmetrical about the y-axis.
d The area under the parabola from 0 to $\frac{1}{2}$ is greater than the area from $\frac{1}{2}$ to 1.

6 **a** -7 **b** 5

7 The area under the line $y = 2x$ from $x = 0$ to $x = 1$ is greater than the area under $y = x$.

8 $\int_{-2}^{1} x^3\,dx$. The area below the x-axis is greater than the area above.

9 **a** i 6 ii -6. The integrals are opposites because the limits have been reversed.
b i 5 ii 5. The factor 20 can be taken out of the integral.
c i 45 ii 30 iii 15. An integral of a sum is the sum of the integrals.
d i 48 ii 3 iii 45. The interval $0 \leq x \leq 2$ can be dissected into the intervals $0 \leq x \leq 1$ and $1 \leq x \leq 2$.
e i 0 ii 0. An integral over an interval of zero width is zero.

10 **a** 0. The interval has zero width. **b** 0. The interval has zero width.
c 0. The integrand is odd. **d** 0. The integrand is odd.
e 0. The integrand is odd. **f** 0. The integrand is odd.

11 **a** The curves meet at $(0, 0)$ and at $(1, 1)$.
b In the interval $0 \leq x \leq 1$, the curve $y = x^3$ is below the curve $y = x^2$.
c $\frac{1}{4}$ and $\frac{1}{3}$

12 **a** $\frac{\pi}{4} - \frac{1}{2} - \frac{1}{4} + \frac{1}{4} + \frac{\pi}{4} = \frac{\pi}{2} - \frac{1}{2}$ **b** $\left(1 - \frac{\pi}{4}\right) + \frac{1}{2} - \frac{1}{2} - \frac{\pi}{4} = 1 - \frac{\pi}{2}$

13 **a** i 5 ii $\frac{25}{2}$
b i 25 ii $17\frac{1}{2}$ iii $27\frac{1}{2}$

14 **a** i $\frac{1}{2}$ ii 18 iii 8
b i $-\frac{1}{2}$ ii -18 iii -8

15 **b** The function is even, so its graph is symmetrical about the y-axis.

16 **a** false **c** false **d** false

17 **b** i 3 ii 4 iii $-\frac{7}{3}$ iv $\frac{10}{3}$ v $60\frac{2}{3}$ vi 33

18 **a** True, as the function is odd.
b True, as $\sin 4x°$ is odd and $\cos 2x°$ is even.
c False, as $2^{-x^2} > 0$ for all x.
d True, as $2^x < 3^x$ for $0 < x < 1$.
e False, as $2^x > 3^x$ for $-1 < x < 0$.
f True, as $t^n \geq t^{n+1}$ for $0 \leq t \leq 1$ and hence $\dfrac{1}{1+t^n} \leq \dfrac{1}{1+t^{n+1}}$.

19 **a** The integral is $-\dfrac{1}{N} + 1$, which converges to 1 as $N \to \infty$.
b The integral is $-1 + \dfrac{1}{\varepsilon}$, which diverges to ∞ as $\varepsilon \to 0^+$.
c The integral is $2\sqrt{N} - 2$, which diverges to ∞ as $N \to \infty$.
d The integral is $2 - 2\sqrt{\varepsilon}$, which converges to 2 as $\varepsilon \to 0^+$.

Integration Part 1 Integration Basics - Homework Answers

Lesson 5C | The definite integral and its properties

Past-Paper Homework - Worked Solutions

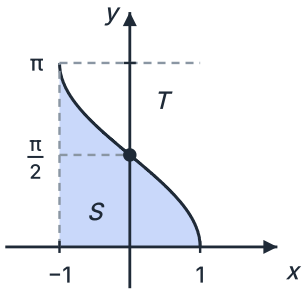
PP 1 **Sydney Technical High School, 2021, Extension 1 Mathematics, Trial HSC, Q6**

The graph is above the x-axis for $0 \leq x < 2$, below it for $2 < x < 6$, and above it again for $6 < x < 8$. Each integral is a sum of signed areas.

- $\int_0^2 f(x)\,dx = \int_0^1 f(x)\,dx + \int_1^2 f(x)\,dx$ (dissection at $x = 1$), and $\int_1^2 f(x)\,dx > 0$ because the curve is above the axis there. So (B) is greater than (A).
- $\int_0^8 f(x)\,dx = \int_0^2 f(x)\,dx - (\text{area from 2 to 6}) + (\text{area from 6 to 8})$. The region below the axis from 2 to 6 is wider (4 units) and deeper (about twice the height) than the small hump from 6 to 8, so the area from 2 to 6 is greater, and (D) is less than (B).
- $\int_0^7 f(x)\,dx = \int_0^8 f(x)\,dx - \int_7^8 f(x)\,dx$, and $\int_7^8 f(x)\,dx > 0$, so (C) is less than (D), and so less than (B).

Adding any area below the axis makes the integral smaller, so the greatest value is reached at $x = 2$, where the curve crosses from above to below the axis. **Answer (B).**

PP 2 **Abbotsleigh, 2023, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q10**



The graph of $y = \cos^{-1} x$ falls from $(-1, \pi)$ through $\left(0, \frac{\pi}{2}\right)$ to $(1, 0)$, and $\cos^{-1}(-x) = \pi - \cos^{-1} x$. Let $g(x) = \cos^{-1} x - \frac{\pi}{2}$. Then

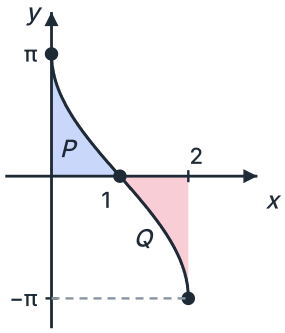
$$\begin{aligned} g(-x) &= \cos^{-1}(-x) - \tfrac{\pi}{2} \\ &= \pi - \cos^{-1} x - \tfrac{\pi}{2} \\ &= \tfrac{\pi}{2} - \cos^{-1} x \\ &= -g(x), \end{aligned}$$

so $g(x)$ is odd, and $\cos^{-1} x = \frac{\pi}{2} + g(x)$.

$$\begin{aligned} \int_{-1}^{1} \cos^{-1} x \, dx &= \int_{-1}^{1} \tfrac{\pi}{2} \, dx + \int_{-1}^{1} g(x) \, dx \qquad \text{(integral of a sum)} \\ &= \tfrac{\pi}{2} \times 2 + 0 \qquad \text{(a rectangle, and } g \text{ is odd)} \\ &= \pi. \end{aligned}$$

Geometrically: the curve cuts the rectangle $-1 \le x \le 1, 0 \le y \le \pi$ (area $2 \times \pi = 2\pi$) into the region S under the curve and the region T above it. A rotation of $180°$ about $\left(0, \frac{\pi}{2}\right)$ maps S onto T, so each has area $\frac{1}{2} \times 2\pi = \pi$. **Answer (C).** Option (D) is the area of the whole rectangle.

PP 3 **Epping Boys High School, 2024, Trial Higher School Certificate Examination, Mathematics Extension 1, Q12(c)(ii)**



Let $g(x) = 2\cos^{-1}(x-1) - \pi$. Its graph is the graph of $y = f(x)$ in part (i) moved down by π: it runs from $(0, \pi)$ through $(1, 0)$ to $(2, -\pi)$.

The graph of $y = g(x)$ has point symmetry about $(1, 0)$, because for $0 \le t \le 1$, using $\cos^{-1}(-t) = \pi - \cos^{-1} t$,

$$\begin{aligned} g(1-t) &= 2\cos^{-1}(-t) - \pi \\ &= 2\left(\pi - \cos^{-1} t\right) - \pi \\ &= \pi - 2\cos^{-1} t \\ &= -\left(2\cos^{-1} t - \pi\right) \\ &= -g(1+t). \end{aligned}$$

(Equivalently, the graph of $y = f(x)$ has point symmetry about $(1, \pi)$.) The region P above the x-axis from $x = 0$ to $x = 1$ and the region Q below the axis from $x = 1$ to $x = 2$ are congruent (a rotation of $180°$ about $(1, 0)$ maps one onto the other), so they have equal areas. Hence, dissecting the interval at $x = 1$,

$$\int_0^2 \left[2\cos^{-1}(x-1) - \pi\right] dx = \int_0^1 g(x)\,dx + \int_1^2 g(x)\,dx = \text{area } P - \text{area } Q = 0.$$

Integration Part 1 Integration Basics - Homework Answers

Lesson 5D | Proving the fundamental theorem

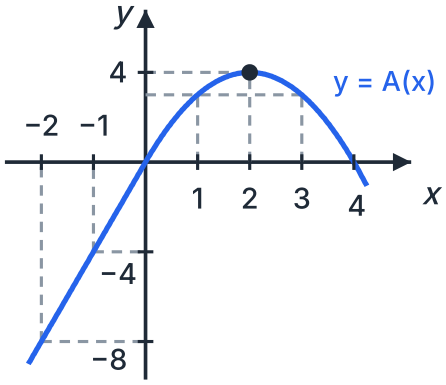
Lesson Examples

C1 **Tutor demo.** The graph is the horizontal line $y = 4$ for $t < 0$, and the line through $(0, 4)$ and $(2, 0)$ for $t \geq 0$, which is at height 2 when $t = 1$, -2 when $t = 3$ and -4 when $t = 4$. Use rectangles for $x < 0$, and a trapezium and triangles for $x > 0$.

- $A(-2) = -(2 \times 4) = -8$ and $A(-1) = -(1 \times 4) = -4$ (rectangles above the axis; the integrals run backwards).
- $A(0) = 0$.
- $A(1)$: a trapezium with parallel sides 4 and 2 and width 1, so $A(1) = \dfrac{4+2}{2} \times 1 = 3$.
- $A(2)$: a triangle with base 2 and height 4, so $A(2) = \frac{1}{2} \times 2 \times 4 = 4$.
- $A(3)$: from 2 to 3 a triangle below the axis with base 1 and height 2, so $A(3) = 4 - \frac{1}{2} \times 1 \times 2 = 4 - 1 = 3$.
- $A(4)$: from 2 to 4 a triangle below the axis with base 2 and height 4, so $A(4) = 4 - \frac{1}{2} \times 2 \times 4 = 4 - 4 = 0$.

x	-2	-1	0	1	2	3	4
$A(x)$	-8	-4	0	3	4	3	0

$A(x)$ is increasing for $x < 2$ because $y > 0$, and decreasing for $x > 2$ because $y < 0$. Its highest point is $(2, 4)$. For $x \leq 0$ the graph is a straight line.



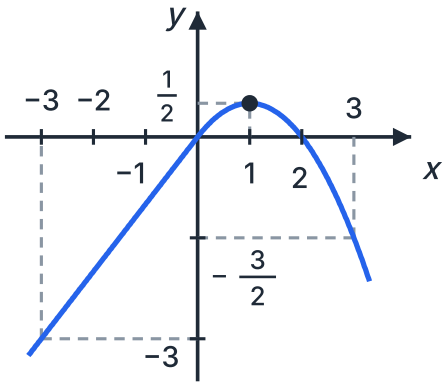
Your turn. The graph is the horizontal line $y = 1$ for $t < 0$, and the line through $(0, 1)$ and $(1, 0)$ for $t \geq 0$, which is at height -1 when $t = 2$ and -2 when $t = 3$.

- $A(-3) = -(3 \times 1) = -3$, $A(-2) = -(2 \times 1) = -2$ and $A(-1) = -(1 \times 1) = -1$ (rectangles above the axis; the integrals run backwards).
- $A(0) = 0$.
- $A(1)$: a triangle with base 1 and height 1, so $A(1) = \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$.

- $A(2)$: from 1 to 2 a triangle below the axis with base 1 and height 1, so $A(2) = \frac{1}{2} - \frac{1}{2} \times 1 \times 1 = \frac{1}{2} - \frac{1}{2} = 0$.
- $A(3)$: from 1 to 3 a triangle below the axis with base 2 and height 2, so $A(3) = \frac{1}{2} - \frac{1}{2} \times 2 \times 2 = \frac{1}{2} - 2 = -\frac{3}{2}$.

x	-3	-2	-1	0	1	2	3
$A(x)$	-3	-2	-1	0	$\frac{1}{2}$	0	$-\frac{3}{2}$

$A(x)$ is increasing for $x < 1$ because $y > 0$, and decreasing for $x > 1$ because $y < 0$. Its highest point is $\left(1, \frac{1}{2}\right)$. For $x \leq 0$ the graph is a straight line.



C2 **Tutor demo.** By the differential form, replace t by x in the integrand.

a $f(t) = (t^3 + 2)^4$, so $\dfrac{d}{dx}\displaystyle\int_0^x (t^3 + 2)^4 \, dt = (x^3 + 2)^4$.

b $f(t) = \sqrt{1 + t^3}$, so $\dfrac{d}{dx}\displaystyle\int_1^x \sqrt{1 + t^3} \, dt = \sqrt{1 + x^3}$.

c $f(t) = e^{-3t^2}$, so $\dfrac{d}{dx}\displaystyle\int_{-2}^x e^{-3t^2} \, dt = e^{-3x^2}$.

Your turn.

a $f(t) = (t^4 - 1)^5$, so $\dfrac{d}{dx}\displaystyle\int_2^x (t^4 - 1)^5 \, dt = (x^4 - 1)^5$.

b $f(t) = \log_e(t^2 + 1)$, so $\dfrac{d}{dx}\displaystyle\int_3^x \log_e(t^2 + 1) \, dt = \log_e(x^2 + 1)$.

c $f(t) = e^{t^3}$, so $\dfrac{d}{dx}\displaystyle\int_{-1}^x e^{t^3} \, dt = e^{x^3}$.

The lower limits do not affect the answers.

C3 **Tutor demo.**

a The primitive of $12t^3$ is $\dfrac{12t^4}{4} = 3t^4$.

$$\int_2^x 12t^3 \, dt = \Big[3t^4\Big]_2^x = 3x^4 - 3 \times 2^4 = 3x^4 - 48,$$

$$\text{so} \quad \frac{d}{dx}\int_2^x 12t^3\,dt = \frac{d}{dx}\left(3x^4 - 48\right)$$
$$= 12x^3, \quad \text{consistent with the differential form.}$$

b The primitive of $t^3 - 6t^2 + 4$ is $\dfrac{t^4}{4} - \dfrac{6t^3}{3} + 4t = \frac{1}{4}t^4 - 2t^3 + 4t$.

$$\int_{-1}^x (t^3 - 6t^2 + 4)\,dt = \left[\tfrac{1}{4}t^4 - 2t^3 + 4t\right]_{-1}^x$$
$$= \left(\tfrac{1}{4}x^4 - 2x^3 + 4x\right) - \left(\tfrac{1}{4}(-1)^4 - 2(-1)^3 + 4(-1)\right)$$
$$= \left(\tfrac{1}{4}x^4 - 2x^3 + 4x\right) - \left(\tfrac{1}{4} + 2 - 4\right)$$
$$= \tfrac{1}{4}x^4 - 2x^3 + 4x + \tfrac{7}{4},$$

$$\text{so} \quad \frac{d}{dx}\int_{-1}^x (t^3 - 6t^2 + 4)\,dt = \frac{d}{dx}\left(\tfrac{1}{4}x^4 - 2x^3 + 4x + \tfrac{7}{4}\right)$$
$$= x^3 - 6x^2 + 4, \quad \text{consistent.}$$

c Write $\dfrac{1}{t^3} = t^{-3}$. Its primitive is $\dfrac{t^{-2}}{-2} = -\frac{1}{2}t^{-2}$.

$$\int_2^x \frac{1}{t^3}\,dt = \left[-\tfrac{1}{2}t^{-2}\right]_2^x = -\tfrac{1}{2}x^{-2} - \left(-\tfrac{1}{2} \times 2^{-2}\right) = -\tfrac{1}{2}x^{-2} + \tfrac{1}{8},$$

$$\text{so} \quad \frac{d}{dx}\int_2^x \frac{1}{t^3}\,dt = \frac{d}{dx}\left(-\tfrac{1}{2}x^{-2} + \tfrac{1}{8}\right)$$
$$= -\tfrac{1}{2} \times (-2)x^{-3}$$
$$= x^{-3}, \quad \text{consistent.}$$

Your turn.

a The primitive of $3t^2$ is $\dfrac{3t^3}{3} = t^3$.

$$\int_1^x 3t^2\,dt = \left[t^3\right]_1^x = x^3 - 1^3 = x^3 - 1,$$

$$\text{so} \quad \frac{d}{dx}\int_1^x 3t^2\,dt = \frac{d}{dx}\left(x^3 - 1\right)$$
$$= 3x^2, \quad \text{consistent with the differential form.}$$

b The primitive of $t^3 + 3t^2 - 2$ is $\dfrac{t^4}{4} + \dfrac{3t^3}{3} - 2t = \frac{1}{4}t^4 + t^3 - 2t$.

$$\int_{-1}^x (t^3 + 3t^2 - 2)\,dt = \left[\tfrac{1}{4}t^4 + t^3 - 2t\right]_{-1}^x$$
$$= \left(\tfrac{1}{4}x^4 + x^3 - 2x\right) - \left(\tfrac{1}{4}(-1)^4 + (-1)^3 - 2(-1)\right)$$
$$= \left(\tfrac{1}{4}x^4 + x^3 - 2x\right) - \left(\tfrac{1}{4} - 1 + 2\right)$$
$$= \tfrac{1}{4}x^4 + x^3 - 2x - \tfrac{5}{4},$$

$$\text{so} \quad \frac{d}{dx}\int_{-1}^{x}(t^3+3t^2-2)\,dt = \frac{d}{dx}\left(\tfrac{1}{4}x^4+x^3-2x-\tfrac{5}{4}\right)$$
$$= x^3+3x^2-2, \quad \textbf{consistent.}$$

c Write $\dfrac{2}{t^2} = 2t^{-2}$. Its primitive is $\dfrac{2t^{-1}}{-1} = -2t^{-1}$.

$$\int_1^x \frac{2}{t^2}\,dt = \Big[-2t^{-1}\Big]_1^x = -2x^{-1}-\left(-2\times 1^{-1}\right) = -2x^{-1}+2,$$

$$\text{so} \quad \frac{d}{dx}\int_1^x \frac{2}{t^2}\,dt = \frac{d}{dx}\left(-2x^{-1}+2\right)$$
$$= -2\times(-1)x^{-2}$$
$$= 2x^{-2}, \quad \textbf{consistent.}$$

C4 **Tutor demo.**

a The domain is all real x, and y is continuous at every real number, so it is a continuous function.

b The domain is $x \neq -3$, and y is continuous at every value in its domain, so it is a continuous function.

c Now -3 lies in the domain, and y is not continuous at $x=-3$ (near $x=-3$ the values $\dfrac{1}{x+3}$ are very large, far from the value 0), so it is not a continuous function.

d The domain is $x \geq -3$, and y is continuous at every value in its domain, so it is a continuous function.

Your turn.

a The domain is all real x, and y is continuous at every real number, so it is a continuous function.

b The domain is $x \neq 0$, and y is continuous at every value in its domain, so it is a continuous function.

c Now 0 lies in the domain, and y is not continuous at $x=0$ (near $x=0$ the values $\dfrac{1}{x^2}$ are very large, far from the value 0), so it is not a continuous function.

d The domain is $x > -1$, and y is continuous at every value in its domain, so it is a continuous function.

Textbook Homework

1 **a** The region is a triangle with base x and height $3x$, so $A(x) = \frac{1}{2}\times x\times 3x = \frac{3}{2}x^2$.
b $A'(x) = \frac{3}{2}\times 2x = 3x$, which is the original function $y=3t$ with t replaced by x.

2 **a** $y = 3$. The region is a rectangle with width x and height 3, so $A(x) = 3 \times x = 3x$, and $A'(x) = 3$.

b $y = 2t$ (gradient $\frac{8}{4} = 2$). The region is a triangle with base x and height $2x$, so $A(x) = \frac{1}{2} \times x \times 2x = x^2$, and $A'(x) = 2x$.

c $y = 2 + t$ (gradient $\frac{6-2}{4} = 1$). The region is a trapezium with parallel sides 2 and $2 + x$ and width x, so

$A(x) = \frac{1}{2}\big(2 + (2 + x)\big) \times x = \frac{1}{2}(4 + x)x = 2x + \frac{1}{2}x^2$, and $A'(x) = 2 + x$.

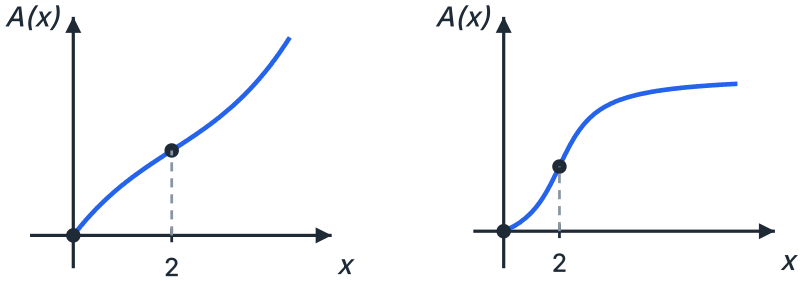
d $y = 5 - t$ (gradient $\frac{1-5}{4} = -1$). The region is a trapezium with parallel sides 5 and $5 - x$ and width x, so

$A(x) = \frac{1}{2}\big(5 + (5 - x)\big) \times x = \frac{1}{2}(10 - x)x = 5x - \frac{1}{2}x^2$, and $A'(x) = 5 - x$.

In each part $A'(x)$ is the original function with t replaced by x.

3 **a** $A(0) = 0$, and $f(t) > 0$, so $A(x)$ is increasing for all $x \geq 0$. In $[0, 2]$, $f(t)$ is decreasing, so $A(x)$ is increasing at a decreasing rate. In $[2, \infty)$, $f(t)$ is increasing, so $A(x)$ is increasing at an increasing rate. It has an inflection at $x = 2$.

b $A(0) = 0$, and $f(t) > 0$, so $A(x)$ is increasing for all $x \geq 0$. In $[0, 2]$, $f(t)$ is increasing, so $A(x)$ is increasing at an increasing rate. In $[2, \infty)$, $f(t)$ is decreasing towards 0, so $A(x)$ is increasing at a decreasing rate, and the graph flattens out. It has an inflection at $x = 2$.



4 **a** $\dfrac{1}{x}$ **b** $\dfrac{1}{1 + x^3}$ **c** $e^{-\frac{1}{2}x^2}$

5 **a** By the differential form, the answer is $3x^2 - 12$. Check:

$$\begin{aligned}\int_1^x (3t^2 - 12)\,dt &= \Big[t^3 - 12t\Big]_1^x \\ &= \big(x^3 - 12x\big) - \big(1^3 - 12 \times 1\big) \\ &= \big(x^3 - 12x\big) - \big(1 - 12\big) \\ &= x^3 - 12x + 11,\end{aligned}$$

and $\dfrac{d}{dx}\big(x^3 - 12x + 11\big) = 3x^2 - 12$.

b By the differential form, the answer is $x^3 + 4x$. Check:

$$\begin{aligned}\int_2^x (t^3+4t)\,dt &= \left[\tfrac{1}{4}t^4+2t^2\right]_2^x \\ &= \left(\tfrac{1}{4}x^4+2x^2\right)-\left(\tfrac{1}{4}\times 2^4+2\times 2^2\right) \\ &= \left(\tfrac{1}{4}x^4+2x^2\right)-(4+8) \\ &= \tfrac{1}{4}x^4+2x^2-12,\end{aligned}$$

and $\dfrac{d}{dx}\left(\tfrac{1}{4}x^4+2x^2-12\right)=x^3+4x$.

c By the differential form, the answer is $\dfrac{1}{x^2}$. Check:

$$\begin{aligned}\int_2^x \frac{1}{t^2}\,dt &= \left[-t^{-1}\right]_2^x \\ &= -x^{-1}-\left(-2^{-1}\right) \\ &= -x^{-1}+\tfrac{1}{2} \\ &= \tfrac{1}{2}-\frac{1}{x},\end{aligned}$$

and $\dfrac{d}{dx}\left(\tfrac{1}{2}-x^{-1}\right)=x^{-2}=\dfrac{1}{x^2}$.

6

a i x^2 ii x^3+3x iii $\dfrac{1}{x}$ iv $(x^3-3)^4$

b i $\displaystyle\int_1^x t^2\,dt=\left[\tfrac{1}{3}t^3\right]_1^x=\tfrac{1}{3}x^3-\tfrac{1}{3}$, and $\dfrac{d}{dx}\left(\tfrac{1}{3}x^3-\tfrac{1}{3}\right)=x^2$.

ii

$$\begin{aligned}\int_2^x (t^3+3t)\,dt &= \left[\tfrac{1}{4}t^4+\tfrac{3}{2}t^2\right]_2^x \\ &= \left(\tfrac{1}{4}x^4+\tfrac{3}{2}x^2\right)-\left(\tfrac{1}{4}\times 2^4+\tfrac{3}{2}\times 2^2\right) \\ &= \left(\tfrac{1}{4}x^4+\tfrac{3}{2}x^2\right)-(4+6) \\ &= \tfrac{1}{4}x^4+\tfrac{3}{2}x^2-10,\end{aligned}$$

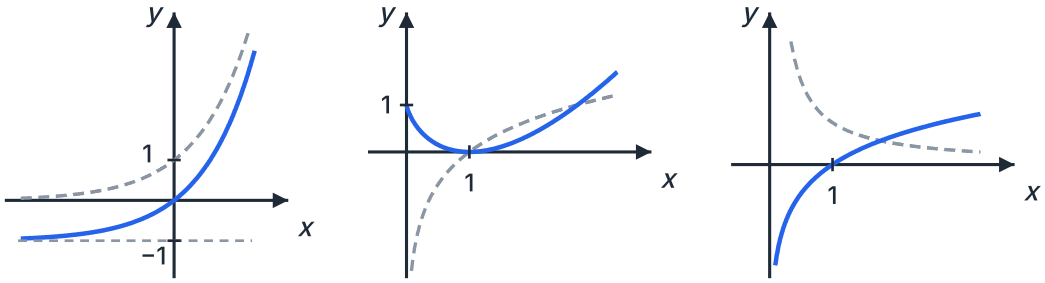
and $\dfrac{d}{dx}\left(\tfrac{1}{4}x^4+\tfrac{3}{2}x^2-10\right)=x^3+3x$.

7 **a** $e^t > 0$ and e^t is increasing, so $A(x)$ is increasing at an increasing rate. $A(0) = 0$, and $A(x) < 0$ for $x < 0$, because the integral runs backwards and the curve is above the axis. The curve $y = A(x)$ looks like $y = e^x - 1$.

b The curve is zero at $x = 1$, and is increasing at an increasing rate for $x > 1$, because $\log_e t$ is positive and increasing there. For $0 < x < 1$, $\log_e t < 0$ and the integral runs backwards, so $A(x) > 0$, and $A(x)$ is decreasing. Its lowest point is $(1, 0)$.

c The curve is zero at $x = 1$, and is increasing at a decreasing rate, because $\frac{1}{t}$ is positive and decreasing. For $0 < x < 1$, $A(x) < 0$, because the integral runs backwards and the curve is above the axis.

In each sketch below, $y = f(t)$ is dashed and $y = A(x)$ is solid.



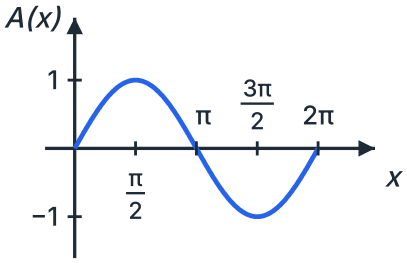
8 **a** By the symmetry of the cosine graph, the regions from $\frac{\pi}{2}$ to π and from π to $\frac{3\pi}{2}$ (below the axis) and from $\frac{3\pi}{2}$ to 2π (above the axis) are all congruent to P, so each has area 1. So $A\left(\frac{\pi}{2}\right) = 1$, $A(\pi) = 1 - 1 = 0$, $A\left(\frac{3\pi}{2}\right) = 0 - 1 = -1$ and $A(2\pi) = -1 + 1 = 0$.

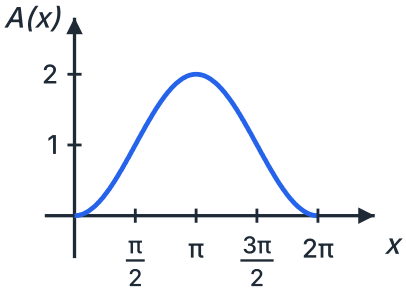
x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$A(x)$	0	1	0	-1	0

The curve looks like $y = \sin x$, and this suggests that $\sin x$ has derivative $\cos x$.
b Because $\sin x = \cos\left(\frac{\pi}{2} - x\right)$, the region under $y = \sin x$ from 0 to $\frac{\pi}{2}$ is the mirror image of P, so it has area 1. Each hump of the sine graph is symmetric about its middle, so the region from $\frac{\pi}{2}$ to π also has area 1. The regions from π to $\frac{3\pi}{2}$ and from $\frac{3\pi}{2}$ to 2π are the same shapes below the axis, so each has area 1 too. So $A\left(\frac{\pi}{2}\right) = 1$, $A(\pi) = 1 + 1 = 2$, $A\left(\frac{3\pi}{2}\right) = 2 - 1 = 1$ and $A(2\pi) = 1 - 1 = 0$.

x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$A(x)$	0	1	2	1	0

The graph looks like $y = 1 - \cos x$, which suggests that the derivative of $1 - \cos x$ is $\sin x$, that is, the derivative of $\cos x$ is $-\sin x$.





9 **a** $A(x)$ is increasing when $f(t)$ is positive, that is, for $x < c$, and is decreasing for $x > c$.

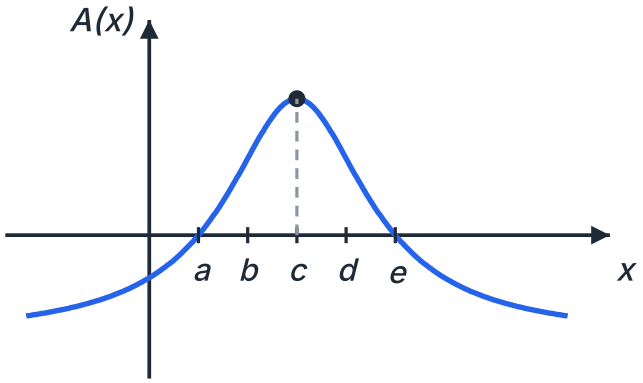
b $A(x)$ has a maximum turning point at $x = c$, and no minimum turning points.

c $A(x)$ has inflections where $f(t)$ changes from increasing to decreasing or back (its turning points), that is, at $x = b$ and $x = d$.

d $A(a) = 0$. Because of the point symmetry of $f(t)$, the area from c to e below the axis equals the area from a to c above it, so $A(e) = 0$. The zeroes of $A(x)$ are $x = a$ and $x = e$.

e $A(x)$ is positive for $a < x < e$, and negative for $x < a$ and for $x > e$.

f



10 **a** The function is continuous at every real number, so it is a continuous function.

b The domain is $x \neq 2$, and y is continuous at every value in its domain, so it is a continuous function.

c Now 2 lies in the domain, and y is not continuous at $x = 2$, so it is not a continuous function.

d The domain is $x \geq 0$, and y is continuous at every value in its domain, so it is a continuous function.

e The domain is $x > 0$, and y is continuous at every value in its domain, so it is a continuous function.

f The domain is $x \geq 0$, and y is not continuous at $x = 0$, so it is not a continuous function.

Integration Part 1 Integration Basics - Homework Answers

Lesson 5E | The indefinite integral

Lesson Examples

C1 **Tutor demo.**

a $7 = 7x^0$, so $\int 7\,dx = 7 \times \dfrac{x^1}{1} + C = 7x + C.$

b $\int 15x^4\,dx = 15 \times \dfrac{x^5}{5} + C = 3x^5 + C$

c $\int (6x^2 - 4x + 3)\,dx = 6 \times \dfrac{x^3}{3} - 4 \times \dfrac{x^2}{2} + 3x + C$

$= 2x^3 - 2x^2 + 3x + C$

Your turn.

a $5 = 5x^0$, so $\int 5\,dx = 5 \times \dfrac{x^1}{1} + C = 5x + C.$

b $\int 20x^3\,dx = 20 \times \dfrac{x^4}{4} + C = 5x^4 + C$

c

$\int (9x^2 + 2x - 4)\,dx = 9 \times \dfrac{x^3}{3} + 2 \times \dfrac{x^2}{2} - 4x + C = 3x^3 + x^2 - 4x + C$

C2 **Tutor demo.**

a Here $n = 4$, $a = 2$ and $b = 1$.

$\int (2x + 1)^4\,dx = \dfrac{(2x + 1)^5}{2 \times 5} + C = \frac{1}{10}(2x + 1)^5 + C$

b Here $n = 2$, $a = -3$ and $b = 7$.

$\int (7 - 3x)^2\,dx = \dfrac{(7 - 3x)^3}{(-3) \times 3} + C = \dfrac{(7 - 3x)^3}{-9} + C = -\frac{1}{9}(7 - 3x)^3 + C$

c Here $n = 5$, $a = 3$ and $b = -2$.

$\int 4(3x - 2)^5\,dx = 4 \times \dfrac{(3x - 2)^6}{3 \times 6} + C = \frac{4}{18}(3x - 2)^6 + C$

$= \frac{2}{9}(3x - 2)^6 + C$

Your turn.

a Here $n = 3$, $a = 5$ and $b = 2$.

$\int (5x + 2)^3\,dx = \dfrac{(5x + 2)^4}{5 \times 4} + C = \frac{1}{20}(5x + 2)^4 + C$

b Here $n = 5$, $a = -3$ and $b = 4$.

$\int (4 - 3x)^5\,dx = \dfrac{(4 - 3x)^6}{(-3) \times 6} + C = \dfrac{(4 - 3x)^6}{-18} + C = -\frac{1}{18}(4 - 3x)^6 + C$

c Here $n = 7$, $a = 4$ and $b = -1$.

$$\int 2(4x-1)^7\,dx = 2 \times \frac{(4x-1)^8}{4 \times 8} + C = \tfrac{2}{32}(4x-1)^8 + C$$
$$= \tfrac{1}{16}(4x-1)^8 + C$$

C3 Tutor demo.

a

$$\int \frac{8}{x^5}\,dx = \int 8x^{-5}\,dx$$
$$= 8 \times \frac{x^{-4}}{-4} + C$$
$$= -2x^{-4} + C$$
$$= -\frac{2}{x^4} + C$$

b Here $n = -3$, $a = 2$ and $b = 5$.

$$\int \frac{dx}{(2x+5)^3} = \int (2x+5)^{-3}\,dx$$
$$= \frac{(2x+5)^{-2}}{2 \times (-2)} + C$$
$$= -\tfrac{1}{4}(2x+5)^{-2} + C$$
$$= -\frac{1}{4(2x+5)^2} + C$$

Your turn.

a

$$\int \frac{6}{x^4}\,dx = \int 6x^{-4}\,dx$$
$$= 6 \times \frac{x^{-3}}{-3} + C$$
$$= -2x^{-3} + C$$
$$= -\frac{2}{x^3} + C$$

b Here $n = -2$, $a = 4$ and $b = -1$.

$$\int \frac{dx}{(4x-1)^2} = \int (4x-1)^{-2}\,dx$$
$$= \frac{(4x-1)^{-1}}{4 \times (-1)} + C$$
$$= -\tfrac{1}{4}(4x-1)^{-1} + C$$
$$= -\frac{1}{4(4x-1)} + C$$

C4 Tutor demo.

a $(x^2+2)^2 = (x^2)^2 + 2 \times x^2 \times 2 + 2^2 = x^4 + 4x^2 + 4$, so

$$\int (x^2+2)^2\,dx = \int (x^4 + 4x^2 + 4)\,dx$$
$$= \frac{x^5}{5} + 4 \times \frac{x^3}{3} + 4x + C$$
$$= \frac{x^5}{5} + \frac{4x^3}{3} + 4x + C$$

b $\left(2 - \dfrac{1}{x^2}\right)\left(2 + \dfrac{1}{x^2}\right) = 2^2 - \left(\dfrac{1}{x^2}\right)^2 = 4 - \dfrac{1}{x^4}$, so

$$\begin{aligned}\int \left(4-\frac{1}{x^4}\right)dx &= \int (4-x^{-4})\,dx \\ &= 4x-\frac{x^{-3}}{-3}+C \\ &= 4x+\tfrac{1}{3}x^{-3}+C \\ &= 4x+\frac{1}{3x^3}+C\end{aligned}$$

Your turn.

a $(x^2-3)^2=(x^2)^2-2\times x^2\times 3+3^2=x^4-6x^2+9$, so

$$\begin{aligned}\int (x^2-3)^2\,dx &= \int (x^4-6x^2+9)\,dx \\ &= \frac{x^5}{5}-6\times\frac{x^3}{3}+9x+C \\ &= \frac{x^5}{5}-2x^3+9x+C\end{aligned}$$

b $\left(\frac{1}{x}-2\right)\left(\frac{1}{x}+2\right)=\left(\frac{1}{x}\right)^2-2^2=\frac{1}{x^2}-4$, so

$$\begin{aligned}\int \left(\frac{1}{x^2}-4\right)dx &= \int (x^{-2}-4)\,dx \\ &= \frac{x^{-1}}{-1}-4x+C \\ &= -\frac{1}{x}-4x+C\end{aligned}$$

C5 Tutor demo.

a

$$\begin{aligned}\int_1^9 \sqrt{x}\,dx &= \int_1^9 x^{\frac{1}{2}}\,dx \\ &= \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right]_1^9 \\ &= \tfrac{2}{3}\left[x^{\frac{3}{2}}\right]_1^9 \\ &= \tfrac{2}{3}\times\left(9^{\frac{3}{2}}-1^{\frac{3}{2}}\right) \\ &= \tfrac{2}{3}\times(27-1) \\ &= \tfrac{52}{3} \\ &= 17\tfrac{1}{3}\end{aligned}$$

b

$$\begin{aligned}\int_1^9 \frac{1}{\sqrt{x}}\,dx &= \int_1^9 x^{-\frac{1}{2}}\,dx \\ &= \left[\frac{x^{\frac{1}{2}}}{\frac{1}{2}}\right]_1^9 \\ &= 2\left[x^{\frac{1}{2}}\right]_1^9 \\ &= 2\times\left(9^{\frac{1}{2}}-1^{\frac{1}{2}}\right) \\ &= 2\times(3-1) \\ &= 4\end{aligned}$$

Your turn.

a

$$\begin{aligned}\int_0^4 \sqrt{x}\,dx &= \int_0^4 x^{\frac{1}{2}}\,dx \\ &= \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right]_0^4 \\ &= \frac{2}{3}\left[x^{\frac{3}{2}}\right]_0^4 \\ &= \frac{2}{3}\times\left(4^{\frac{3}{2}}-0^{\frac{3}{2}}\right) \\ &= \frac{2}{3}\times(8-0) \\ &= \frac{16}{3} \\ &= 5\frac{1}{3}\end{aligned}$$

b

$$\begin{aligned}\int_1^{16} \frac{1}{\sqrt{x}}\,dx &= \int_1^{16} x^{-\frac{1}{2}}\,dx \\ &= \left[\frac{x^{\frac{1}{2}}}{\frac{1}{2}}\right]_1^{16} \\ &= 2\left[x^{\frac{1}{2}}\right]_1^{16} \\ &= 2\times\left(16^{\frac{1}{2}}-1^{\frac{1}{2}}\right) \\ &= 2\times(4-1) \\ &= 6\end{aligned}$$

C6 Tutor demo.

a $\dfrac{1}{\sqrt{5-4x}} = (5-4x)^{-\frac{1}{2}}$

b Here $n=-\frac{1}{2}$, $a=-4$ and $b=5$, so the new index is $\frac{1}{2}$.

$$\begin{aligned}\int \frac{dx}{\sqrt{5-4x}} &= \int (5-4x)^{-\frac{1}{2}}\,dx \\ &= \frac{(5-4x)^{\frac{1}{2}}}{-4\times\frac{1}{2}}+C \\ &= \frac{(5-4x)^{\frac{1}{2}}}{-2}+C \\ &= -\tfrac{1}{2}\sqrt{5-4x}+C\end{aligned}$$

Your turn.

a $\dfrac{1}{\sqrt{2x+7}} = (2x+7)^{-\frac{1}{2}}$

b Here $n=-\frac{1}{2}$, $a=2$ and $b=7$, so the new index is $\frac{1}{2}$.

$$\begin{aligned}\int \frac{dx}{\sqrt{2x+7}} &= \int (2x+7)^{-\frac{1}{2}}\,dx \\ &= \frac{(2x+7)^{\frac{1}{2}}}{2\times\frac{1}{2}}+C \\ &= \frac{(2x+7)^{\frac{1}{2}}}{1}+C \\ &= \sqrt{2x+7}+C\end{aligned}$$

Textbook Homework

Each answer shows the standard form applied, then the simplified primitive. C is a constant.

1

a $4 \times \dfrac{x^1}{1} + C = 4x + C$ **b** $\dfrac{x^1}{1} + C = x + C$ **c** C

d $-2 \times \dfrac{x^1}{1} + C = -2x + C$ **e** $\dfrac{x^2}{2} + C$ **f** $\dfrac{x^3}{3} + C$ **g** $\dfrac{x^4}{4} + C$

h $\dfrac{x^8}{8} + C$

In part c, the derivative of a constant is zero, so the primitive of 0 is a constant.

2

a $2 \times \dfrac{x^2}{2} + C = x^2 + C$ **b** $4 \times \dfrac{x^2}{2} + C = 2x^2 + C$

c $3 \times \dfrac{x^3}{3} + C = x^3 + C$ **d** $4 \times \dfrac{x^4}{4} + C = x^4 + C$

e $10 \times \dfrac{x^{10}}{10} + C = x^{10} + C$ **f** $2 \times \dfrac{x^4}{4} + C = \dfrac{x^4}{2} + C$

g $4 \times \dfrac{x^6}{6} + C = \dfrac{2x^6}{3} + C$ **h** $3 \times \dfrac{x^9}{9} + C = \dfrac{x^9}{3} + C$

3

a $\dfrac{x^2}{2} + \dfrac{x^3}{3} + C$ **b** $\dfrac{x^5}{5} - \dfrac{x^4}{4} + C$ **c** $\dfrac{x^8}{8} + \dfrac{x^{11}}{11} + C$

d $2 \times \dfrac{x^2}{2} + 5 \times \dfrac{x^5}{5} + C = x^2 + x^5 + C$

e $9 \times \dfrac{x^9}{9} - 11x + C = x^9 - 11x + C$

f $7 \times \dfrac{x^{14}}{14} + 3 \times \dfrac{x^9}{9} + C = \dfrac{x^{14}}{2} + \dfrac{x^9}{3} + C$

g $4x - 3 \times \dfrac{x^2}{2} + C = 4x - \dfrac{3x^2}{2} + C$ **h** $x - \dfrac{x^3}{3} + \dfrac{x^5}{5} + C$

i $3 \times \dfrac{x^3}{3} - 8 \times \dfrac{x^4}{4} + 7 \times \dfrac{x^5}{5} + C = x^3 - 2x^4 + \dfrac{7x^5}{5} + C$

4

a $\dfrac{x^{-1}}{-1} + C = -x^{-1} + C$ **b** $\dfrac{x^{-2}}{-2} + C = -\frac{1}{2}x^{-2} + C$

c $\dfrac{x^{-7}}{-7} + C = -\frac{1}{7}x^{-7} + C$ **d** $3 \times \dfrac{x^{-3}}{-3} + C = -x^{-3} + C$

e $9 \times \dfrac{x^{-9}}{-9} + C = -x^{-9} + C$ **f** $10 \times \dfrac{x^{-5}}{-5} + C = -2x^{-5} + C$

5

a $\dfrac{x^{\frac{3}{2}}}{\frac{3}{2}} + C = \frac{2}{3}x^{\frac{3}{2}} + C$ **b** $\dfrac{x^{\frac{4}{3}}}{\frac{4}{3}} + C = \frac{3}{4}x^{\frac{4}{3}} + C$

c $\dfrac{x^{\frac{5}{4}}}{\frac{5}{4}} + C = \frac{4}{5}x^{\frac{5}{4}} + C$ **d** $\dfrac{x^{\frac{5}{3}}}{\frac{5}{3}} + C = \frac{3}{5}x^{\frac{5}{3}} + C$

e $\dfrac{x^{\frac{1}{2}}}{\frac{1}{2}} + C = 2x^{\frac{1}{2}} + C$ **f** $4 \times \dfrac{x^{\frac{3}{2}}}{\frac{3}{2}} + C = \frac{8}{3}x^{\frac{3}{2}} + C$

6

a $x(x+2) = x^2 + 2x$, so the integral is $\dfrac{x^3}{3} + 2 \times \dfrac{x^2}{2} + C = \frac{1}{3}x^3 + x^2 + C.$

b $x(4 - x^2) = 4x - x^3$, so the integral is

$4 \times \dfrac{x^2}{2} - \dfrac{x^4}{4} + C = 2x^2 - \frac{1}{4}x^4 + C.$

c $x^2(5 - 3x) = 5x^2 - 3x^3$, so the integral is

$5 \times \dfrac{x^3}{3} - 3 \times \dfrac{x^4}{4} + C = \frac{5}{3}x^3 - \frac{3}{4}x^4 + C.$

d $x^3(x - 5) = x^4 - 5x^3$, so the integral is

$\dfrac{x^5}{5} - 5 \times \dfrac{x^4}{4} + C = \frac{1}{5}x^5 - \frac{5}{4}x^4 + C.$

e $(x - 3)^2 = x^2 - 6x + 9$, so the integral is

$\dfrac{x^3}{3} - 6 \times \dfrac{x^2}{2} + 9x + C = \frac{1}{3}x^3 - 3x^2 + 9x + C.$

f $(2x + 1)^2 = 4x^2 + 4x + 1$, so the integral is

$4 \times \dfrac{x^3}{3} + 4 \times \dfrac{x^2}{2} + x + C = \frac{4}{3}x^3 + 2x^2 + x + C.$

g $(1 - x^2)^2 = 1 - 2x^2 + x^4$, so the integral is

$x - 2 \times \dfrac{x^3}{3} + \dfrac{x^5}{5} + C = x - \frac{2}{3}x^3 + \frac{1}{5}x^5 + C.$

h $(2 - 3x)(2 + 3x) = 4 - 9x^2$, so the integral is

$4x - 9 \times \dfrac{x^3}{3} + C = 4x - 3x^3 + C.$

i $(x^2 - 3)(1 - 2x) = x^2 - 2x^3 - 3 + 6x$, so the integral is

$\dfrac{x^3}{3} - 2 \times \dfrac{x^4}{4} - 3x + 6 \times \dfrac{x^2}{2} + C = \frac{1}{3}x^3 - \frac{1}{2}x^4 - 3x + 3x^2 + C.$

7

a $\dfrac{x^2+2x}{x} = \dfrac{x^2}{x} + \dfrac{2x}{x} = x+2$, so the integral is

$\dfrac{x^2}{2} + 2x + C = \frac{1}{2}x^2 + 2x + C.$

b $\dfrac{x^7+x^8}{x^6} = \dfrac{x^7}{x^6} + \dfrac{x^8}{x^6} = x + x^2$, so the integral is $\frac{1}{2}x^2 + \frac{1}{3}x^3 + C.$

c $\dfrac{2x^3 - x^4}{4x} = \dfrac{2x^3}{4x} - \dfrac{x^4}{4x} = \frac{1}{2}x^2 - \frac{1}{4}x^3$, so the integral is

$\frac{1}{2} \times \dfrac{x^3}{3} - \frac{1}{4} \times \dfrac{x^4}{4} + C = \frac{1}{6}x^3 - \frac{1}{16}x^4 + C.$

8

a $\dfrac{1}{x^2} = x^{-2};\ \dfrac{x^{-1}}{-1} + C = -\dfrac{1}{x} + C$

b $\dfrac{1}{x^3} = x^{-3};\ \dfrac{x^{-2}}{-2} + C = -\dfrac{1}{2x^2} + C$

c $\dfrac{1}{x^5} = x^{-5};\ \dfrac{x^{-4}}{-4} + C = -\dfrac{1}{4x^4} + C$

d $\dfrac{1}{x^{10}} = x^{-10};\ \dfrac{x^{-9}}{-9} + C = -\dfrac{1}{9x^9} + C$

e $\dfrac{3}{x^4} = 3x^{-4};\ 3 \times \dfrac{x^{-3}}{-3} + C = -\dfrac{1}{x^3} + C$

f $\dfrac{5}{x^6} = 5x^{-6};\ 5 \times \dfrac{x^{-5}}{-5} + C = -\dfrac{1}{x^5} + C$

g $\dfrac{7}{x^8} = 7x^{-8};\ 7 \times \dfrac{x^{-7}}{-7} + C = -\dfrac{1}{x^7} + C$

h $\dfrac{1}{3x^2} = \frac{1}{3}x^{-2};\ \frac{1}{3} \times \dfrac{x^{-1}}{-1} + C = -\dfrac{1}{3x} + C$

i $\dfrac{1}{7x^5} = \frac{1}{7}x^{-5};\ \frac{1}{7} \times \dfrac{x^{-4}}{-4} + C = -\dfrac{1}{28x^4} + C$

j $-\dfrac{1}{5x^3} = -\frac{1}{5}x^{-3};\ -\frac{1}{5} \times \dfrac{x^{-2}}{-2} + C = \dfrac{1}{10x^2} + C$

k $\dfrac{1}{x^2} - \dfrac{1}{x^5} = x^{-2} - x^{-5};\ \dfrac{x^{-1}}{-1} - \dfrac{x^{-4}}{-4} + C = \dfrac{1}{4x^4} - \dfrac{1}{x} + C$

l $\dfrac{1}{x^3} + \dfrac{1}{x^4} = x^{-3} + x^{-4};\ \dfrac{x^{-2}}{-2} + \dfrac{x^{-3}}{-3} + C = -\dfrac{1}{2x^2} - \dfrac{1}{3x^3} + C$

9

a $\sqrt{x} = x^{\frac{1}{2}};\ \dfrac{x^{\frac{3}{2}}}{\frac{3}{2}} + C = \frac{2}{3}x^{\frac{3}{2}} + C$ b $\sqrt[3]{x} = x^{\frac{1}{3}};\ \dfrac{x^{\frac{4}{3}}}{\frac{4}{3}} + C = \frac{3}{4}x^{\frac{4}{3}} + C$

c $\dfrac{1}{\sqrt{x}} = x^{-\frac{1}{2}};\ \dfrac{x^{\frac{1}{2}}}{\frac{1}{2}} + C = 2x^{\frac{1}{2}} + C = 2\sqrt{x} + C$

d $\sqrt[3]{x^2} = x^{\frac{2}{3}};\ \dfrac{x^{\frac{5}{3}}}{\frac{5}{3}} + C = \frac{3}{5}x^{\frac{5}{3}} + C$

10

a $\frac{2}{3}\left[x^{\frac{3}{2}}\right]_0^9 = \frac{2}{3} \times (27 - 0) = 18$ b $\frac{3}{4}\left[x^{\frac{4}{3}}\right]_0^8 = \frac{3}{4} \times (16 - 0) = 12$

c $2\left[x^{\frac{1}{2}}\right]_{25}^{49} = 2 \times (7 - 5) = 4$ d $\frac{3}{5}\left[x^{\frac{5}{3}}\right]_0^1 = \frac{3}{5} \times (1 - 0) = \frac{3}{5}$

11

a $\dfrac{(x+1)^6}{1 \times 6} + C = \frac{1}{6}(x+1)^6 + C$ b $\dfrac{(x+2)^4}{1 \times 4} + C = \frac{1}{4}(x+2)^4 + C$

c $\dfrac{(4-x)^5}{(-1) \times 5} + C = -\frac{1}{5}(4-x)^5 + C$

d $\dfrac{(3-x)^3}{(-1) \times 3} + C = -\frac{1}{3}(3-x)^3 + C$

e $\dfrac{(3x+1)^5}{3 \times 5} + C = \frac{1}{15}(3x+1)^5 + C$

f $\dfrac{(4x-3)^8}{4 \times 8} + C = \frac{1}{32}(4x-3)^8 + C$

g $\dfrac{(5-2x)^7}{(-2) \times 7} + C = -\frac{1}{14}(5-2x)^7 + C$

h $\dfrac{(1-5x)^8}{(-5) \times 8} + C = -\frac{1}{40}(1-5x)^8 + C$

i $\dfrac{(2x+9)^{12}}{2 \times 12} + C = \frac{1}{24}(2x+9)^{12} + C$

j $3 \times \dfrac{(2x-1)^{11}}{2 \times 11} + C = \frac{3}{22}(2x-1)^{11} + C$

k $4 \times \dfrac{(5x-4)^7}{5 \times 7} + C = \frac{4}{35}(5x-4)^7 + C$

l $7 \times \dfrac{(3-2x)^4}{(-2) \times 4} + C = -\frac{7}{8}(3-2x)^4 + C$

12

a $\dfrac{\left(\frac{1}{3}x-7\right)^5}{\frac{1}{3}\times 5}+C=\frac{3}{5}\left(\frac{1}{3}x-7\right)^5+C$

b $\dfrac{\left(\frac{1}{4}x-7\right)^7}{\frac{1}{4}\times 7}+C=\frac{4}{7}\left(\frac{1}{4}x-7\right)^7+C$

c $\dfrac{\left(1-\frac{1}{5}x\right)^4}{\left(-\frac{1}{5}\right)\times 4}+C=-\frac{5}{4}\left(1-\frac{1}{5}x\right)^4+C$

13

a $\dfrac{(x+1)^{-2}}{1\times(-2)}+C=-\dfrac{1}{2(x+1)^2}+C$

b $\dfrac{(x-5)^{-3}}{1\times(-3)}+C=-\dfrac{1}{3(x-5)^3}+C$

c $\dfrac{(3x-4)^{-1}}{3\times(-1)}+C=-\dfrac{1}{3(3x-4)}+C$

d $\dfrac{(2-x)^{-4}}{(-1)\times(-4)}+C=\dfrac{1}{4(2-x)^4}+C$

e $3\times\dfrac{(x-7)^{-5}}{1\times(-5)}+C=-\dfrac{3}{5(x-7)^5}+C$

f $8\times\dfrac{(4x+1)^{-4}}{4\times(-4)}+C=-\dfrac{1}{2(4x+1)^4}+C$

g $2\times\dfrac{(3-5x)^{-3}}{(-5)\times(-3)}+C=\dfrac{2}{15(3-5x)^3}+C$

h $\frac{4}{5}\times\dfrac{(1-4x)^{-1}}{(-4)\times(-1)}+C=\dfrac{1}{5(1-4x)}+C=\dfrac{1}{5-20x}+C$

i $\frac{7}{8}\times\dfrac{(3x+2)^{-4}}{3\times(-4)}+C=-\dfrac{7}{96(3x+2)^4}+C$

14

a $\sqrt{x}\left(3\sqrt{x}-x\right)=3x-x^{\frac{3}{2}}$, so the integral is

$3\times\dfrac{x^2}{2}-\dfrac{x^{\frac{5}{2}}}{\frac{5}{2}}+C=\frac{3}{2}x^2-\frac{2}{5}x^{\frac{5}{2}}+C.$

b $\left(\sqrt{x}-2\right)\left(\sqrt{x}+2\right)=x-4$, so the integral is $\frac{1}{2}x^2-4x+C.$

c $\left(2\sqrt{x}-1\right)^2=4x-4x^{\frac{1}{2}}+1$, so the integral is

$4\times\dfrac{x^2}{2}-4\times\dfrac{x^{\frac{3}{2}}}{\frac{3}{2}}+x+C=2x^2-\frac{8}{3}x^{\frac{3}{2}}+x+C.$

15 **a** (i) $\frac{2}{3}\left[x^{\frac{3}{2}}\right]_0^1 = \frac{2}{3} \times (1-0) = \frac{2}{3}$ (ii) $2\left[x^{\frac{1}{2}}\right]_1^4 = 2 \times (2-1) = 2$

(iii) $\frac{3}{4}\left[x^{\frac{4}{3}}\right]_0^8 = \frac{3}{4} \times (16-0) = 12$

b (i) $\frac{2}{3}\left[x^{\frac{3}{2}}\right]_0^4 = \frac{2}{3} \times (8-0) = \frac{16}{3} = 5\frac{1}{3}$

(ii) $x\sqrt{x} = x^{\frac{3}{2}}$; $\frac{2}{5}\left[x^{\frac{5}{2}}\right]_1^9 = \frac{2}{5} \times (243-1) = \frac{484}{5} = 96\frac{4}{5}$

(iii) $2\left[x^{\frac{1}{2}}\right]_1^9 = 2 \times (3-1) = 4$

16 **a** $\left(2-\sqrt{x}\right)\left(2+\sqrt{x}\right) = 4-x$;

$\left[4x - \frac{1}{2}x^2\right]_2^4 = (16-8) - (8-2) = 8-6 = 2$

b $\sqrt{x}\left(\sqrt{x}-4\right) = x - 4x^{\frac{1}{2}}$; $\left[\frac{1}{2}x^2 - \frac{8}{3}x^{\frac{3}{2}}\right]_0^1 = \left(\frac{1}{2} - \frac{8}{3}\right) - (0-0) = -\frac{13}{6}$

c $\left(\sqrt{x}-1\right)^2 = x - 2x^{\frac{1}{2}} + 1$;

$\left[\frac{1}{2}x^2 - \frac{4}{3}x^{\frac{3}{2}} + x\right]_4^9 = \left(\frac{81}{2} - 36 + 9\right) - \left(8 - \frac{32}{3} + 4\right)$

$= \frac{27}{2} - \frac{4}{3} = \frac{73}{6} = 12\frac{1}{6}$

17 The standard form would give $\int x^{-1}\,dx = \dfrac{x^0}{0} + C$, which is meaningless, because we cannot divide by zero. Chapter 6 (Lesson 6I in Part 2 of this course) deals with the resolution of this problem.

18 **a** $\dfrac{(2x-1)^{\frac{3}{2}}}{2 \times \frac{3}{2}} + C = \frac{1}{3}(2x-1)^{\frac{3}{2}} + C$

b $\dfrac{(7-4x)^{\frac{3}{2}}}{(-4) \times \frac{3}{2}} + C = -\frac{1}{6}(7-4x)^{\frac{3}{2}} + C$

c $\dfrac{(4x-1)^{\frac{4}{3}}}{4 \times \frac{4}{3}} + C = \frac{3}{16}(4x-1)^{\frac{4}{3}} + C$

d $\dfrac{(3x+5)^{\frac{1}{2}}}{3 \times \frac{1}{2}} + C = \frac{2}{3}\sqrt{3x+5} + C$

19

a $\left[\frac{1}{5}(x+1)^5\right]_0^2 = \frac{1}{5}\times 3^5 - \frac{1}{5}\times 1^5 = \frac{243}{5} - \frac{1}{5} = \frac{242}{5}$

b $\left[\frac{1}{8}(2x-5)^4\right]_2^3 = \frac{1}{8}\times 1^4 - \frac{1}{8}\times(-1)^4 = \frac{1}{8} - \frac{1}{8} = 0$

c $\left[-\frac{1}{6}(1-x)^6\right]_{-2}^2 = -\frac{1}{6}\times(-1)^6 - \left(-\frac{1}{6}\times 3^6\right)$
$= -\frac{1}{6} + \frac{729}{6} = \frac{728}{6} = 121\frac{1}{3}$

d $\left[-\left(1-\frac{x}{5}\right)^5\right]_0^5 = -0^5 - \left(-1^5\right) = 0 + 1 = 1$

e
$\left[-\frac{1}{12}(9-8x)^{\frac{3}{2}}\right]_0^1 = -\frac{1}{12}\times 1^{\frac{3}{2}} - \left(-\frac{1}{12}\times 9^{\frac{3}{2}}\right) = -\frac{1}{12} + \frac{27}{12} = \frac{26}{12} = \frac{13}{6}$

f $\left[2(x+2)^{\frac{1}{2}}\right]_2^7 = 2\times 9^{\frac{1}{2}} - 2\times 4^{\frac{1}{2}} = 6 - 4 = 2$

g $\left[\frac{3}{4}(x+1)^{\frac{4}{3}}\right]_{-2}^0 = \frac{3}{4}\times 1^{\frac{4}{3}} - \frac{3}{4}\times(-1)^{\frac{4}{3}} = \frac{3}{4} - \frac{3}{4} = 0$, because
$(-1)^{\frac{4}{3}} = \left(\sqrt[3]{-1}\right)^4 = (-1)^4 = 1$

h $\left[\frac{2}{9}(3x+1)^{\frac{3}{2}}\right]_1^5 = \frac{2}{9}\times 16^{\frac{3}{2}} - \frac{2}{9}\times 4^{\frac{3}{2}} = \frac{128}{9} - \frac{16}{9} = \frac{112}{9}$

i $\left[-\frac{2}{15}(1-5x)^{\frac{3}{2}}\right]_{-3}^0 = -\frac{2}{15}\times 1^{\frac{3}{2}} - \left(-\frac{2}{15}\times 16^{\frac{3}{2}}\right)$
$= -\frac{2}{15} + \frac{128}{15} = \frac{126}{15} = 8\frac{2}{5}$

20

b (i) Take $u = x$ and $\frac{dv}{dx} = (x-1)^4$, so $\frac{du}{dx} = 1$ and $v = \frac{1}{5}(x-1)^5$. Then
$\int x(x-1)^4\,dx = \frac{1}{5}x(x-1)^5 - \int \frac{1}{5}(x-1)^5\,dx$
$= \frac{1}{5}x(x-1)^5 - \frac{1}{30}(x-1)^6 + C.$

(ii) Take $u = x$ and $\frac{dv}{dx} = \sqrt{1+x}$, so $\frac{du}{dx} = 1$ and $v = \frac{2}{3}(1+x)^{\frac{3}{2}}$. Then
$\int x\sqrt{1+x}\,dx = \frac{2}{3}x(1+x)^{\frac{3}{2}} - \int \frac{2}{3}(1+x)^{\frac{3}{2}}\,dx$
$= \frac{2}{3}x(1+x)^{\frac{3}{2}} - \frac{4}{15}(1+x)^{\frac{5}{2}} + C.$

Integration Part 1 Integration Basics - Homework Answers

Lesson 5F | Finding areas by integration

Lesson Examples

C1 **Tutor demo.** The curve meets the x-axis where $9 - x^2 = 0$, at $(3, 0)$ and $(-3, 0)$. The region lies entirely above the x-axis, and the relevant integral is

$$\begin{aligned}\int_{-3}^{3} (9 - x^2)\, dx &= \left[9x - \frac{x^3}{3}\right]_{-3}^{3} \\ &= (27 - 9) - (-27 + 9) \\ &= 18 - (-18) \\ &= 36,\end{aligned}$$

which is positive. Hence the required area is 36 square units.

Your turn. The curve meets the x-axis where $1 - x^2 = 0$, at $(1, 0)$ and $(-1, 0)$. The region lies entirely above the x-axis, and

$$\begin{aligned}\int_{-1}^{1} (1 - x^2)\, dx &= \left[x - \frac{x^3}{3}\right]_{-1}^{1} \\ &= \left(1 - \tfrac{1}{3}\right) - \left(-1 + \tfrac{1}{3}\right) \\ &= \tfrac{2}{3} - \left(-\tfrac{2}{3}\right) \\ &= 1\tfrac{1}{3}.\end{aligned}$$

Hence the required area is $1\frac{1}{3}$ square units.

C2 **Tutor demo.** The curve meets the x-axis where $x^2 - 4 = 0$, at $(2, 0)$ and $(-2, 0)$. The region lies entirely below the x-axis, and

$$\begin{aligned}\int_{-2}^{2} (x^2 - 4)\, dx &= \left[\frac{x^3}{3} - 4x\right]_{-2}^{2} \\ &= \left(\tfrac{8}{3} - 8\right) - \left(-\tfrac{8}{3} + 8\right) \\ &= -\tfrac{16}{3} - \tfrac{16}{3} \\ &= -10\tfrac{2}{3},\end{aligned}$$

which is negative, because the curve lies below the x-axis. Hence the required area is $10\frac{2}{3}$ square units.

Your turn. The curve meets the x-axis at $(3, 0)$ and $(-3, 0)$, and the region lies entirely below the x-axis.

$$\begin{aligned}\int_{-3}^{3} (x^2 - 9)\, dx &= \left[\frac{x^3}{3} - 9x\right]_{-3}^{3} \\ &= (9 - 27) - (-9 + 27) \\ &= -18 - 18 \\ &= -36,\end{aligned}$$

so the required area is 36 square units.

C3 **Tutor demo.**

a The x-intercepts are $x=-1$, $x=0$ and $x=3$. The cubic has a positive leading coefficient, so it is above the axis for $-1<x<0$ and below it for $0<x<3$.

b Expanding, $y=x(x^2-3x+x-3)=x(x^2-2x-3)=x^3-2x^2-3x$. For the region above the x-axis,

$$\begin{aligned}\int_{-1}^{0}(x^3-2x^2-3x)\,dx &= \left[\frac{x^4}{4}-\frac{2x^3}{3}-\frac{3x^2}{2}\right]_{-1}^{0}\\ &=(0-0-0)-\left(\tfrac{1}{4}+\tfrac{2}{3}-\tfrac{3}{2}\right)\\ &=-\left(\tfrac{3}{12}+\tfrac{8}{12}-\tfrac{18}{12}\right)\\ &=\tfrac{7}{12},\end{aligned}$$

so area above $=\frac{7}{12}$ square units. For the region below the x-axis,

$$\begin{aligned}\int_{0}^{3}(x^3-2x^2-3x)\,dx &= \left[\frac{x^4}{4}-\frac{2x^3}{3}-\frac{3x^2}{2}\right]_{0}^{3}\\ &=\left(\tfrac{81}{4}-18-\tfrac{27}{2}\right)-(0-0-0)\\ &=\tfrac{81}{4}-\tfrac{72}{4}-\tfrac{54}{4}\\ &=-11\tfrac{1}{4},\end{aligned}$$

so area below $=11\frac{1}{4}$ square units. Total area
$=\frac{7}{12}+11\frac{1}{4}=\frac{7}{12}+\frac{135}{12}=\frac{142}{12}=11\frac{5}{6}$ square units.

c

$$\begin{aligned}\int_{-1}^{3}x(x+1)(x-3)\,dx &= \left[\frac{x^4}{4}-\frac{2x^3}{3}-\frac{3x^2}{2}\right]_{-1}^{3}\\ &=-11\tfrac{1}{4}+\tfrac{7}{12}\\ &=-\tfrac{135}{12}+\tfrac{7}{12}\\ &=-10\tfrac{2}{3}.\end{aligned}$$

This integral is the difference $11\frac{1}{4}-\frac{7}{12}=10\frac{2}{3}$ of the two areas, and is negative because the area below is larger than the area above.

Your turn.

a The x-intercepts are $x=-1$, $x=0$ and $x=4$; the curve is above the axis for $-1<x<0$ and below it for $0<x<4$.

b $y=x(x^2-3x-4)=x^3-3x^2-4x$, with primitive $\frac{x^4}{4}-x^3-2x^2$.

$$\begin{aligned}\int_{-1}^{0}(x^3-3x^2-4x)\,dx &= \left[\frac{x^4}{4}-x^3-2x^2\right]_{-1}^{0}\\ &=0-\left(\tfrac{1}{4}+1-2\right)\\ &=\tfrac{3}{4},\end{aligned}$$

$$\begin{aligned}\int_0^4 (x^3 - 3x^2 - 4x)\,dx &= \left[\frac{x^4}{4} - x^3 - 2x^2\right]_0^4 \\ &= (64 - 64 - 32) - 0 \\ &= -32,\end{aligned}$$

so the areas are $\frac{3}{4}$ and 32, and the total area is $32\frac{3}{4}$ square units.

c $\int_{-1}^4 x(x+1)(x-4)\,dx = -32 + \frac{3}{4} = -31\frac{1}{4}$. It is the difference $32 - \frac{3}{4}$ of the two areas, negative because the area below is larger.

C4

Tutor demo.

a Let $f(x) = x^3 - 9x$. Then $f(-x) = (-x)^3 - 9(-x) = -x^3 + 9x = -f(x)$, so $f(x)$ is odd.

b $y = x(x^2 - 9) = x(x-3)(x+3)$, so the x-intercepts are $x = -3$, $x = 0$ and $x = 3$. The two regions have equal areas because the function is odd.

$$\begin{aligned}\int_0^3 (x^3 - 9x)\,dx &= \left[\frac{x^4}{4} - \frac{9x^2}{2}\right]_0^3 \\ &= \left(\tfrac{81}{4} - \tfrac{81}{2}\right) - (0 - 0) \\ &= -\tfrac{81}{4},\end{aligned}$$

so area below the x-axis $= \frac{81}{4}$ square units. Doubling, total area $= \frac{81}{2} = 40\frac{1}{2}$ square units.

Your turn.

a $f(-x) = (-x)^3 - 4(-x) = -x^3 + 4x = -f(x)$, so $f(x) = x^3 - 4x$ is odd.

b $y = x(x-2)(x+2)$, with x-intercepts -2, 0 and 2.

$$\begin{aligned}\int_0^2 (x^3 - 4x)\,dx &= \left[\frac{x^4}{4} - 2x^2\right]_0^2 \\ &= (4 - 8) - (0 - 0) \\ &= -4,\end{aligned}$$

so area below $= 4$ square units, and doubling, total area $= 8$ square units.

C5

Tutor demo.

a The lines meet where $2x + 1 = 7$, that is, at $(3, 7)$. The line $y = 2x + 1$ crosses the y-axis at $(0, 1)$. The region is the triangle with vertices $(0, 1)$, $(0, 7)$ and $(3, 7)$.

b $y = 2x + 1$, so $2x = y - 1$ and $x = \frac{1}{2}y - \frac{1}{2}$.

c

$$\begin{aligned}\int_1^7 \left(\tfrac{1}{2}y - \tfrac{1}{2}\right) dy &= \left[\frac{y^2}{4} - \frac{y}{2}\right]_1^7 \\ &= \left(\frac{49}{4} - \frac{7}{2}\right) - \left(\frac{1}{4} - \frac{1}{2}\right) \\ &= \frac{35}{4} - \left(-\frac{1}{4}\right) \\ &= 9,\end{aligned}$$

which is positive, because the region is to the right of the y-axis. The area is $9\ \text{u}^2$.

d Area of triangle $= \frac{1}{2} \times 3 \times 6 = 9\ \text{u}^2$ (base $3 - 0 = 3$ along $y = 7$, height $7 - 1 = 6$).

Your turn.

a The lines meet where $x + 2 = 5$, at $(3, 5)$; the line crosses the y-axis at $(0, 2)$.

b $x = y - 2$.

c

$$\begin{aligned}\int_2^5 (y - 2)\, dy &= \left[\frac{y^2}{2} - 2y\right]_2^5 \\ &= \left(\frac{25}{2} - 10\right) - (2 - 4) \\ &= \tfrac{5}{2} + 2 \\ &= 4\tfrac{1}{2},\end{aligned}$$

so the area is $4\frac{1}{2}\ \text{u}^2$.

d Area of triangle $= \frac{1}{2} \times 3 \times 3 = 4\frac{1}{2}\ \text{u}^2$.

C6

Tutor demo.

a $y = x^3$, so $x^3 = y$ and $x = y^{\frac{1}{3}}$.

b For the region to the right of the y-axis,

$$\begin{aligned}\int_0^{27} y^{\frac{1}{3}}\, dy &= \tfrac{3}{4}\left[y^{\frac{4}{3}}\right]_0^{27} \\ &= \tfrac{3}{4} \times (81 - 0) \qquad \left(27^{\frac{4}{3}} = 3^4 = 81\right) \\ &= 60\tfrac{3}{4},\end{aligned}$$

so area $= 60\frac{3}{4}$ square units. For the region to the left of the y-axis,

$$\begin{aligned}\int_{-8}^0 y^{\frac{1}{3}}\, dy &= \tfrac{3}{4}\left[y^{\frac{4}{3}}\right]_{-8}^0 \\ &= \tfrac{3}{4} \times (0 - 16) \qquad \left((-8)^{\frac{4}{3}} = (-2)^4 = 16\right) \\ &= -12,\end{aligned}$$

so area $= 12$ square units.

c Total area $= 60\frac{3}{4} + 12 = 72\frac{3}{4}$ square units.

Your turn.

a $x^5 = y$, so $x = y^{\frac{1}{5}}$.

b A primitive of $y^{\frac{1}{5}}$ is $\dfrac{y^{\frac{6}{5}}}{\frac{6}{5}} = \frac{5}{6}y^{\frac{6}{5}}$.

$$\begin{aligned}\int_0^1 y^{\frac{1}{5}}\,dy &= \tfrac{5}{6}\left[y^{\frac{6}{5}}\right]_0^1\\ &= \tfrac{5}{6}\times(1-0)\\ &= \tfrac{5}{6},\end{aligned}$$

$$\begin{aligned}\int_{-1}^0 y^{\frac{1}{5}}\,dy &= \tfrac{5}{6}\left[y^{\frac{6}{5}}\right]_{-1}^0\\ &= \tfrac{5}{6}\times(0-1)\\ &= -\tfrac{5}{6},\end{aligned}$$

using $(-1)^{\frac{6}{5}} = (-1)^6 = 1$. The areas are $\frac{5}{6}$ and $\frac{5}{6}$ square units.

c Total area $= \frac{5}{6} + \frac{5}{6} = 1\frac{2}{3}$ square units.

Textbook Homework

1

a $\int_0^2 2x\,dx = 4\text{ u}^2$

b $\int_1^3 3x^2\,dx = 26\text{ u}^2$

c $\int_0^3 4x^3\,dx = 81\text{ u}^2$

d $\int_{-1}^2 (3x^2+1)\,dx = 12\text{ u}^2$

e $\int_0^3 x^2\,dx = 9\text{ u}^2$

f $\int_2^4 (x^2-2x)\,dx = 6\frac{2}{3}\text{ u}^2$

g $\int_0^{16} \sqrt{x}\,dx = \frac{128}{3}\text{ u}^2$

h $\int_1^3 (5-x)\,dx = 6\text{ u}^2$

i $\int_{-1}^0 (x^3-x)\,dx = \frac{1}{4}\text{ u}^2$

j $\int_{-4}^3 (12-x-x^2)\,dx = 57\frac{1}{6}\text{ u}^2$

k $\int_{-1}^2 (5x^4+1)\,dx = 36\text{ u}^2$

l $\int_1^{27} \sqrt[3]{x}\,dx = 60\text{ u}^2$

2

a $\int_0^5 2y\,dy = 25\text{ u}^2$

b $\int_{-2}^0 3y^2\,dy = 8\text{ u}^2$

c $\int_2^4 (2y-4)\,dy = 4\text{ u}^2$

d $\int_{-3}^3 (27-3y^2)\,dy = 108\text{ u}^2$

e $\int_0^3 y\,dy = \frac{9}{2}\text{ u}^2$

f $\int_3^5 (y^2+1)\,dy = 34\frac{2}{3}\text{ u}^2$

g $\int_0^9 \sqrt{y}\,dy = 18\text{ u}^2$

h $\int_1^4 y^{-\frac{1}{2}}\,dy = 2\text{ u}^2$

3

a $\int_1^3 (x^2-4x+3)\,dx = -\frac{4}{3}$, area $\frac{4}{3}\text{ u}^2$

b $\int_{-3}^0 3x\,dx = -\frac{27}{2}$, area $\frac{27}{2}\text{ u}^2$

c $\int_{-3}^0 x^3\,dx = -\frac{81}{4}$, area $\frac{81}{4}\text{ u}^2$

d $\int_1^3 (1-x^4)\,dx = -46\frac{2}{5}$, area $46\frac{2}{5}\text{ u}^2$

4

a $\int_1^4 (1-y)\,dy = -\frac{9}{2}$, area $\frac{9}{2}\text{ u}^2$

b $\int_2^4 (y^2-6y+8)\,dy = -\frac{4}{3}$, area $\frac{4}{3}\text{ u}^2$

c $\int_{-8}^{-1} y^{\frac{1}{3}}\,dy = -\frac{45}{4}$, area $\frac{45}{4}\text{ u}^2$

d $\int_0^3 (-y^2)\,dy = -9$, area 9 u^2

5

b $4\frac{1}{2}\text{ u}^2$ **c** 2 u^2 **d** $6\frac{1}{2}\text{ u}^2$

e $2\frac{1}{2}$. This is the area above the x-axis minus the area below it.

6 **b** $10\frac{2}{3}\ \text{u}^2$ **c** $2\frac{1}{3}\ \text{u}^2$ **d** $13\ \text{u}^2$
e $-8\frac{1}{3}$. This is the area above the x-axis minus the area below it.

7 **b** $2\frac{2}{3}\ \text{u}^2$ **c** $\frac{5}{12}\ \text{u}^2$ **d** $3\frac{1}{12}\ \text{u}^2$
e $-2\frac{1}{4}$. This is the area above the x-axis minus the area below it.

8 **a** $11\frac{2}{3}\ \text{u}^2$ **b** $128\frac{1}{2}\ \text{u}^2$ **c** $4\ \text{u}^2$ **d** $8\frac{1}{2}\ \text{u}^2$ **e** $32\frac{3}{4}\ \text{u}^2$ **f** $11\frac{1}{3}\ \text{u}^2$

9 **a** $13\ \text{u}^2$ **b** $2\frac{1}{2}\ \text{u}^2$ **c** $9\frac{1}{3}\ \text{u}^2$ **d** $7\frac{1}{3}\ \text{u}^2$

10 **a** (i) $64\ \text{u}^2$ (ii) $128\ \text{u}^2$ (iii) $64\frac{4}{5}\ \text{u}^2$ **b** (i) $50\ \text{u}^2$ (ii) $18\ \text{u}^2$ (iii) $\frac{32}{3}\ \text{u}^2$

11 $8\ \text{u}^2$

12 **a** $(2, 0)$, $(0, 4\sqrt{2})$, $(0, -4\sqrt{2})$ **b** $\dfrac{16\sqrt{2}}{3}\ \text{u}^2$ (both methods); in (ii), $x = 2 - \dfrac{y^2}{16}$

13 **a** $y = \frac{1}{3}x^3 - 2x^2 + 3x$
b The curve passes through the origin, $\left(1, 1\frac{1}{3}\right)$ is a maximum turning point and $(3, 0)$ is a minimum turning point.
c $\frac{4}{3}\ \text{u}^2$

15 **a** $2 : n + 1$ **b** $1 : n + 1$

16 **b** $a^2 = \frac{1}{2}\left(3 + \sqrt{5}\right)$, $a^4 = \frac{1}{2}\left(7 + 3\sqrt{5}\right)$, $a^5 = \frac{1}{2}\left(11 + 5\sqrt{5}\right)$
c Areas are $\frac{1}{5}\ \text{u}^2$, $\frac{1}{10}\ \text{u}^2$ and $\frac{1}{10}\ \text{u}^2$.

17 **a** Straight line segments from $(0, 4)$ through $(3, 0)$ down to $(6, -4)$, then up through $(10, 0)$ to $(12, 2)$.
b maximum at $(3, 6)$, minimum at $(10, -8)$ **c** $x = 0$, $x = 6$
d The curve starts at the origin, rises to the maximum $(3, 6)$, falls through $(6, 0)$ to the minimum $(10, -8)$, then rises to the endpoint $(12, -6)$. **e** $24\ \text{u}^2$

18 **a** $-\dfrac{1}{n+1}$ **b** $\dfrac{1}{n+1}$
c For $n < -1$, the region under $y = x^n$ to the right of $x = 1$, although unbounded, has the finite area $-\dfrac{1}{n+1}$. For $n > -1$, the region under $y = x^n$ from $x = 0$ to $x = 1$ has the finite area $\dfrac{1}{n+1}$, even when the curve has an asymptote at $x = 0$.

Integration Part 1 Integration Basics - Homework Answers

Lesson 5F | Finding areas by integration

Past-Paper Homework - Worked Solutions

PP 1 Normanhurst Boys High School, 2022, Mathematics Extension 1, Year 12 Course Assessment Task 4 (Trial Examination), Q10

The region is bounded by the y-axis and the horizontal lines $y = 1$ and $y = 2$, so integrate with respect to y, with limits $y = 1$ and $y = 2$ (Box 16). Write x as a function of y:

$$\begin{aligned} y &= x^3 - 1 \\ x^3 &= y + 1 \\ x &= \sqrt[3]{y+1}. \end{aligned}$$

For $1 \leq y \leq 2$, $y + 1$ is positive, so $x = \sqrt[3]{y+1}$ is positive and the region lies to the right of the y-axis. Hence the integral is the area itself:

$$\text{area} = \int_1^2 \sqrt[3]{y+1}\, dy.$$

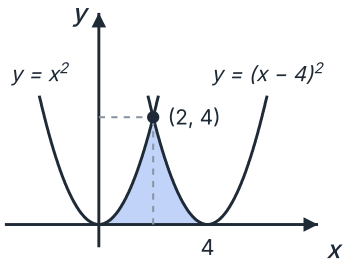
Answer (B). Option (A) still has the integrand written in terms of x. Options (C) and (D) are not the inverse of $y = x^3 - 1$: for example, (D) uses $x = y + 1$, which is the inverse of $y = x - 1$.

Integration Part 1 Integration Basics - Homework Answers

Lesson 5G | Areas of compound regions

Lesson Examples

C1 **Tutor demo.**



a Meeting point: solve $x^2 = (x-4)^2$.

$$\begin{aligned} x^2 &= x^2 - 8x + 16 \\ 8x &= 16 \\ x &= 2, \end{aligned}$$

and $y = 2^2 = 4$, so the curves meet at $(2, 4)$. The sketch is on the right.

b The region is under $y = x^2$ from $x = 0$ to $x = 2$, and under $y = (x-4)^2$ from $x = 2$ to $x = 4$.

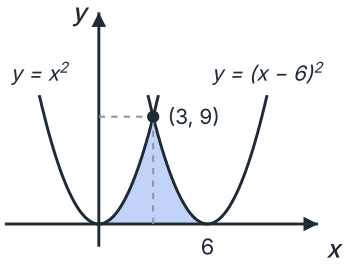
c The region is above the x-axis. Cut it at $x = 2$.

$$\begin{aligned} \int_0^2 x^2\,dx &= \left[\frac{x^3}{3}\right]_0^2 \\ &= \tfrac{8}{3} - 0 \\ &= \tfrac{8}{3} \end{aligned}$$

$$\begin{aligned} \int_2^4 (x-4)^2\,dx &= \left[\frac{(x-4)^3}{3}\right]_2^4 \\ &= \frac{0^3}{3} - \frac{(-2)^3}{3} \\ &= 0 - \left(-\tfrac{8}{3}\right) \\ &= \tfrac{8}{3} \end{aligned}$$

$$\begin{aligned} \text{area} &= \tfrac{8}{3} + \tfrac{8}{3} \\ &= \tfrac{16}{3} \\ &= 5\tfrac{1}{3} \text{ square units.} \end{aligned}$$

Your turn.



a Meeting point: solve $x^2 = (x-6)^2$.

$$\begin{aligned} x^2 &= x^2 - 12x + 36 \\ 12x &= 36 \\ x &= 3, \end{aligned}$$

and $y = 3^2 = 9$, so the curves meet at $(3, 9)$. The sketch is on the right.

b The region is under $y = x^2$ from $x = 0$ to $x = 3$, and under $y = (x-6)^2$ from $x = 3$ to $x = 6$.

c Cut the region at $x = 3$.

$$\begin{aligned}\int_0^3 x^2\,dx &= \left[\frac{x^3}{3}\right]_0^3 \\ &= \tfrac{27}{3} - 0 \\ &= 9\end{aligned}$$

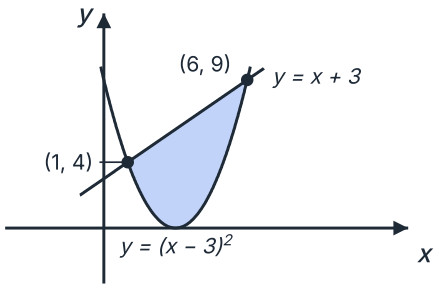
$$\begin{aligned}\int_3^6 (x-6)^2\,dx &= \left[\frac{(x-6)^3}{3}\right]_3^6 \\ &= \frac{0^3}{3} - \frac{(-3)^3}{3} \\ &= 0 - (-9) \\ &= 9\end{aligned}$$

$$\begin{aligned}\text{area} &= 9 + 9 \\ &= 18 \text{ square units.}\end{aligned}$$

C2 **Tutor demo.**

a Substituting $y = x + 3$ into $y = (x-3)^2$ gives

$$\begin{aligned}(x-3)^2 &= x + 3 \\ x^2 - 6x + 9 &= x + 3 \\ x^2 - 7x + 6 &= 0 \\ (x-1)(x-6) &= 0 \\ x &= 1 \text{ or } 6.\end{aligned}$$



When $x = 1$, $y = 1 + 3 = 4$. When $x = 6$, $y = 6 + 3 = 9$. So the graphs meet at $(1, 4)$ and $(6, 9)$.

b The sketch is on the right.

c At $x = 2$, the line gives $y = 5$ and the parabola gives $y = 1$, so the line is above the parabola.

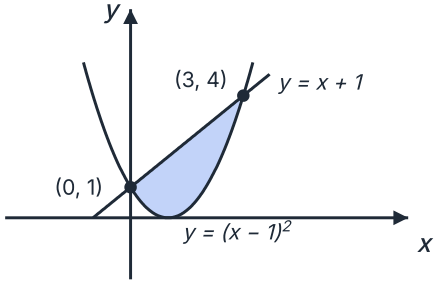
$$\begin{aligned}\text{area} &= \int_1^6 \left((x+3) - (x-3)^2\right) dx \\ &= \int_1^6 \left(x + 3 - (x^2 - 6x + 9)\right) dx \\ &= \int_1^6 (-x^2 + 7x - 6)\,dx \\ &= \left[-\frac{x^3}{3} + \frac{7x^2}{2} - 6x\right]_1^6 \\ &= \left(-72 + 126 - 36\right) - \left(-\tfrac{1}{3} + 3\tfrac{1}{2} - 6\right) \\ &= 18 - \left(-2\tfrac{5}{6}\right) \\ &= 20\tfrac{5}{6} \text{ square units.}\end{aligned}$$

(At $x = 6$: $-\frac{216}{3} = -72$ and $\frac{7\times36}{2} = 126$. At $x = 1$: $-\frac{2}{6} + \frac{21}{6} - \frac{36}{6} = -\frac{17}{6} = -2\frac{5}{6}$.)

Your turn.

a Substituting $y = x + 1$ into $y = (x-1)^2$ gives

$$\begin{aligned}(x-1)^2 &= x+1\\ x^2 - 2x + 1 &= x+1\\ x^2 - 3x &= 0\\ x(x-3) &= 0\\ x &= 0 \text{ or } 3.\end{aligned}$$



When $x = 0$, $y = 1$. When $x = 3$, $y = 4$. So the graphs meet at $(0, 1)$ and $(3, 4)$.

b The sketch is on the right.

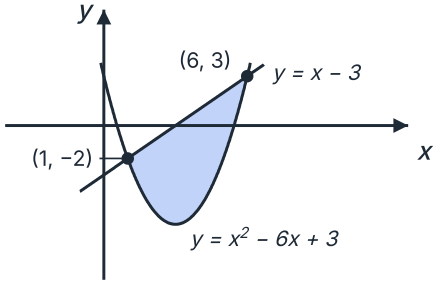
c At $x = 1$, the line gives $y = 2$ and the parabola gives $y = 0$, so the line is above the parabola.

$$\begin{aligned}\text{area} &= \int_0^3 \left((x+1) - (x-1)^2\right) dx\\ &= \int_0^3 \left(x + 1 - (x^2 - 2x + 1)\right) dx\\ &= \int_0^3 (-x^2 + 3x)\, dx\\ &= \left[-\frac{x^3}{3} + \frac{3x^2}{2}\right]_0^3\\ &= \left(-9 + 13\tfrac{1}{2}\right) - (0+0)\\ &= 4\tfrac{1}{2} \text{ square units.}\end{aligned}$$

C3 **Tutor demo.**

a Substituting $y = x - 3$ into $y = x^2 - 6x + 3$ gives

$$\begin{aligned}x^2 - 6x + 3 &= x - 3\\ x^2 - 7x + 6 &= 0\\ (x-1)(x-6) &= 0\\ x &= 1 \text{ or } 6.\end{aligned}$$



When $x = 1$, $y = -2$. When $x = 6$, $y = 3$. So the curves meet at $(1, -2)$ and $(6, 3)$.

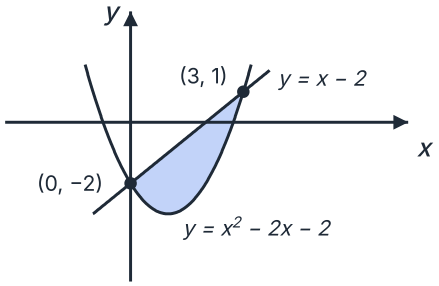
b The sketch is on the right. The line is above the parabola: at $x = 2$, the line gives $y = -1$ and the parabola gives $y = -5$. The region crosses the x-axis, but Box 17 still applies.

$$\begin{aligned}\text{area} &= \int_1^6 \big((x-3)-(x^2-6x+3)\big)\,dx \\ &= \int_1^6 (x-3-x^2+6x-3)\,dx \\ &= \int_1^6 (-x^2+7x-6)\,dx \\ &= \Big[-\frac{x^3}{3}+\frac{7x^2}{2}-6x\Big]_1^6 \\ &= \big(-72+126-36\big)-\big(-\tfrac{1}{3}+3\tfrac{1}{2}-6\big) \\ &= 18+2\tfrac{5}{6} \\ &= 20\tfrac{5}{6}\ \text{square units},\end{aligned}$$

the same as in C2, because both curves are those of C2 shifted down 6.

Your turn.

a Substituting $y = x - 2$ into $y = x^2 - 2x - 2$ gives



$$\begin{aligned}x^2-2x-2 &= x-2 \\ x^2-3x &= 0 \\ x(x-3) &= 0 \\ x &= 0 \text{ or } 3.\end{aligned}$$

When $x = 0$, $y = -2$. When $x = 3$, $y = 1$. So the curves meet at $(0, -2)$ and $(3, 1)$.

b The sketch is on the right. The line is above the parabola: at $x = 1$, the line gives $y = -1$ and the parabola gives $y = -3$.

$$\begin{aligned}\text{area} &= \int_0^3 \big((x-2)-(x^2-2x-2)\big)\,dx \\ &= \int_0^3 (x-2-x^2+2x+2)\,dx \\ &= \int_0^3 (-x^2+3x)\,dx \\ &= \Big[-\frac{x^3}{3}+\frac{3x^2}{2}\Big]_0^3 \\ &= \big(-9+13\tfrac{1}{2}\big)-(0+0) \\ &= 4\tfrac{1}{2}\ \text{square units}.\end{aligned}$$

C4 **Tutor demo.** In the left-hand region $0 \le x \le 1$, the second curve is above the first: at $x = 0$, the second curve gives $y = 7$ and the first gives $y = -1$.

$$\begin{aligned}\text{area} &= \int_0^1 \left((x^2 - 8x + 7) - (-x^2 + 2x - 1)\right) dx \\ &= \int_0^1 (x^2 - 8x + 7 + x^2 - 2x + 1)\, dx \\ &= \int_0^1 (2x^2 - 10x + 8)\, dx \\ &= \left[\frac{2x^3}{3} - 5x^2 + 8x\right]_0^1 \\ &= \left(\tfrac{2}{3} - 5 + 8\right) - (0 - 0 + 0) \\ &= 3\tfrac{2}{3}\ \text{u}^2\end{aligned}$$

In the right-hand region $1 \leq x \leq 4$, the first curve is above the second: at $x = 2$, the first curve gives $y = -1$ and the second gives $y = -5$.

$$\begin{aligned}\text{area} &= \int_1^4 \left((-x^2 + 2x - 1) - (x^2 - 8x + 7)\right) dx \\ &= \int_1^4 (-x^2 + 2x - 1 - x^2 + 8x - 7)\, dx \\ &= \int_1^4 (-2x^2 + 10x - 8)\, dx \\ &= \left[-\frac{2x^3}{3} + 5x^2 - 8x\right]_1^4 \\ &= \left(-42\tfrac{2}{3} + 80 - 32\right) - \left(-\tfrac{2}{3} + 5 - 8\right) \\ &= 5\tfrac{1}{3} - \left(-3\tfrac{2}{3}\right) \\ &= 9\ \text{u}^2\end{aligned}$$

$$\begin{aligned}\text{total area} &= 3\tfrac{2}{3} + 9 \\ &= 12\tfrac{2}{3}\ \text{u}^2.\end{aligned}$$

Your turn. In the left-hand region $0 \leq x \leq 1$, the second curve is above the first: at $x = 0$, the second curve gives $y = 5$ and the first gives $y = -1$.

$$\begin{aligned}\text{area} &= \int_0^1 \left((x^2 - 6x + 5) - (-x^2 + 2x - 1)\right) dx \\ &= \int_0^1 (2x^2 - 8x + 6)\, dx \\ &= \left[\frac{2x^3}{3} - 4x^2 + 6x\right]_0^1 \\ &= \left(\tfrac{2}{3} - 4 + 6\right) - (0 - 0 + 0) \\ &= 2\tfrac{2}{3}\ \text{u}^2\end{aligned}$$

In the right-hand region $1 \leq x \leq 3$, the first curve is above the second: at $x = 2$, the first curve gives $y = -1$ and the second gives $y = -3$.

$$\begin{aligned}
\text{area} &= \int_1^3 \left((-x^2+2x-1)-(x^2-6x+5)\right)dx \\
&= \int_1^3 (-2x^2+8x-6)\,dx \\
&= \left[-\frac{2x^3}{3}+4x^2-6x\right]_1^3 \\
&= \left(-18+36-18\right)-\left(-\tfrac{2}{3}+4-6\right) \\
&= 0-\left(-2\tfrac{2}{3}\right) \\
&= 2\tfrac{2}{3}\text{ u}^2 \\
\text{total area} &= 2\tfrac{2}{3}+2\tfrac{2}{3} \\
&= 5\tfrac{1}{3}\text{ u}^2.
\end{aligned}$$

Textbook Homework

1 a $\frac{1}{6}$ u^2 b $\frac{1}{4}$ u^2 c $\frac{3}{10}$ u^2 d $\frac{1}{12}$ u^2 e $\frac{2}{35}$ u^2 f $20\frac{5}{6}$ u^2 g 36 u^2 h $20\frac{5}{6}$ u^2

2 a $\frac{4}{3}$ u^2 b $\frac{1}{6}$ u^2 c $\frac{4}{3}$ u^2 d $4\frac{1}{2}$ u^2

3 a $5\frac{1}{3}$ u^2 b $\frac{9}{4}$ u^2

4 a $16\frac{2}{3}$ u^2 b $9\frac{1}{3}$ u^2

5 c $4\frac{1}{2}$ u^2

6 c $\frac{4}{3}$ u^2

7 c 36 u^2

8 a $4\frac{1}{2}$ u^2 b $20\frac{5}{6}$ u^2 c $2\frac{2}{3}$ u^2

9 c 36 u^2

10 c $\frac{4}{3}$ u^2

11 a $4\frac{1}{2}$ u^2 b $20\frac{5}{6}$ u^2 c $21\frac{1}{3}$ u^2

12 c $\frac{1}{3}$ u^2

13 b $y=x-2$ c $5\frac{1}{3}$ u^2

14 c 108 u^2

15 a The points are $(-4,-67)$, $(1,-2)$ and $(2,5)$. c $73\frac{5}{6}$ u^2

16 a $(0,0)$, $\left(\frac{1}{2},\frac{1}{8}\right)$, $(1,0)$ b $\frac{1}{16}$ u^2

17 a $-1<x<1$ or $x>4$ b $21\frac{1}{12}$ u^2

18 b $y=2x-7$ c $\frac{7}{12}$ u^2

19 $k=1-\dfrac{1}{\sqrt[3]{2}}$

Integration Part 1 Integration Basics - Homework Answers

Lesson 5H | The trapezoidal rule

Lesson Examples

C1

Tutor demo. Table of values for $f(x) = \dfrac{12}{x}$:

x	1	2	3	4
$\dfrac{12}{x}$	12	6	4	3

a One subinterval uses the two end values.

$$\begin{aligned}\int_1^4 \frac{12}{x}\,dx &\doteqdot \frac{4-1}{2} \times \Big(f(1)+f(4)\Big)\\ &= \tfrac{3}{2} \times (12+3)\\ &= \tfrac{3}{2} \times 15\\ &= 22\tfrac{1}{2}\end{aligned}$$

b Three subintervals of width 1 use the four values: $\int_1^4 = \int_1^2 + \int_2^3 + \int_3^4$, so

$$\begin{aligned}\int_1^4 \frac{12}{x}\,dx &\doteqdot \tfrac{1}{2}\Big(f(1)+f(2)\Big) + \tfrac{1}{2}\Big(f(2)+f(3)\Big) + \tfrac{1}{2}\Big(f(3)+f(4)\Big)\\ &= \tfrac{1}{2}(12+6) + \tfrac{1}{2}(6+4) + \tfrac{1}{2}(4+3)\\ &= 9+5+3\tfrac{1}{2}\\ &= 17\tfrac{1}{2}\end{aligned}$$

Your turn. Table of values for $f(x) = \dfrac{60}{x}$:

x	2	3	4	5
$\dfrac{60}{x}$	30	20	15	12

a
$$\begin{aligned}\int_2^5 \frac{60}{x}\,dx &\doteqdot \frac{5-2}{2} \times \Big(f(2)+f(5)\Big)\\ &= \tfrac{3}{2} \times (30+12)\\ &= \tfrac{3}{2} \times 42\\ &= 63\end{aligned}$$

b $\displaystyle\int_2^5 \frac{60}{x}\,dx \doteqdot \tfrac{1}{2}\Big(f(2)+f(3)\Big)+\tfrac{1}{2}\Big(f(3)+f(4)\Big)+\tfrac{1}{2}\Big(f(4)+f(5)\Big)$

$$\begin{aligned}&= \tfrac{1}{2}(30+20)+\tfrac{1}{2}(20+15)+\tfrac{1}{2}(15+12)\\&= 25+17\tfrac{1}{2}+13\tfrac{1}{2}\\&= 56\end{aligned}$$

C2 **Tutor demo. a** $f(1)=50\times 1-1^4=50-1=49$ and $f(3)=50\times 3-3^4=150-81=69$.

$$\begin{aligned}\int_1^3 (50x-x^4)\,dx &\doteqdot \frac{3-1}{2}\times\Big(f(1)+f(3)\Big)\\&= 1\times(49+69)\\&= 118\end{aligned}$$

b $y=50x-x^4$, so $y'=50-4x^3$ and $y''=-12x^2$. Since $x^2>0$, $y''<0$ throughout $1\le x\le 3$, so the curve is concave down, the chord is below the curve, and the trapezoidal rule underestimates the integral.

c

$$\begin{aligned}\int_1^3 (50x-x^4)\,dx &= \Big[25x^2-\tfrac{1}{5}x^5\Big]_1^3\\&= \big(25\times 9-\tfrac{1}{5}\times 243\big)-\big(25\times 1-\tfrac{1}{5}\times 1\big)\\&= (225-48.6)-(25-0.2)\\&= 176.4-24.8\\&= 151.6,\end{aligned}$$

which is greater than 118, as part **b** predicts.

Your turn. a $f(1)=1^3+1=2$ and $f(3)=3^3+1=28$.

$$\begin{aligned}\int_1^3 (x^3+1)\,dx &\doteqdot \frac{3-1}{2}\times\Big(f(1)+f(3)\Big)\\&= 1\times(2+28)\\&= 30\end{aligned}$$

b $y=x^3+1$, so $y'=3x^2$ and $y''=6x$. Since $x>0$, $y''>0$ throughout $1\le x\le 3$, so the curve is concave up, the chord is above the curve, and the trapezoidal rule overestimates the integral.

c

$$\begin{aligned}\int_1^3 (x^3+1)\,dx &= \Big[\tfrac{1}{4}x^4+x\Big]_1^3\\&= \big(\tfrac{1}{4}\times 81+3\big)-\big(\tfrac{1}{4}\times 1+1\big)\\&= 23\tfrac{1}{4}-1\tfrac{1}{4}\\&= 22,\end{aligned}$$

which is less than 30, as part **b** predicts.

C3 **Tutor demo. a** Multiply by $60 \times 60 = 3600$ to get cubic metres per hour, then divide by $1\,000\,000$ to get millions. For example, $50 \times 3600 = 180\,000$, which is 0.18 million, and $900 \times 3600 = 3\,240\,000$, which is 3.24 million.

t	0	3	6	9	12	15	18
R	0.18	1.44	3.24	2.52	1.8	1.08	0.72

b Here $h = 3$ and $n = 6$, with $a = x_0 = 0$, $x_1 = 3$, …, $x_5 = 15$, $x_6 = b = 18$.

$$\begin{aligned}\int_0^{18} R\,dt &\doteqdot \frac{3}{2}\Big(f(0) + 2f(3) + 2f(6) + 2f(9) \\ &\qquad +2f(12) + 2f(15) + f(18)\Big) \\ &= 1.5\,(0.18 + 2.88 + 6.48 + 5.04 + 3.6 + 2.16 + 0.72) \\ &= 1.5 \times 21.06 \\ &= 31.59\end{aligned}$$

Using the second formula, $\dfrac{b-a}{2n} = \dfrac{18-0}{12} = 1.5$, and

$$\begin{aligned}\int_0^{18} R\,dt &\doteqdot \frac{18-0}{12}\Big(f(0) + f(18) \\ &\qquad +2\Big(f(3) + f(6) + f(9) + f(12) + f(15)\Big)\Big) \\ &= 1.5\,\big(0.18 + 0.72 + 2(1.44 + 3.24 + 2.52 + 1.8 + 1.08)\big) \\ &= 1.5\,(0.9 + 2 \times 10.08) \\ &= 1.5 \times 21.06 \\ &= 31.59\end{aligned}$$

About 31.6 million cubic metres of water flowed into the dam.

Your turn. The rate is in litres per minute and the time is in minutes, so no conversion is needed. Here $h = 10$ and $n = 4$.

$$\begin{aligned}\int_0^{40} (\text{rate})\,dt &\doteqdot \frac{10}{2}\Big(f(0) + 2f(10) + 2f(20) + 2f(30) + f(40)\Big) \\ &= 5\,(20 + 70 + 80 + 60 + 15) \\ &= 5 \times 245 \\ &= 1225\end{aligned}$$

$$\begin{aligned}\int_0^{40} (\text{rate})\,dt &\doteqdot \frac{40-0}{8}\Big(f(0) + f(40) + 2\Big(f(10) + f(20) + f(30)\Big)\Big) \\ &= 5\,\big(20 + 15 + 2(35 + 40 + 30)\big) \\ &= 5\,(35 + 210) \\ &= 1225\end{aligned}$$

About 1225 litres flowed into the tank.

C4

Tutor demo. Here $h = \dfrac{0.5 - 0}{10} = 0.05$. Set up the sheet as in Worked example 35, with these changes: in step 1, enter 0 into A2 and type `=0.05+A2` into B2; in step 6, type `=L4*0.025`, because $\frac{1}{2}h = 0.025$. Values to six decimal places:

x	$\phi(x)$	row 4
0	0.398942	0.398942
0.05	0.398444	0.796888
0.1	0.396953	0.793905
0.15	0.394479	0.788959
0.2	0.391043	0.782085
0.25	0.386668	0.773336
0.3	0.381388	0.762776
0.35	0.375240	0.750481
0.4	0.368270	0.736540
0.45	0.360527	0.721054
0.5	0.352065	0.352065

The sum of row 4 is $L4 \fallingdotseq 7.657031$, so

$$\int_0^{0.5} \phi(x)\,dx \fallingdotseq 0.025 \times 7.657031 \fallingdotseq 0.191426 \fallingdotseq 0.1914.$$

For $0 \le x \le 0.5$, $x^2 - 1 < 0$ and $\phi(x) > 0$, so $\phi''(x) < 0$. The curve is concave down, so this is an underestimate. (With 100 subintervals the estimate is 0.1915.)

Your turn. Here $h = \dfrac{2 - 1}{10} = 0.1$: enter 1 into A2 and keep `=0.1+A2` in B2 and `=L4*0.05` in L5.

x	$\phi(x)$	row 4
1	0.241971	0.241971
1.1	0.217852	0.435704
1.2	0.194186	0.388372
1.3	0.171369	0.342737
1.4	0.149727	0.299455
1.5	0.129518	0.259035
1.6	0.110921	0.221842
1.7	0.094049	0.188098
1.8	0.078950	0.157900
1.9	0.065616	0.131232
2	0.053991	0.053991

The sum of row 4 is $L4 \fallingdotseq 2.720337$, so

$$\int_1^2 \phi(x)\,dx \fallingdotseq 0.05 \times 2.720337 \fallingdotseq 0.136017 \fallingdotseq 0.1360.$$

For $1 < x \le 2$, $x^2 - 1 > 0$ and $\phi(x) > 0$, so $\phi''(x) > 0$. The curve is concave up, so this is an overestimate. (With 100 subintervals the estimate is 0.1359.)

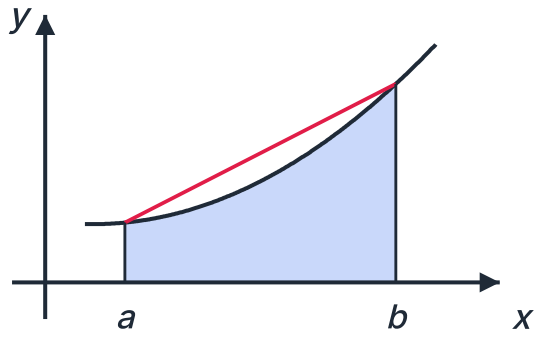
Textbook Homework

1 **a** $\frac{1}{2}(8 + 12) \times 4 = 40$ **b** $\frac{1}{2}(6.2 + 4.8) \times 4 = 22$ **c** $\frac{1}{2}(-4 - 9) \times 4 = -26$

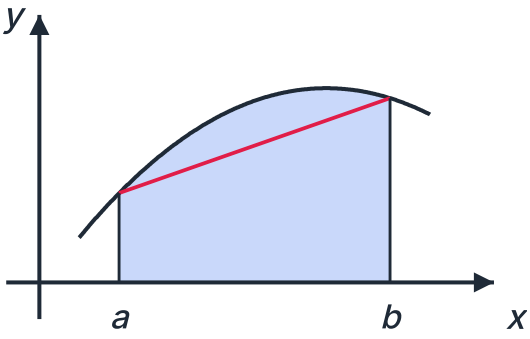
2 **a** $\frac{1}{2}(12+20)\times 4+\frac{1}{2}(20+30)\times 4=64+100=164$
b $h=4$: $\frac{4}{2}(12+2\times 20+30)=2\times 82=164$, the same.

3 $h=5$: $\frac{5}{2}(2.4+2\times 2.6+4.4)=2.5\times 12=30$

4 **a** The curve is concave up, so the chord is above the curve, and the area under the chord will be greater than the area under the curve.

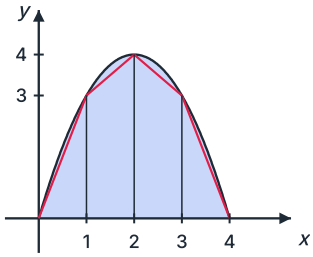


b The curve is concave down, so the chord is underneath the curve, and the area under the chord will be less than the area under the curve.



5 **a**

x	0	1	2	3	4
y	0	3	4	3	0



b $h=1$:
$\frac{1}{2}(0+2\times 3+2\times 4+2\times 3+0)=\frac{1}{2}\times 20=10$
c

$$\begin{aligned}\int_0^4 (4x-x^2)\,dx &= \left[2x^2-\tfrac{1}{3}x^3\right]_0^4\\ &= \left(32-\tfrac{64}{3}\right)-0\\ &= 10\tfrac{2}{3}\end{aligned}$$

The curve is concave down ($y''=-2<0$), so each chord is below the curve.
d The error is $10\frac{2}{3}-10=\frac{2}{3}$, and
$\frac{2}{3}\div 10\frac{2}{3}\times 100\%=\frac{1}{16}\times 100\%=6\frac{1}{4}\%$.

6 **a**

x	1	2	3	4	5
y	6	3	2	$1\frac{1}{2}$	$1\frac{1}{5}$

b $h=1$: $\frac{1}{2}\left(6+2\times 3+2\times 2+2\times 1\frac{1}{2}+1\frac{1}{5}\right)=\frac{1}{2}\times 20\frac{1}{5}=10\frac{1}{10}$
c $y=6x^{-1}$, so $y'=-6x^{-2}$ and $y''=12x^{-3}$. Then y'' is positive in the interval $1\le x\le 5$, so the curve is concave up.

8 **a** $h = \frac{1}{2}$: $\frac{1}{4}\left(1 + 2 \times 0.7071 + 0.5\right) \fallingdotseq 0.7286 \fallingdotseq 0.73$

b $h = 1$: $\frac{1}{2}(4 + 2 \times 2 + 1) = 4.5$

c $h = 1$:

$\frac{1}{2}\left(\sqrt[3]{7} + 2\sqrt[3]{5} + \sqrt[3]{3}\right) \fallingdotseq \frac{1}{2}(1.9129 + 3.4200 + 1.4422) \fallingdotseq 3.3876 \fallingdotseq 3.4$

d $h = 6$: $3\left(\sqrt{16} + 2\sqrt{10} + \sqrt{4}\right) \fallingdotseq 3(4 + 6.3246 + 2) \fallingdotseq 36.97 \fallingdotseq 37$

9 **a** $h = 1$: $\frac{1}{2}\left(\frac{1}{2} + 2 \times \frac{1}{3} + 2 \times \frac{1}{4} + 2 \times \frac{1}{5} + \frac{1}{6}\right)$

$= \frac{67}{60} \fallingdotseq 1.12$

b $h = \frac{1}{2}$: $\frac{1}{4}\left(0.5 + 2(0.36940 + 0.33333 + 0.31010) + 0.29289\right)$

$\fallingdotseq 0.70464 \fallingdotseq 0.705$

c $h = 1$: $\frac{1}{2}\left(\sqrt{13} + 2(\sqrt{22} + \sqrt{33} + \sqrt{46}) + \sqrt{61}\right)$

$\fallingdotseq \frac{1}{2}(3.6056 + 34.4346 + 7.8102) \fallingdotseq 22.925 \fallingdotseq 22.9$

d $h = \frac{1}{4}$: $\frac{1}{8}\left(0 + 2(0.09691 + 0.17609 + 0.24304) + 0.30103\right)$

$\fallingdotseq 0.16664 \fallingdotseq 0.167$

10 $h = 1$: $\frac{1}{2}\left(1.5 + 2(1.3 + 1.4 + 2.0 + 2.4) + 2.7\right) = \frac{1}{2} \times 18.4 = 9.2$ metres

11 The widths at 0 and 40 are zero. $h = 10$:

$5\left(0 + 2(20 + 18 + 17) + 0\right) = 5 \times 110 = 550 \text{ m}^2$

12 $h = 50$: $25\left(5 + 2(10 + 13 + 14 + 11 + 7) + 3\right) = 25 \times 118 = 2950 \text{ m}^2$

13 **a** $h = \frac{1}{4}$:

$\frac{1}{8}\left(1 + 2(0.968246 + 0.866025 + 0.661438) + 0\right) \fallingdotseq 0.748927 \fallingdotseq 0.7489$

b The integral is the area of a quarter of a unit circle, $\frac{\pi}{4}$, so $\pi \fallingdotseq 4 \times 0.7489 \fallingdotseq 3.0$.

The approximation is less than the integral, because the curve is concave down.

14 $h = 1$:

$$\begin{aligned}\int_1^5 \ln x \, dx &\fallingdotseq \tfrac{1}{2}\left(\ln 1 + 2\ln 2 + 2\ln 3 + 2\ln 4 + \ln 5\right) \\ &= \tfrac{1}{2}\ln\left(1 \times 2^2 \times 3^2 \times 4^2 \times 5\right) \\ &= \tfrac{1}{2}\ln 2880 \\ &= \ln\sqrt{2880} \\ &\fallingdotseq \ln 54, \text{ since } \sqrt{2880} \fallingdotseq 53.7.\end{aligned}$$

15 **a** $y' = \dfrac{1}{2\sqrt{x}} > 0$ for $x > 0$.

b On each subinterval $[k-1, k]$, the curve is increasing, so it lies below the upper rectangle of height $\sqrt{k}$ and width 1. Adding the n rectangles,

$$\sqrt{1} + \cdots + \sqrt{n} \geq \int_0^n \sqrt{x}\,dx = \left[\tfrac{2}{3}x^{\frac{3}{2}}\right]_0^n = \tfrac{2}{3}n\sqrt{n}.$$

c With $h = 1$, the trapezoidal rule gives $\frac{1}{2}\left(0 + 2\sqrt{1} + \cdots + 2\sqrt{n-1} + \sqrt{n}\right) = \sqrt{1} + \cdots + \sqrt{n} - \frac{1}{2}\sqrt{n}$. Since $y'' = -\frac{1}{4}x^{-\frac{3}{2}} < 0$, the curve is concave down and this underestimates the integral, so $\sqrt{1} + \cdots + \sqrt{n} - \frac{1}{2}\sqrt{n} \leq \frac{2}{3}n\sqrt{n}$, that is,

$$\sqrt{1} + \cdots + \sqrt{n} \leq \tfrac{2}{3}n\sqrt{n} + \tfrac{1}{2}\sqrt{n} = \frac{\sqrt{n}\,(4n+3)}{6}.$$

d With $n = 12\,000$, $876\,356 \leq \text{sum} \leq 876\,411$, so the sum is $876\,400$ to the nearest hundred.

16 The estimates settle down as the number of subintervals increases (values to 4 decimal places):

subintervals	5	10	20	100
a	2.6655	2.4744	2.4181	2.3987
b	16.1003	16.3034	16.3581	16.3761
c	1.0366	0.8863	0.8862	0.8862

In parts **a** and **c** the estimates decrease, and in part **b** they increase.

17 Readings from the graph (yours may differ a little), then $R = 3.6 \times \text{flow rate}$.

a Flow rates 2.4, 2.2, 2.1 and 2.3 m³/s give

t	17:40	23:40	05:40	11:40
R	8.64	7.92	7.56	8.28

With $h = 6$: $\frac{6}{2}\left(8.64 + 2 \times 7.92 + 2 \times 7.56 + 8.28\right) = 3 \times 47.88 = 143.64$, so about 144 thousand cubic metres.

b Flow rates 3.3, 4.0, 7.6, 12.7, 11.0, 7.4 and 6.3 m³/s give

t	17:40	20:40	23:40	02:40	05:40	08:40	11:40
R	11.88	14.4	27.36	45.72	39.6	26.64	22.68

With $h = 3$: $\frac{3}{2}\left(11.88 + 2(14.4 + 27.36 + 45.72 + 39.6 + 26.64) + 22.68\right)$
$= 1.5 \times 342 = 513$, so about 513 thousand cubic metres. The flow changes quickly in this period, with a sharp peak near 06/05/2004 03:40, so narrower subintervals are needed for the chords to follow the curve. In part **a** the flow is almost steady, so 6 hours is narrow enough.

c $513 \div 143.64 \fallingdotseq 3.6$, so about 3.6 times as much water. Light rain fell in the morning of 05/05/2004 with little effect. Steady rain from about 17:40 on 05/05/2004 and very heavy rain just after 23:40 were followed, within a few hours, by a steep rise to the peak. After the rain stopped, the flow fell quickly at first and then slowly over the next day.

Integration Part 1 Integration Basics - Homework Answers

Lesson 5H | The trapezoidal rule

Past-Paper Homework - Worked Solutions

PP 1 **Sydney Grammar School, 2023, Trial Examination, Form VI Mathematics Extension 1, Q3**

The shaded region runs from $x = 2$ to $x = 4$, so one application of the trapezoidal rule uses the two heights $\log_a 2$ and $\log_a 4$, with width $4 - 2 = 2$.

$$\begin{aligned}\text{Area} &\doteqdot \frac{4-2}{2}\left(\log_a 2 + \log_a 4\right)\\ &= \log_a 2 + \log_a 4\\ &= \log_a 2 + \log_a 2^2\\ &= \log_a 2 + 2\log_a 2\\ &= 3\log_a 2.\end{aligned}$$

Answer (A). Option (C) is the average of the two heights, without multiplying by the width 2. Option (B) is wrong because $\log_a 2 + \log_a 4 = \log_a(2 \times 4) = \log_a 8$, not $\log_a 6$. Option (D) does not come from the two heights.

Integration Part 1 Integration Basics - Homework Answers

Lesson 5I | The reverse chain rule

Lesson Examples

C1 **Tutor demo.**

a $\text{Let } y = (x^4 - 3)^5.$

$$\begin{aligned}\text{Then } \frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} \\ &= 5(x^4 - 3)^4 \times 4x^3 \\ &= 20x^3(x^4 - 3)^4.\end{aligned}$$

$$\begin{aligned}\text{Let } u &= x^4 - 3. \\ \text{Then } y &= u^5. \\ \text{Hence } \frac{du}{dx} &= 4x^3 \\ \text{and } \frac{dy}{du} &= 5u^4.\end{aligned}$$

b By part **a**, $\dfrac{d}{dx}(x^4 - 3)^5 = 20x^3(x^4 - 3)^4$. Reversing this,

$\displaystyle\int 20x^3(x^4 - 3)^4\,dx = (x^4 - 3)^5$, so $(x^4 - 3)^5$ is a primitive.

c $x^3(x^4 - 3)^4$ is $20x^3(x^4 - 3)^4$ divided by 20. Dividing by 20,

$\displaystyle\int x^3(x^4 - 3)^4\,dx = \tfrac{1}{20}(x^4 - 3)^5 + C.$

Your turn.

a $\text{Let } y = (x^2 + 7)^3.$

$$\begin{aligned}\text{Then } \frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} \\ &= 3(x^2 + 7)^2 \times 2x \\ &= 6x(x^2 + 7)^2.\end{aligned}$$

$$\begin{aligned}\text{Let } u &= x^2 + 7. \\ \text{Then } y &= u^3. \\ \text{Hence } \frac{du}{dx} &= 2x \\ \text{and } \frac{dy}{du} &= 3u^2.\end{aligned}$$

b By part **a**, $\dfrac{d}{dx}(x^2 + 7)^3 = 6x(x^2 + 7)^2$. Reversing this,

$\displaystyle\int 6x(x^2 + 7)^2\,dx = (x^2 + 7)^3$, so $(x^2 + 7)^3$ is a primitive.

c $x(x^2 + 7)^2$ is $6x(x^2 + 7)^2$ divided by 6. Dividing by 6,

$\displaystyle\int x(x^2 + 7)^2\,dx = \tfrac{1}{6}(x^2 + 7)^3 + C.$

C2 Tutor demo.

a Let $y = \dfrac{1}{4+x^3}$.

$$\begin{aligned}\text{Then } \frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} \\ &= -(4+x^3)^{-2} \times 3x^2 \\ &= \frac{-3x^2}{(4+x^3)^2}.\end{aligned}$$

Let $u = 4+x^3$.
Then $y = u^{-1}$.
Hence $\dfrac{du}{dx} = 3x^2$
and $\dfrac{dy}{du} = -u^{-2}$.

b By part **a**, $\dfrac{d}{dx}\left(\dfrac{1}{4+x^3}\right) = \dfrac{-3x^2}{(4+x^3)^2}$. Reversing this,

$$\int \frac{-3x^2}{(4+x^3)^2}\,dx = \frac{1}{4+x^3} + C.$$

Dividing by -3, $\displaystyle\int \frac{x^2}{(4+x^3)^2}\,dx = \frac{-1}{3(4+x^3)} + C.$

Your turn.

a Let $y = \dfrac{1}{9+x^2}$.

$$\begin{aligned}\text{Then } \frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} \\ &= -(9+x^2)^{-2} \times 2x \\ &= \frac{-2x}{(9+x^2)^2}.\end{aligned}$$

Let $u = 9+x^2$.
Then $y = u^{-1}$.
Hence $\dfrac{du}{dx} = 2x$
and $\dfrac{dy}{du} = -u^{-2}$.

b By part **a**, $\dfrac{d}{dx}\left(\dfrac{1}{9+x^2}\right) = \dfrac{-2x}{(9+x^2)^2}$. Reversing this,

$$\int \frac{-2x}{(9+x^2)^2}\,dx = \frac{1}{9+x^2} + C.$$

Dividing by -2, $\displaystyle\int \frac{x}{(9+x^2)^2}\,dx = \frac{-1}{2(9+x^2)} + C.$

C3 **Tutor demo.** The derivative of x^4+2 is $4x^3$, and $x^3 = \frac{1}{4} \times 4x^3$.

$$\begin{aligned}\int x^3(x^4+2)^5\,dx &= \tfrac{1}{4}\int 4x^3(x^4+2)^5\,dx \\ &= \tfrac{1}{4} \times \tfrac{1}{6}(x^4+2)^6 + C \\ &= \tfrac{1}{24}(x^4+2)^6 + C\end{aligned}$$

Let $u = x^4+2$.

Then $\dfrac{du}{dx} = 4x^3$.

Here $\displaystyle\int u^5 \frac{du}{dx}\,dx = \tfrac{1}{6}u^6$.

Your turn. The derivative of x^2-3 is $2x$, and $x = \frac{1}{2} \times 2x$.

$$\begin{aligned}\int x(x^2-3)^4\,dx &= \tfrac{1}{2}\int 2x(x^2-3)^4\,dx \\ &= \tfrac{1}{2} \times \tfrac{1}{5}(x^2-3)^5 + C \\ &= \tfrac{1}{10}(x^2-3)^5 + C\end{aligned}$$

Let $u = x^2-3$.

Then $\dfrac{du}{dx} = 2x$.

Here $\displaystyle\int u^4 \frac{du}{dx}\,dx = \tfrac{1}{5}u^5$.

C4 **Tutor demo.**

a $$\begin{aligned}\int x\sqrt{9-x^2}\,dx &= -\tfrac{1}{2}\int(-2x)\times(9-x^2)^{\frac{1}{2}}\,dx \\ &= -\tfrac{1}{2}\times\tfrac{2}{3}(9-x^2)^{\frac{3}{2}}+C \\ &= -\tfrac{1}{3}(9-x^2)^{\frac{3}{2}}+C\end{aligned}$$

Let $u = 9-x^2$.
Then $\dfrac{du}{dx} = -2x$.
Here $\displaystyle\int u^{\frac{1}{2}}\frac{du}{dx}\,dx = \tfrac{2}{3}u^{\frac{3}{2}}$.

b Using the primitive from part **a**, and $9^{\frac{3}{2}} = \left(\sqrt{9}\right)^3 = 3^3 = 27$,

$$\begin{aligned}\int_0^3 x\sqrt{9-x^2}\,dx &= -\tfrac{1}{3}\left[(9-x^2)^{\frac{3}{2}}\right]_0^3 \\ &= -\tfrac{1}{3}\left((9-9)^{\frac{3}{2}}-(9-0)^{\frac{3}{2}}\right) \\ &= -\tfrac{1}{3}(0-27) \\ &= 9\end{aligned}$$

c The definite integral is meaningless because $\sqrt{9-x^2}$ is undefined for $x > 3$, and the interval $0 \le x \le 4$ includes the values $3 < x \le 4$.

Your turn.

a $$\begin{aligned}\int x\sqrt{4-x^2}\,dx &= -\tfrac{1}{2}\int(-2x)\times(4-x^2)^{\frac{1}{2}}\,dx \\ &= -\tfrac{1}{2}\times\tfrac{2}{3}(4-x^2)^{\frac{3}{2}}+C \\ &= -\tfrac{1}{3}(4-x^2)^{\frac{3}{2}}+C\end{aligned}$$

Let $u = 4-x^2$.
Then $\dfrac{du}{dx} = -2x$.
Here $\displaystyle\int u^{\frac{1}{2}}\frac{du}{dx}\,dx = \tfrac{2}{3}u^{\frac{3}{2}}$.

b Using the primitive from part **a**, and $4^{\frac{3}{2}} = \left(\sqrt{4}\right)^3 = 2^3 = 8$,

$$\begin{aligned}\int_0^2 x\sqrt{4-x^2}\,dx &= -\tfrac{1}{3}\left[(4-x^2)^{\frac{3}{2}}\right]_0^2 \\ &= -\tfrac{1}{3}\left((4-4)^{\frac{3}{2}}-(4-0)^{\frac{3}{2}}\right) \\ &= -\tfrac{1}{3}(0-8) \\ &= \tfrac{8}{3}\end{aligned}$$

c The definite integral is meaningless because $\sqrt{4-x^2}$ is undefined for $x > 2$, and the interval $0 \le x \le 3$ includes the values $2 < x \le 3$.

Textbook Homework

C is a constant. In each part **b**, the integrand is compared with the derivative in part **a**.

1

a $\dfrac{d}{dx}(2x+3)^4 = 4(2x+3)^3 \times 2 = 8(2x+3)^3$

b i $\displaystyle\int 8(2x+3)^3\,dx = (2x+3)^4 + C$

ii $16(2x+3)^3 = 2 \times 8(2x+3)^3$, so $\displaystyle\int 16(2x+3)^3\,dx = 2(2x+3)^4 + C.$

2

a $\dfrac{d}{dx}(3x-5)^3 = 3(3x-5)^2 \times 3 = 9(3x-5)^2$

b i $\displaystyle\int 9(3x-5)^2\,dx = (3x-5)^3 + C$

ii $27(3x-5)^2 = 3 \times 9(3x-5)^2$, so $\displaystyle\int 27(3x-5)^2\,dx = 3(3x-5)^3 + C.$

3

a $\dfrac{d}{dx}(1+4x)^5 = 5(1+4x)^4 \times 4 = 20(1+4x)^4$

b i $\displaystyle\int 20(1+4x)^4\,dx = (1+4x)^5 + C$

ii $10(1+4x)^4 = \frac{1}{2} \times 20(1+4x)^4$, so $\displaystyle\int 10(1+4x)^4\,dx = \frac{1}{2}(1+4x)^5 + C$

4

a $\dfrac{d}{dx}(1-2x)^4 = 4(1-2x)^3 \times (-2) = -8(1-2x)^3$

b i $\displaystyle\int -8(1-2x)^3\,dx = (1-2x)^4 + C$

ii $-2(1-2x)^3 = \frac{1}{4} \times \big(-8(1-2x)^3\big)$, so
$\displaystyle\int -2(1-2x)^3\,dx = \frac{1}{4}(1-2x)^4 + C.$

5

a $\dfrac{d}{dx}(4x+3)^{-1} = -(4x+3)^{-2} \times 4 = -4(4x+3)^{-2}$

b i $\displaystyle\int -4(4x+3)^{-2}\,dx = (4x+3)^{-1} + C$

ii $(4x+3)^{-2} = -\frac{1}{4} \times \big(-4(4x+3)^{-2}\big)$, so
$\displaystyle\int (4x+3)^{-2}\,dx = -\frac{1}{4}(4x+3)^{-1} + C.$

6 a $\dfrac{d}{dx}(2x-5)^{\frac{1}{2}} = \frac{1}{2}(2x-5)^{-\frac{1}{2}} \times 2 = (2x-5)^{-\frac{1}{2}}$

b i $\displaystyle\int (2x-5)^{-\frac{1}{2}}\,dx = (2x-5)^{\frac{1}{2}} + C$

ii $\displaystyle\int \tfrac{1}{3}(2x-5)^{-\frac{1}{2}}\,dx = \tfrac{1}{3}(2x-5)^{\frac{1}{2}} + C$

7 a $\dfrac{d}{dx}(x^2+3)^4 = 4(x^2+3)^3 \times 2x = 8x(x^2+3)^3$

b i $\displaystyle\int 8x(x^2+3)^3\,dx = (x^2+3)^4 + C$

ii $40x(x^2+3)^3 = 5 \times 8x(x^2+3)^3$, so

$\displaystyle\int 40x(x^2+3)^3\,dx = 5(x^2+3)^4 + C.$

8 a $\dfrac{d}{dx}(x^3-1)^5 = 5(x^3-1)^4 \times 3x^2 = 15x^2(x^3-1)^4$

b i $\displaystyle\int 15x^2(x^3-1)^4\,dx = (x^3-1)^5 + C$

ii $3x^2(x^3-1)^4 = \frac{1}{5} \times 15x^2(x^3-1)^4$, so

$\displaystyle\int 3x^2(x^3-1)^4\,dx = \tfrac{1}{5}(x^3-1)^5 + C.$

9 a $\dfrac{d}{dx}\sqrt{2x^2+3} = \dfrac{d}{dx}(2x^2+3)^{\frac{1}{2}} = \frac{1}{2}(2x^2+3)^{-\frac{1}{2}} \times 4x = \dfrac{2x}{\sqrt{2x^2+3}}$

b i $\displaystyle\int \frac{2x}{\sqrt{2x^2+3}}\,dx = \sqrt{2x^2+3} + C$

ii $\dfrac{x}{\sqrt{2x^2+3}} = \frac{1}{2} \times \dfrac{2x}{\sqrt{2x^2+3}}$, so $\displaystyle\int \frac{x}{\sqrt{2x^2+3}}\,dx = \tfrac{1}{2}\sqrt{2x^2+3} + C.$

10 a $\dfrac{d}{dx}\sqrt{x} = \dfrac{d}{dx}x^{\frac{1}{2}} = \frac{1}{2}x^{-\frac{1}{2}} = \dfrac{1}{2\sqrt{x}}$, so

$\dfrac{d}{dx}\left(\sqrt{x}+1\right)^3 = 3\left(\sqrt{x}+1\right)^2 \times \dfrac{1}{2\sqrt{x}} = \dfrac{3\left(\sqrt{x}+1\right)^2}{2\sqrt{x}}$

b i $\displaystyle\int \frac{3\left(\sqrt{x}+1\right)^2}{2\sqrt{x}}\,dx = \left(\sqrt{x}+1\right)^3 + C$

ii $\dfrac{\left(\sqrt{x}+1\right)^2}{\sqrt{x}} = \frac{2}{3} \times \dfrac{3\left(\sqrt{x}+1\right)^2}{2\sqrt{x}}$, so

$\displaystyle\int \frac{\left(\sqrt{x}+1\right)^2}{\sqrt{x}}\,dx = \tfrac{2}{3}\left(\sqrt{x}+1\right)^3 + C.$

11 **a** $\dfrac{d}{dx}(x^3+3x^2+5)^4 = 4(x^3+3x^2+5)^3 \times (3x^2+6x)$

$= 4(x^3+3x^2+5)^3 \times 3(x^2+2x) = 12(x^2+2x)(x^3+3x^2+5)^3$

b i $\displaystyle\int 12(x^2+2x)(x^3+3x^2+5)^3\,dx = (x^3+3x^2+5)^4 + C$

ii Dividing by 12, $\displaystyle\int (x^2+2x)(x^3+3x^2+5)^3\,dx = \tfrac{1}{12}(x^3+3x^2+5)^4 + C.$

12 **a**

$\dfrac{d}{dx}(5-x^2-x)^7 = 7(5-x^2-x)^6 \times (-2x-1) = -7(2x+1)(5-x^2-x)^6$

b i $-14x-7 = -7(2x+1)$, so

$\displaystyle\int (-14x-7)(5-x^2-x)^6\,dx = (5-x^2-x)^7 + C.$

ii Dividing by -7, $\displaystyle\int (2x+1)(5-x^2-x)^6\,dx = -\tfrac{1}{7}(5-x^2-x)^7 + C.$

13 **a** $f(x) = 5x+4$, $f'(x) = 5$, $n = 3$:

$\displaystyle\int 5(5x+4)^3\,dx = \frac{(5x+4)^4}{4} + C = \tfrac{1}{4}(5x+4)^4 + C$

b $f(x) = 1-3x$, $f'(x) = -3$, $n = 5$:

$\displaystyle\int -3(1-3x)^5\,dx = \frac{(1-3x)^6}{6} + C = \tfrac{1}{6}(1-3x)^6 + C$

c $f(x) = x^2-5$, $f'(x) = 2x$, $n = 7$:

$\displaystyle\int 2x(x^2-5)^7\,dx = \frac{(x^2-5)^8}{8} + C = \tfrac{1}{8}(x^2-5)^8 + C$

d $f(x) = x^3+7$, $f'(x) = 3x^2$, $n = 4$:

$\displaystyle\int 3x^2(x^3+7)^4\,dx = \frac{(x^3+7)^5}{5} + C = \tfrac{1}{5}(x^3+7)^5 + C$

e $f(x) = 3x^2+2$, $f'(x) = 6x$, $n = -2$:

$\displaystyle\int \frac{6x}{(3x^2+2)^2}\,dx = \int 6x(3x^2+2)^{-2}\,dx$

$\displaystyle= \frac{(3x^2+2)^{-1}}{-1} + C = \frac{-1}{3x^2+2} + C$

f $f(x) = 9-2x^3$, $f'(x) = -6x^2$, $n = -\tfrac{1}{2}$:

$\displaystyle\int \frac{-6x^2}{\sqrt{9-2x^3}}\,dx = \int -6x^2(9-2x^3)^{-\frac{1}{2}}\,dx = \frac{(9-2x^3)^{\frac{1}{2}}}{\frac{1}{2}} + C$

$= 2\sqrt{9-2x^3} + C$

14

a $u = 5x^2 + 3$, $\dfrac{du}{dx} = 10x$, $n = 2$: $\displaystyle\int 10x(5x^2+3)^2\,dx = \tfrac{1}{3}(5x^2+3)^3 + C$

b $u = x^2 + 1$, $\dfrac{du}{dx} = 2x$, $n = 3$: $\displaystyle\int 2x(x^2+1)^3\,dx = \tfrac{1}{4}(x^2+1)^4 + C$

c $u = 1 + 4x^3$, $\dfrac{du}{dx} = 12x^2$, $n = 5$: $\displaystyle\int 12x^2(1+4x^3)^5\,dx = \tfrac{1}{6}(1+4x^3)^6 + C$

d $u = 1 + 3x^2$, $\dfrac{du}{dx} = 6x$, $n = 4$: $\displaystyle\int x(1+3x^2)^4\,dx = \tfrac{1}{6}\int 6x(1+3x^2)^4\,dx$

$= \tfrac{1}{6} \times \tfrac{1}{5}(1+3x^2)^5 + C = \tfrac{1}{30}(1+3x^2)^5 + C$

e $u = 1 - x^4$, $\dfrac{du}{dx} = -4x^3$, $n = 7$:

$\displaystyle\int x^3(1-x^4)^7\,dx = -\tfrac{1}{4}\int(-4x^3)(1-x^4)^7\,dx = -\tfrac{1}{4} \times \tfrac{1}{8}(1-x^4)^8 + C$

$= -\tfrac{1}{32}(1-x^4)^8 + C$

f $u = x^3 - 1$, $\dfrac{du}{dx} = 3x^2$, $n = \tfrac{1}{2}$: $\displaystyle\int 3x^2\sqrt{x^3-1}\,dx = \int 3x^2(x^3-1)^{\frac{1}{2}}\,dx$

$= \dfrac{(x^3-1)^{\frac{3}{2}}}{\frac{3}{2}} + C = \tfrac{2}{3}(x^3-1)^{\frac{3}{2}} + C$

g $u = 5x^2 + 1$, $\dfrac{du}{dx} = 10x$, $n = \tfrac{1}{2}$:

$\displaystyle\int x\sqrt{5x^2+1}\,dx = \tfrac{1}{10}\int 10x(5x^2+1)^{\frac{1}{2}}\,dx = \tfrac{1}{10} \times \tfrac{2}{3}(5x^2+1)^{\frac{3}{2}} + C$

$= \tfrac{1}{15}(5x^2+1)^{\frac{3}{2}} + C$

h $u = x^2 + 3$, $\dfrac{du}{dx} = 2x$, $n = -\tfrac{1}{2}$:

$\displaystyle\int \frac{2x}{\sqrt{x^2+3}}\,dx = \int 2x(x^2+3)^{-\frac{1}{2}}\,dx = \frac{(x^2+3)^{\frac{1}{2}}}{\frac{1}{2}} + C = 2\sqrt{x^2+3} + C$

i $u = 4x^2 + 8x + 1$, $\dfrac{du}{dx} = 8x + 8 = 8(x+1)$, $n = -\tfrac{1}{2}$:

$\displaystyle\int \frac{x+1}{\sqrt{4x^2+8x+1}}\,dx = \tfrac{1}{8}\int 8(x+1)(4x^2+8x+1)^{-\frac{1}{2}}\,dx$

$= \tfrac{1}{8} \times 2(4x^2+8x+1)^{\frac{1}{2}} + C = \tfrac{1}{4}\sqrt{4x^2+8x+1} + C$

j $u = x^2 + 5$, $\dfrac{du}{dx} = 2x$, $n = -3$: $\displaystyle\int \frac{x}{(x^2+5)^3}\,dx = \tfrac{1}{2}\int 2x(x^2+5)^{-3}\,dx$

$= \tfrac{1}{2} \times \dfrac{(x^2+5)^{-2}}{-2} + C = -\dfrac{1}{4(x^2+5)^2} + C$

15

a $u = x^3 + 1$, $\dfrac{du}{dx} = 3x^2$, $n = 4$:

$$\begin{aligned}\int_{-1}^{1} x^2(x^3+1)^4\,dx &= \tfrac{1}{3}\int_{-1}^{1} 3x^2(x^3+1)^4\,dx \\ &= \tfrac{1}{3} \times \tfrac{1}{5}\Big[(x^3+1)^5\Big]_{-1}^{1} \\ &= \tfrac{1}{15}\left((1+1)^5 - (-1+1)^5\right) = \tfrac{1}{15}(32-0) = \tfrac{32}{15}\end{aligned}$$

b $u = 5x^2 + 1$, $\dfrac{du}{dx} = 10x$, $n = -3$:

$$\begin{aligned}\int_{0}^{1} \frac{x}{(5x^2+1)^3}\,dx &= \tfrac{1}{10}\int_{0}^{1} 10x(5x^2+1)^{-3}\,dx \\ &= \tfrac{1}{10} \times \left(-\tfrac{1}{2}\right)\Big[(5x^2+1)^{-2}\Big]_{0}^{1} \\ &= -\tfrac{1}{20}\left(6^{-2} - 1^{-2}\right) = -\tfrac{1}{20}\left(\tfrac{1}{36} - 1\right) \\ &= -\tfrac{1}{20} \times \left(-\tfrac{35}{36}\right) = \tfrac{35}{720} = \tfrac{7}{144}\end{aligned}$$

c $u = 1 - 4x^2$, $\dfrac{du}{dx} = -8x$, $n = \frac{1}{2}$:

$$\begin{aligned}\int_{0}^{\frac{1}{2}} x\sqrt{1-4x^2}\,dx &= -\tfrac{1}{8}\int_{0}^{\frac{1}{2}} (-8x)(1-4x^2)^{\frac{1}{2}}\,dx \\ &= -\tfrac{1}{8} \times \tfrac{2}{3}\Big[(1-4x^2)^{\frac{3}{2}}\Big]_{0}^{\frac{1}{2}} \\ &= -\tfrac{1}{12}\left(0^{\frac{3}{2}} - 1^{\frac{3}{2}}\right) = -\tfrac{1}{12}(0-1) = \tfrac{1}{12}\end{aligned}$$

d $u = x^2 + 10x + 3$, $\dfrac{du}{dx} = 2x + 10 = 2(x+5)$, $n = 2$:

$$\begin{aligned}\int_{-3}^{-1} (x+5)(x^2+10x+3)^2\,dx &= \tfrac{1}{2}\int_{-3}^{-1} 2(x+5)(x^2+10x+3)^2\,dx \\ &= \tfrac{1}{2} \times \tfrac{1}{3}\Big[(x^2+10x+3)^3\Big]_{-3}^{-1} \\ &= \tfrac{1}{6}\left((1-10+3)^3 - (9-30+3)^3\right) \\ &= \tfrac{1}{6}\left((-6)^3 - (-18)^3\right) \\ &= \tfrac{1}{6}(-216+5832) = \tfrac{1}{6} \times 5616 = 936\end{aligned}$$

16 **a** $u = 1 - \dfrac{1}{x} = 1 - x^{-1}$, $\dfrac{du}{dx} = x^{-2} = \dfrac{1}{x^2}$, $n = 5$:

$$\int \frac{\left(1 - \frac{1}{x}\right)^5}{x^2}\,dx = \int \left(1 - \frac{1}{x}\right)^5 \times \frac{1}{x^2}\,dx = \tfrac{1}{6}\left(1 - \frac{1}{x}\right)^6 + C$$

b $u = 1 + \sqrt{x}$, $\dfrac{du}{dx} = \dfrac{1}{2\sqrt{x}}$, $n = -2$:

$$\int_1^4 \frac{1}{\sqrt{x}\left(1+\sqrt{x}\right)^2}\,dx = 2\int_1^4 \left(1+\sqrt{x}\right)^{-2} \times \frac{1}{2\sqrt{x}}\,dx$$

$$= 2\left[\frac{\left(1+\sqrt{x}\right)^{-1}}{-1}\right]_1^4 = -2\left[\left(1+\sqrt{x}\right)^{-1}\right]_1^4$$

$$= -2\left((1+2)^{-1} - (1+1)^{-1}\right)$$

$$= -2\left(\tfrac{1}{3} - \tfrac{1}{2}\right) = -2 \times \left(-\tfrac{1}{6}\right) = \tfrac{1}{3}$$

17 **a** $\sqrt{x^2-1}$ needs $x^2 - 1 \geq 0$, so the domain is $x \leq -1$ or $x \geq 1$.

b By the product rule, with $\dfrac{d}{dx}(x^2-1)^{\frac{1}{2}} = \tfrac{1}{2}(x^2-1)^{-\frac{1}{2}} \times 2x$,

$$f'(x) = 1 \times \sqrt{x^2-1} + x \times \tfrac{1}{2}(x^2-1)^{-\frac{1}{2}} \times 2x$$

$$= \sqrt{x^2-1} + \frac{x^2}{\sqrt{x^2-1}}$$

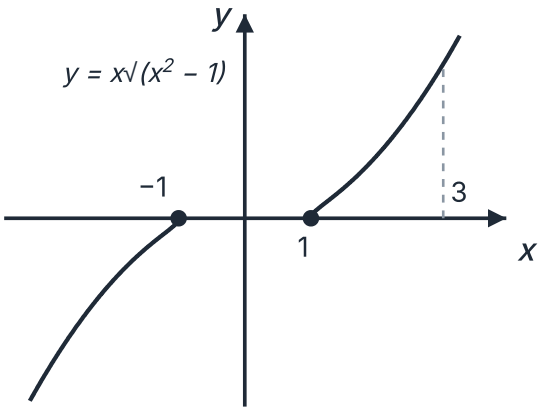
$$= \frac{(x^2-1)+x^2}{\sqrt{x^2-1}}$$

$$= \frac{2x^2-1}{\sqrt{x^2-1}}$$

$f'(x) = 0$ only when $2x^2 - 1 = 0$, that is, $x = \pm\dfrac{1}{\sqrt{2}}$. These values are not in the domain, because $\dfrac{1}{\sqrt{2}} < 1$. (At $x = \pm 1$, $f'(x)$ is undefined.) So there are no stationary points.

c $f(-x) = (-x)\sqrt{(-x)^2 - 1} = -x\sqrt{x^2 - 1} = -f(x)$, so f is odd and its graph has point symmetry about the origin.

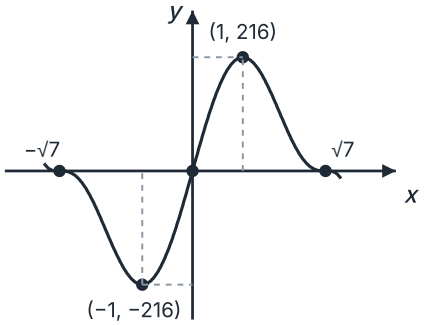
For $x \geq 1$, $f(x) \geq 0$, with $f(1) = 0$, and $f'(x) > 0$ for $x > 1$ (since $2x^2 - 1 > 0$), so the curve rises. The branch for $x \leq -1$ is the rotation of this branch through $180°$ about the origin.



d On $1 \leq x \leq 3$, $f(x) \geq 0$, and $8^{\frac{3}{2}} = 8\sqrt{8} = 8 \times 2\sqrt{2} = 16\sqrt{2}$. $u = x^2 - 1$, $\frac{du}{dx} = 2x$, $n = \frac{1}{2}$.

$$\begin{aligned}\text{Area} &= \int_1^3 x\sqrt{x^2 - 1}\,dx = \tfrac{1}{2}\int_1^3 2x(x^2 - 1)^{\frac{1}{2}}\,dx \\ &= \tfrac{1}{2} \times \tfrac{2}{3}\left[(x^2 - 1)^{\frac{3}{2}}\right]_1^3 \\ &= \tfrac{1}{3}\left(8^{\frac{3}{2}} - 0^{\frac{3}{2}}\right) = \tfrac{1}{3} \times 16\sqrt{2} = \tfrac{16\sqrt{2}}{3}\ \text{u}^2\end{aligned}$$

18 **a** Intercepts: $y = 0$ when $x = 0$ or $7 - x^2 = 0$, that is, at $(0, 0)$, $(\sqrt{7}, 0)$ and $(-\sqrt{7}, 0)$. By the product rule,



$$\begin{aligned} y' &= 1 \times (7 - x^2)^3 + x \times 3(7 - x^2)^2 \times (-2x) \\ &= (7 - x^2)^2\big((7 - x^2) - 6x^2\big) \\ &= (7 - x^2)^2(7 - 7x^2) \\ &= 7(7 - x^2)^2(1 - x^2) \end{aligned}$$

So $y' = 0$ at $x = \pm 1$ and $x = \pm\sqrt{7}$. At $x = 1$, $y = 1 \times 6^3 = 216$; at $x = -1$, $y = -216$.
Since $(7 - x^2)^2 \geq 0$, y' has the sign of $1 - x^2$: positive for $-1 < x < 1$, and negative for $|x| > 1$ (apart from the zeroes at $\pm\sqrt{7}$).
So $(1, 216)$ is a maximum turning point and $(-1, -216)$ is a minimum turning point. At $x = \pm\sqrt{7}$, y' is negative on both sides, so $(\sqrt{7}, 0)$ and $(-\sqrt{7}, 0)$ are horizontal points of inflection.

b The curve is odd: $y \geq 0$ for $0 \leq x \leq \sqrt{7}$ and $y \leq 0$ for $-\sqrt{7} \leq x \leq 0$, so the two parts have equal area. $u = 7 - x^2$, $\dfrac{du}{dx} = -2x$, $n = 3$.

$$\begin{aligned} \text{Area} &= 2\int_0^{\sqrt{7}} x(7 - x^2)^3\,dx \\ &= 2 \times \left(-\tfrac{1}{2}\right)\int_0^{\sqrt{7}} (-2x)(7 - x^2)^3\,dx \\ &= -\tfrac{1}{4}\Big[(7 - x^2)^4\Big]_0^{\sqrt{7}} \\ &= -\tfrac{1}{4}\left(0^4 - 7^4\right) = \tfrac{1}{4} \times 2401 = \tfrac{2401}{4} = 600\tfrac{1}{4}\ \text{u}^2 \end{aligned}$$

Integration Part 1 Integration Basics - Homework Answers

Lesson 5I | The reverse chain rule

Past-Paper Homework - Worked Solutions

PP 1 **Ascham School, 2020, Mathematics Extension 1 Trial HSC Examination, Q5**

The derivative of $1-x^2$ is $-2x$, and $x=-\frac{1}{2}\times(-2x)$. $u=1-x^2$, $\frac{du}{dx}=-2x$, $n=-\frac{1}{2}$.

$$\begin{aligned}\int\frac{x}{\sqrt{1-x^2}}\,dx&=-\tfrac{1}{2}\int(-2x)(1-x^2)^{-\frac{1}{2}}\,dx\\&=-\tfrac{1}{2}\times\frac{(1-x^2)^{\frac{1}{2}}}{\frac{1}{2}}+c\\&=-\tfrac{1}{2}\times2(1-x^2)^{\frac{1}{2}}+c\\&=-\sqrt{1-x^2}+c\end{aligned}$$

The answer is **D**. Option C is half of D, so its derivative is $\frac{x}{2\sqrt{1-x^2}}$, half the integrand.

Options A and B are not $-\sqrt{1-x^2}$ plus a constant.

PP 2 **Sydney Boys High School, 2020, Year 12 Term 3 Trial HSC, Mathematics Extension 1, Q10**

The derivative of $9-x^2$ is $-2x$, and $x=-\frac{1}{2}\times(-2x)$. $u=9-x^2$, $\frac{du}{dx}=-2x$, $n=-\frac{1}{2}$.

$$\begin{aligned}\int\frac{x}{\sqrt{9-x^2}}\,dx&=-\tfrac{1}{2}\int(-2x)(9-x^2)^{-\frac{1}{2}}\,dx\\&=-\tfrac{1}{2}\times\frac{(9-x^2)^{\frac{1}{2}}}{\frac{1}{2}}+C\\&=-\tfrac{1}{2}\times2(9-x^2)^{\frac{1}{2}}+C\\&=-\sqrt{9-x^2}+C\end{aligned}$$

The answer is **A**. Check: $\frac{d}{dx}\left(-\sqrt{9-x^2}\right)=-\frac{1}{2}(9-x^2)^{-\frac{1}{2}}\times(-2x)=\frac{x}{\sqrt{9-x^2}}$.

Option B differentiates to twice the integrand, option D to -2 times it, and option C to the opposite of the integrand.

Integration Part 1 Integration Basics - Homework Answers

Review | Chapter review exercise

1

a $\int_0^1 3x^2\,dx = \left[x^3\right]_0^1 = 1 - 0 = 1$ b $\int_1^2 x\,dx = \left[\frac{x^2}{2}\right]_1^2 = 2 - \frac{1}{2} = \frac{3}{2}$

c $\int_2^5 4x^3\,dx = \left[x^4\right]_2^5 = 625 - 16 = 609$

d $\int_{-1}^1 x^4\,dx = \left[\frac{x^5}{5}\right]_{-1}^1 = \frac{1}{5} - \left(-\frac{1}{5}\right) = \frac{2}{5}$

e $\int_{-4}^{-2} 2x\,dx = \left[x^2\right]_{-4}^{-2} = 4 - 16 = -12$

f $\int_{-3}^{-1} x^2\,dx = \left[\frac{x^3}{3}\right]_{-3}^{-1} = -\frac{1}{3} - (-9) = 8\frac{2}{3}$

g $\int_0^2 (x+3)\,dx = \left[\frac{x^2}{2} + 3x\right]_0^2 = (2+6) - 0 = 8$

h $\int_{-1}^4 (2x-5)\,dx = \left[x^2 - 5x\right]_{-1}^4 = (16-20) - (1+5) = -10$

i $\int_{-3}^1 (x^2 - 2x + 1)\,dx = \left[\frac{x^3}{3} - x^2 + x\right]_{-3}^1 = \left(\frac{1}{3} - 1 + 1\right) - (-9 - 9 - 3)$
$= \frac{1}{3} + 21 = 21\frac{1}{3}$

2

a $x(x-1) = x^2 - x$, so
$\int_1^3 (x^2 - x)\,dx = \left[\frac{x^3}{3} - \frac{x^2}{2}\right]_1^3 = \left(9 - \frac{9}{2}\right) - \left(\frac{1}{3} - \frac{1}{2}\right) = \frac{9}{2} + \frac{1}{6} = 4\frac{2}{3}$

b $(x+1)(x-3) = x^2 - 2x - 3$, so
$\int_{-1}^0 (x^2 - 2x - 3)\,dx = \left[\frac{x^3}{3} - x^2 - 3x\right]_{-1}^0 = 0 - \left(-\frac{1}{3} - 1 + 3\right) = -1\frac{2}{3}$

c $(2x-1)^2 = 4x^2 - 4x + 1$, so
$\int_0^1 (4x^2 - 4x + 1)\,dx = \left[\frac{4x^3}{3} - 2x^2 + x\right]_0^1 = \left(\frac{4}{3} - 2 + 1\right) - 0 = \frac{1}{3}$

3 **a** $\dfrac{x^2-3x}{x} = x-3$, so

$$\int_1^2 (x-3)\,dx = \left[\frac{x^2}{2}-3x\right]_1^2 = (2-6)-\left(\tfrac{1}{2}-3\right) = -4+\tfrac{5}{2} = -1\tfrac{1}{2}$$

b $\dfrac{3x^4-4x^2}{x^2} = 3x^2-4$, so

$$\int_2^3 (3x^2-4)\,dx = \Big[x^3-4x\Big]_2^3 = (27-12)-(8-8) = 15$$

c $\dfrac{x^3-2x^4}{x^2} = x-2x^2$, so

$$\int_{-2}^{-1} (x-2x^2)\,dx = \left[\frac{x^2}{2}-\frac{2x^3}{3}\right]_{-2}^{-1} = \left(\tfrac{1}{2}+\tfrac{2}{3}\right)-\left(2+\tfrac{16}{3}\right)$$
$$= \tfrac{7}{6}-\tfrac{44}{6} = -\tfrac{37}{6} = -6\tfrac{1}{6}$$

4 **a ii** $5k-20=10$, so $5k=30$ and $k=6$. **b ii** $k^2-k=6$, so $k^2-k-6=0$, $(k-3)(k+2)=0$, and $k=3$ or $k=-2$. The positive value is $k=3$.

5 **a** 0. The integral has zero width: the upper and lower limits are both 3.
b 0. The integrand is odd, $(-x)^3=-x^3$, and the interval $-2\le x\le 2$ is symmetric about $x=0$.
c 0. The integrand is odd, $(-x)^3-9(-x)=-(x^3-9x)$, and the interval $-3\le x\le 3$ is symmetric about $x=0$.

6 **a** A triangle from $x=0$ to $x=2$ with height 4, and a rectangle from $x=2$ to $x=3$ with height 4: $\displaystyle\int_0^3 f(x)\,dx = \tfrac{1}{2}\times 2\times 4 + 1\times 4 = 4+4 = 8$.
b A triangle above the x-axis from $x=0$ to $x=2$ with height 2, and a triangle below the x-axis from $x=2$ to $x=3$ with depth 1, which counts as negative:

$$\int_0^3 f(x)\,dx = \tfrac{1}{2}\times 2\times 2 - \tfrac{1}{2}\times 1\times 1 = 2-\tfrac{1}{2} = \tfrac{3}{2}.$$

7 **a i** $A(x) = \Big[4t-\tfrac{1}{2}t^2\Big]_{-2}^{x} = \left(4x-\tfrac{1}{2}x^2\right)-(-8-2) = 4x-\tfrac{1}{2}x^2+10$

a ii $A(x) = \Big[-t^{-1}\Big]_2^x = -x^{-1}-\left(-\tfrac{1}{2}\right) = \tfrac{1}{2}-x^{-1}$

b i $\dfrac{d}{dx}\left(4x-\tfrac{1}{2}x^2+10\right) = 4-x$ **b ii** $\dfrac{d}{dx}\left(\tfrac{1}{2}-x^{-1}\right) = x^{-2}$

c i x^5-5x^3+1 **c ii** $\dfrac{x^2+4}{x^2-1}$ (the integrand with t replaced by x)

8

a $\dfrac{x^2}{2}+2x+C$

b $\dfrac{x^4}{4}+3\times\dfrac{x^3}{3}-5\times\dfrac{x^2}{2}+x+C=\dfrac{x^4}{4}+x^3-\dfrac{5x^2}{2}+x+C$

c $x(x-1)=x^2-x$, so the integral is $\dfrac{x^3}{3}-\dfrac{x^2}{2}+C$.

d $(x-3)(2-x)=-x^2+5x-6$, so the integral is $-\dfrac{x^3}{3}+\dfrac{5x^2}{2}-6x+C$.

e $\dfrac{x^{-1}}{-1}+C=-x^{-1}+C$ **f** $\displaystyle\int x^{-7}\,dx=\frac{x^{-6}}{-6}+C=-\frac{1}{6x^6}+C$

g $\displaystyle\int x^{\frac{1}{2}}\,dx=\frac{x^{\frac{3}{2}}}{\frac{3}{2}}+C=\frac{2x^{\frac{3}{2}}}{3}+C$ **h** $\dfrac{(x+1)^5}{5}+C=\frac{1}{5}(x+1)^5+C$

i $\dfrac{(2x-3)^6}{2\times6}+C=\frac{1}{12}(2x-3)^6+C$

9

a $\displaystyle\int_{-3}^{1}x^2\,dx=\left[\frac{x^3}{3}\right]_{-3}^{1}=\frac{1}{3}-(-9)=9\frac{1}{3}$, so the area is $9\frac{1}{3}\ \text{u}^2$.

b $\displaystyle\int_{-2}^{0}(x^3-4x)\,dx=\left[\frac{x^4}{4}-2x^2\right]_{-2}^{0}=0-(4-8)=4$, so the area is $4\ \text{u}^2$.

c $\displaystyle\int_{1}^{3}(x^2-4x+3)\,dx=\left[\frac{x^3}{3}-2x^2+3x\right]_{1}^{3}$
$=(9-18+9)-\left(\frac{1}{3}-2+3\right)=-\frac{4}{3}$. The region is below the x-axis, so the area is $\frac{4}{3}\ \text{u}^2$.

d Integrate with respect to y:
$\displaystyle\int_{3}^{4}(2y-6)\,dy=\Big[y^2-6y\Big]_{3}^{4}=(16-24)-(9-18)=-8+9=1$, so the area is $1\ \text{u}^2$.

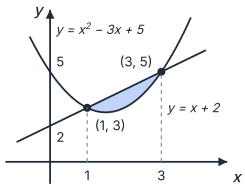
e $\displaystyle\int_{0}^{1}(x-x^2)\,dx=\left[\frac{x^2}{2}-\frac{x^3}{3}\right]_{0}^{1}=\frac{1}{2}-\frac{1}{3}=\frac{1}{6}$, so the area is $\frac{1}{6}\ \text{u}^2$.

f $\displaystyle\int_{-1}^{1}(x^2-x^4)\,dx=\left[\frac{x^3}{3}-\frac{x^5}{5}\right]_{-1}^{1}$
$=\left(\frac{1}{3}-\frac{1}{5}\right)-\left(-\frac{1}{3}+\frac{1}{5}\right)=\frac{2}{15}+\frac{2}{15}=\frac{4}{15}$, so the area is $\frac{4}{15}\ \text{u}^2$.

g $\int_1^2 (3x - 2 - x^2)\,dx = \left[\frac{3x^2}{2} - 2x - \frac{x^3}{3}\right]_1^2$
$= \left(6 - 4 - \frac{8}{3}\right) - \left(\frac{3}{2} - 2 - \frac{1}{3}\right) = -\frac{2}{3} + \frac{5}{6} = \frac{1}{6}$, so the area is $\frac{1}{6}$ u^2.

h
$\int_{-2}^1 ((1 - x^2) - (x - 1))\,dx = \int_{-2}^1 (2 - x - x^2)\,dx = \left[2x - \frac{x^2}{2} - \frac{x^3}{3}\right]_{-2}^1$
$= \left(2 - \frac{1}{2} - \frac{1}{3}\right) - \left(-4 - 2 + \frac{8}{3}\right) = \frac{7}{6} + \frac{10}{3} = \frac{27}{6} = 4\frac{1}{2}$, so the area is $4\frac{1}{2}$ u^2.

10 **b** Between $x = 1$ and $x = 3$ the line is above the parabola, and $(x + 2) - (x^2 - 3x + 5) = -x^2 + 4x - 3$.



$\text{Area} = \int_1^3 (-x^2 + 4x - 3)\,dx = \left[-\frac{x^3}{3} + 2x^2 - 3x\right]_1^3$
$= (-9 + 18 - 9) - \left(-\frac{1}{3} + 2 - 3\right) = 0 + \frac{4}{3} = \frac{4}{3}$ u^2

11 **a** Three function values give two subintervals, so $h = 1$.

x	1	2	3
2^x	2	4	8

$\int_1^3 2^x\,dx \div \frac{h}{2}(f(1) + 2f(2) + f(3)) = \frac{1}{2}(2 + 2 \times 4 + 8) = \frac{1}{2} \times 18 = 9$

b Five function values give four subintervals, so $h = \frac{1}{2}$. The values of $\log_{10} x$ are given to 4 decimal places.

x	1	1.5	2	2.5	3
$\log_{10} x$	0	0.1761	0.3010	0.3979	0.4771

$\int_1^3 \log_{10} x\,dx \div \frac{h}{2}(f(1) + 2f(1.5) + 2f(2) + 2f(2.5) + f(3))$
$\div \frac{1}{4}(0 + 2 \times (0.1761 + 0.3010 + 0.3979) + 0.4771)$
$\div \frac{1}{4} \times 2.2271 \div 0.5568 \div 0.56$ (to two significant figures)

12 **a** $\frac{d}{dx}(3x + 4)^6 = 6(3x + 4)^5 \times 3 = 18(3x + 4)^5$

b i $\int 18(3x + 4)^5\,dx = (3x + 4)^6 + C$

ii $9(3x + 4)^5 = \frac{1}{2} \times 18(3x + 4)^5$, so $\int 9(3x + 4)^5\,dx = \frac{1}{2}(3x + 4)^6 + C$.

13

a $\dfrac{d}{dx}(x^2-1)^3 = 3(x^2-1)^2 \times 2x = 6x(x^2-1)^2$

b i $\displaystyle\int 6x(x^2-1)^2\,dx = (x^2-1)^3 + C$

ii Dividing by 6, $\displaystyle\int x(x^2-1)^2\,dx = \tfrac{1}{6}(x^2-1)^3 + C.$

14

a $u = x^3+1$, $\dfrac{du}{dx} = 3x^2$, $n = 4$: $\displaystyle\int 3x^2(x^3+1)^4\,dx = \tfrac{1}{5}(x^3+1)^5 + C$

b $u = x^2-5$, $\dfrac{du}{dx} = 2x$, $n = -3$: $\displaystyle\int \frac{2x}{(x^2-5)^3}\,dx = \int 2x(x^2-5)^{-3}\,dx$

$$= \frac{(x^2-5)^{-2}}{-2} + C = -\frac{1}{2(x^2-5)^2} + C$$